

Gravity Model

5.4

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1 Module Index	1
1.1 Modules	1
2 Namespace Index	3
2.1 Namespace List	3
3 Hierarchical Index	5
3.1 Class Hierarchy	5
4 Data Structure Index	7
4.1 Data Structures	7
5 File Index	9
5.1 File List	9
6 Module Documentation	11
6.1 Models	11
6.1.1 Detailed Description	11
6.2 Environment	11
6.2.1 Detailed Description	11
6.3 Gravity	11
6.3.1 Detailed Description	12
7 Namespace Documentation	13
7.1 jeod Namespace Reference	13
7.1.1 Detailed Description	14
7.1.2 Variable Documentation	14
7.1.2.1 speed_of_light_sq	14
8 Data Structure Documentation	15
8.1 jeod::GravityControls Class Reference	15
8.1.1 Detailed Description	17
8.1.2 Constructor & Destructor Documentation	17
8.1.2.1 GravityControls() [1/2]	17
8.1.2.2 ~GravityControls()	17
8.1.2.3 GravityControls() [2/2]	17
8.1.3 Member Function Documentation	17
8.1.3.1 accel_mag_less_ptr()	17
8.1.3.2 calc_nonspherical()	18
8.1.3.3 calc_relativistic()	18
8.1.3.4 calc_spherical()	20
8.1.3.5 gravitation() [1/2]	20
8.1.3.6 gravitation() [2/2]	21
8.1.3.7 initialize_control()	21
8.1.3.8 JEOD_CLASS_ESTABLISH_FRIENDS2()	22

8.1.3.9 operator=()	22
8.1.3.10 reset_control()	22
8.1.4 Field Documentation	22
8.1.4.1 active	23
8.1.4.2 battin_method	23
8.1.4.3 body	23
8.1.4.4 gradient	24
8.1.4.5 grav_accel	24
8.1.4.6 grav_accel_magsq	24
8.1.4.7 grav_grad	24
8.1.4.8 grav_manager	25
8.1.4.9 grav_pot	25
8.1.4.10 perturbing_only	25
8.1.4.11 relativistic	25
8.1.4.12 skip_spherical	26
8.1.4.13 source_name	26
8.1.4.14 spherical	26
8.1.4.15 subscribed_to_inertial	26
8.1.4.16 subscribed_to_pfix	27
8.2 jeod::GravityIntegFrame Class Reference	27
8.2.1 Detailed Description	28
8.2.2 Constructor & Destructor Documentation	28
8.2.2.1 GravityIntegFrame()	28
8.2.2.2 ~GravityIntegFrame()	28
8.2.3 Member Function Documentation	28
8.2.3.1 p()	28
8.2.4 Field Documentation	28
8.2.4.1 accel	28
8.2.4.2 is_third_body	29
8.2.4.3 pos	29
8.2.4.4 ref_frame	29
8.2.4.5 time	29
8.3 jeod::GravityInteraction Class Reference	30
8.3.1 Detailed Description	31
8.3.2 Constructor & Destructor Documentation	31
8.3.2.1 GravityInteraction() [1/2]	31
8.3.2.2 ~GravityInteraction()	31
8.3.2.3 GravityInteraction() [2/2]	31
8.3.3 Member Function Documentation	31
8.3.3.1 add_control()	31
8.3.3.2 initialize_controls()	32
8.3.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()	32

8.3.3.4 operator=()	32
8.3.3.5 remove_control()	32
8.3.3.6 reset_controls()	33
8.3.3.7 set_integ_frame()	33
8.3.3.8 sort_controls()	34
8.3.4 Field Documentation	34
8.3.4.1 grav_accel	34
8.3.4.2 grav_controls	34
8.3.4.3 grav_grad	35
8.3.4.4 grav_pot	35
8.3.4.5 integ_frame_index	35
8.4 jeod::GravityManager Class Reference	35
8.4.1 Detailed Description	36
8.4.2 Constructor & Destructor Documentation	36
8.4.2.1 GravityManager() [1/2]	36
8.4.2.2 ~GravityManager()	37
8.4.2.3 GravityManager() [2/2]	37
8.4.3 Member Function Documentation	37
8.4.3.1 add_grav_source()	37
8.4.3.2 find_grav_source()	37
8.4.3.3 get_bodies()	38
8.4.3.4 gravitation() [1/2]	38
8.4.3.5 gravitation() [2/2]	39
8.4.3.6 initialize_model()	39
8.4.3.7 initialize_state()	40
8.4.3.8 operator=()	40
8.4.3.9 p()	40
8.4.4 Field Documentation	41
8.4.4.1 sources	41
8.5 jeod::GravityMessages Class Reference	41
8.5.1 Detailed Description	42
8.5.2 Constructor & Destructor Documentation	42
8.5.2.1 GravityMessages() [1/2]	42
8.5.2.2 GravityMessages() [2/2]	42
8.5.3 Member Function Documentation	42
8.5.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()	42
8.5.3.2 operator=()	42
8.5.4 Field Documentation	42
8.5.4.1 domain_error	43
8.5.4.2 duplicate_entry	43
8.5.4.3 invalid_limit	43
8.5.4.4 invalid_name	43

8.5.4.5 invalid_object	44
8.5.4.6 missing_entry	44
8.5.4.7 null_pointer	44
8.6 jeod::GravitySource Class Reference	45
8.6.1 Detailed Description	45
8.6.2 Constructor & Destructor Documentation	46
8.6.2.1 GravitySource() [1/2]	46
8.6.2.2 ~GravitySource()	46
8.6.2.3 GravitySource() [2/2]	46
8.6.3 Member Function Documentation	46
8.6.3.1 initialize_state()	46
8.6.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	47
8.6.3.3 operator=()	47
8.6.4 Field Documentation	47
8.6.4.1 frames	47
8.6.4.2 inertial	47
8.6.4.3 mu	48
8.6.4.4 name	48
8.6.4.5 pfix	49
8.7 jeod::SphericalHarmonicsDeltaCoeffs Class Reference	49
8.7.1 Detailed Description	50
8.7.2 Constructor & Destructor Documentation	50
8.7.2.1 SphericalHarmonicsDeltaCoeffs()	50
8.7.2.2 ~SphericalHarmonicsDeltaCoeffs()	50
8.7.3 Member Function Documentation	50
8.7.3.1 initialize()	50
8.7.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	51
8.7.3.3 update()	51
8.7.4 Field Documentation	51
8.7.4.1 dC20	52
8.7.4.2 degree	52
8.7.4.3 delta_Cnm	52
8.7.4.4 delta_Snm	52
8.7.4.5 grav_source	53
8.7.4.6 order	53
8.8 jeod::SphericalHarmonicsDeltaCoeffsInit Class Reference	53
8.8.1 Detailed Description	54
8.8.2 Constructor & Destructor Documentation	54
8.8.2.1 SphericalHarmonicsDeltaCoeffsInit()	54
8.8.2.2 ~SphericalHarmonicsDeltaCoeffsInit()	54
8.8.3 Member Function Documentation	54
8.8.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()	55

8.8.4 Field Documentation	55
8.8.4.1 degree	55
8.8.4.2 delta_Cnm	55
8.8.4.3 delta_Snm	55
8.8.4.4 order	56
8.9 jeod::SphericalHarmonicsDeltaControls Class Reference	56
8.9.1 Detailed Description	57
8.9.2 Constructor & Destructor Documentation	57
8.9.2.1 SphericalHarmonicsDeltaControls()	57
8.9.2.2 ~SphericalHarmonicsDeltaControls()	57
8.9.3 Member Function Documentation	57
8.9.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()	57
8.9.4 Field Documentation	57
8.9.4.1 active	57
8.9.4.2 degree	58
8.9.4.3 first_order_only	58
8.9.4.4 grav_effect	58
8.9.4.5 grav_source	58
8.9.4.6 order	59
8.10 jeod::SphericalHarmonicsGravityControls Class Reference	59
8.10.1 Detailed Description	61
8.10.2 Constructor & Destructor Documentation	61
8.10.2.1 SphericalHarmonicsGravityControls() [1/2]	61
8.10.2.2 ~SphericalHarmonicsGravityControls()	62
8.10.2.3 SphericalHarmonicsGravityControls() [2/2]	62
8.10.3 Member Function Documentation	62
8.10.3.1 add_deltacontrol()	62
8.10.3.2 calc_nonspherical()	62
8.10.3.3 check_validity()	64
8.10.3.4 disable_min_radius_warnings()	64
8.10.3.5 get_degree()	65
8.10.3.6 get_degree_order()	65
8.10.3.7 get_grad_degree()	65
8.10.3.8 get_grad_degree_order()	66
8.10.3.9 get_grad_order()	67
8.10.3.10 get_order()	67
8.10.3.11 initialize_control()	67
8.10.3.12 JEOD_CLASS_ESTABLISH_FRIENDS2()	68
8.10.3.13 operator=()	68
8.10.3.14 set_degree()	68
8.10.3.15 set_degree_order()	68
8.10.3.16 set_grad_degree()	69

8.10.3.17 set_grad_degree_order()	69
8.10.3.18 set_grad_order()	70
8.10.3.19 set_order()	70
8.10.3.20 sum_deltacoeffs()	70
8.10.3.21 update_deltacoeffs()	71
8.10.4 Field Documentation	71
8.10.4.1 degree	71
8.10.4.2 delta_Cnm	71
8.10.4.3 delta_degree	72
8.10.4.4 delta_order	72
8.10.4.5 delta_Snm	72
8.10.4.6 gradient_degree	72
8.10.4.7 gradient_order	73
8.10.4.8 harmonics_source	73
8.10.4.9 min_radius_warn	73
8.10.4.10 order	74
8.10.4.11 Pnm	74
8.10.4.12 total_dC20	74
8.10.4.13 var_effects	74
8.11 jeod::SphericalHarmonicsGravitySource Class Reference	75
8.11.1 Detailed Description	76
8.11.2 Constructor & Destructor Documentation	76
8.11.2.1 SphericalHarmonicsGravitySource() [1/2]	76
8.11.2.2 ~SphericalHarmonicsGravitySource()	77
8.11.2.3 SphericalHarmonicsGravitySource() [2/2]	77
8.11.3 Member Function Documentation	77
8.11.3.1 add_deltacoeff()	77
8.11.3.2 find_deltacoeff()	77
8.11.3.3 initialize_body()	78
8.11.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()	78
8.11.3.5 operator=()	78
8.11.4 Field Documentation	78
8.11.4.1 a_by_rad	79
8.11.4.2 alpha	79
8.11.4.3 beta	79
8.11.4.4 Cnm	79
8.11.4.5 degree	80
8.11.4.6 delta_coeffs	80
8.11.4.7 eta	80
8.11.4.8 int_to_double	81
8.11.4.9 nrdiag	81
8.11.4.10 order	81

8.11.4.11 radius	82
8.11.4.12 Snm	82
8.11.4.13 tide_free	82
8.11.4.14 tide_free_delta	83
8.11.4.15 epsilon	83
8.11.4.16 xi	83
8.11.4.17 zeta	84
8.12 jeod::SphericalHarmonicsGravitySource_default_data Class Reference	84
8.12.1 Detailed Description	85
8.12.2 Constructor & Destructor Documentation	85
8.12.2.1 ~SphericalHarmonicsGravitySource_default_data()	85
8.12.3 Member Function Documentation	85
8.12.3.1 initialize()	85
8.13 jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data Class Reference	85
8.13.1 Detailed Description	86
8.13.2 Member Function Documentation	86
8.13.2.1 initialize()	86
8.14 jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data Class Reference	86
8.14.1 Detailed Description	87
8.14.2 Member Function Documentation	87
8.14.2.1 initialize()	87
8.15 jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data Class Reference	87
8.15.1 Detailed Description	88
8.15.2 Member Function Documentation	88
8.15.2.1 initialize()	88
8.16 jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data Class Reference	88
8.16.1 Detailed Description	89
8.16.2 Member Function Documentation	89
8.16.2.1 initialize()	89
8.17 jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data Class Reference	89
8.17.1 Detailed Description	89
8.17.2 Member Function Documentation	90
8.17.2.1 initialize()	90
8.18 jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data Class Reference	90
8.18.1 Detailed Description	90
8.18.2 Member Function Documentation	90
8.18.2.1 initialize()	91
8.19 jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data Class Reference	91
8.19.1 Detailed Description	91
8.19.2 Member Function Documentation	91
8.19.2.1 initialize()	92
8.20 jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data Class Reference	92

8.20.1 Detailed Description	92
8.20.2 Member Function Documentation	92
8.20.2.1 initialize()	93
8.21 jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data Class Reference	93
8.21.1 Detailed Description	93
8.21.2 Member Function Documentation	93
8.21.2.1 initialize()	94
8.22 jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data Class Reference	94
8.22.1 Detailed Description	94
8.22.2 Member Function Documentation	94
8.22.2.1 initialize()	95
8.23 jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data Class Reference	95
8.23.1 Detailed Description	95
8.23.2 Member Function Documentation	95
8.23.2.1 initialize()	96
8.24 jeod::SphericalHarmonicsSolidBodyTides Class Reference	96
8.24.1 Detailed Description	97
8.24.2 Constructor & Destructor Documentation	97
8.24.2.1 SphericalHarmonicsSolidBodyTides()	97
8.24.2.2 ~SphericalHarmonicsSolidBodyTides()	97
8.24.3 Member Function Documentation	97
8.24.3.1 initialize()	97
8.24.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	98
8.24.3.3 update()	98
8.25 jeod::SphericalHarmonicsSolidBodyTidesInit Class Reference	98
8.25.1 Detailed Description	99
8.25.2 Constructor & Destructor Documentation	99
8.25.2.1 SphericalHarmonicsSolidBodyTidesInit()	99
8.25.2.2 ~SphericalHarmonicsSolidBodyTidesInit()	99
8.25.3 Member Function Documentation	99
8.25.3.1 p()	99
8.26 jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data Class Reference	100
8.26.1 Detailed Description	100
8.26.2 Member Function Documentation	100
8.26.2.1 initialize()	100
8.27 jeod::SphericalHarmonicsTidalEffects Class Reference	100
8.27.1 Detailed Description	101
8.27.2 Constructor & Destructor Documentation	101
8.27.2.1 SphericalHarmonicsTidalEffects()	102
8.27.2.2 ~SphericalHarmonicsTidalEffects()	102
8.27.3 Member Function Documentation	102
8.27.3.1 initialize()	102

8.27.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	103
8.27.3.3 update()	103
8.27.4 Field Documentation	103
8.27.4.1 k2	103
8.27.4.2 Knm	103
8.27.4.3 num_tidal_bodies	104
8.27.4.4 pfix	104
8.27.4.5 tidal_bodies	104
8.27.4.6 tidal_bodies_inertial	104
8.27.4.7 xp	105
8.27.4.8 yp	105
8.28 jeod::SphericalHarmonicsTidalEffectsInit Class Reference	105
8.28.1 Detailed Description	106
8.28.2 Constructor & Destructor Documentation	106
8.28.2.1 SphericalHarmonicsTidalEffectsInit()	106
8.28.2.2 ~SphericalHarmonicsTidalEffectsInit()	106
8.28.3 Member Function Documentation	106
8.28.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()	107
8.28.4 Field Documentation	107
8.28.4.1 k2	107
8.28.4.2 Knm	107
8.28.4.3 tidal_body_names	107
8.28.4.4 xp	108
8.28.4.5 yp	108
9 File Documentation	109
9.1 class_declarations.hh File Reference	109
9.1.1 Detailed Description	109
9.2 earth_GEMT1.cc File Reference	109
9.2.1 Macro Definition Documentation	110
9.2.1.1 JEOD_FRIEND_CLASS	110
9.3 earth_GEMT1.hh File Reference	110
9.4 earth_GGM02C.cc File Reference	110
9.4.1 Macro Definition Documentation	111
9.4.1.1 JEOD_FRIEND_CLASS	111
9.5 earth_GGM02C.hh File Reference	111
9.6 earth_GGM05C.cc File Reference	111
9.6.1 Macro Definition Documentation	112
9.6.1.1 JEOD_FRIEND_CLASS	112
9.7 earth_GGM05C.hh File Reference	112
9.8 earth_solid_tides.cc File Reference	112
9.8.1 Macro Definition Documentation	113

9.8.1.1 JEOD_FRIEND_CLASS	113
9.9 earth_solid_tides.hh File Reference	113
9.10 earth_spherical.cc File Reference	113
9.10.1 Macro Definition Documentation	113
9.10.1.1 JEOD_FRIEND_CLASS	114
9.11 earth_spherical.hh File Reference	114
9.12 gravity_controls.cc File Reference	114
9.12.1 Detailed Description	115
9.13 gravity_controls.hh File Reference	115
9.13.1 Detailed Description	115
9.14 gravity_integ_frame.hh File Reference	115
9.14.1 Detailed Description	116
9.15 gravity_interaction.cc File Reference	116
9.15.1 Detailed Description	116
9.16 gravity_interaction.hh File Reference	117
9.16.1 Detailed Description	117
9.17 gravity_manager.cc File Reference	117
9.17.1 Detailed Description	118
9.18 gravity_manager.hh File Reference	118
9.18.1 Detailed Description	118
9.19 gravity_messages.cc File Reference	118
9.19.1 Detailed Description	119
9.19.2 Macro Definition Documentation	119
9.19.2.1 MAKE_GRAVITY_MESSAGE_CODE	119
9.20 gravity_messages.hh File Reference	119
9.20.1 Detailed Description	119
9.21 gravity_source.cc File Reference	120
9.21.1 Detailed Description	120
9.22 gravity_source.hh File Reference	120
9.22.1 Detailed Description	121
9.23 jupiter_spherical.cc File Reference	121
9.23.1 Macro Definition Documentation	121
9.23.1.1 JEOD_FRIEND_CLASS	121
9.24 jupiter_spherical.hh File Reference	121
9.25 mars_MRO110B2.cc File Reference	122
9.25.1 Macro Definition Documentation	122
9.25.1.1 JEOD_FRIEND_CLASS	122
9.26 mars_MRO110B2.hh File Reference	122
9.27 mars_spherical.cc File Reference	123
9.27.1 Macro Definition Documentation	123
9.27.1.1 JEOD_FRIEND_CLASS	123
9.28 mars_spherical.hh File Reference	123

9.29 moon_GRAIL150.cc File Reference	124
9.29.1 Macro Definition Documentation	124
9.29.1.1 JEOD_FRIEND_CLASS	124
9.30 moon_GRAIL150.hh File Reference	124
9.31 moon_LP150Q.cc File Reference	125
9.31.1 Macro Definition Documentation	125
9.31.1.1 JEOD_FRIEND_CLASS	125
9.32 moon_LP150Q.hh File Reference	125
9.33 moon_spherical.cc File Reference	126
9.33.1 Macro Definition Documentation	126
9.33.1.1 JEOD_FRIEND_CLASS	126
9.34 moon_spherical.hh File Reference	126
9.35 spherical_harmonics_calc_nonspherical.cc File Reference	127
9.35.1 Detailed Description	127
9.36 spherical_harmonics_delta_coeffs.cc File Reference	127
9.36.1 Detailed Description	127
9.37 spherical_harmonics_delta_coeffs.hh File Reference	128
9.37.1 Detailed Description	128
9.38 spherical_harmonics_delta_coeffs_init.hh File Reference	128
9.38.1 Detailed Description	128
9.39 spherical_harmonics_delta_controls.hh File Reference	129
9.39.1 Detailed Description	129
9.40 spherical_harmonics_gravity_controls.cc File Reference	129
9.40.1 Detailed Description	129
9.41 spherical_harmonics_gravity_controls.hh File Reference	130
9.41.1 Detailed Description	130
9.42 spherical_harmonics_gravity_source.cc File Reference	130
9.42.1 Detailed Description	131
9.43 spherical_harmonics_gravity_source.hh File Reference	131
9.43.1 Detailed Description	131
9.44 spherical_harmonics_gravity_source_default_data.hh File Reference	131
9.45 spherical_harmonics_solid_body_tides.cc File Reference	132
9.45.1 Detailed Description	132
9.46 spherical_harmonics_solid_body_tides.hh File Reference	132
9.46.1 Detailed Description	132
9.47 spherical_harmonics_solid_body_tides_init.hh File Reference	133
9.47.1 Detailed Description	133
9.48 spherical_harmonics_tidal_effects.cc File Reference	133
9.48.1 Detailed Description	134
9.49 spherical_harmonics_tidal_effects.hh File Reference	134
9.49.1 Detailed Description	134
9.50 spherical_harmonics_tidal_effects_init.hh File Reference	134

9.50.1 Detailed Description	135
9.51 sun_spherical.cc File Reference	135
9.51.1 Macro Definition Documentation	135
9.51.1.1 JEOD_FRIEND_CLASS	135
9.52 sun_spherical.hh File Reference	135
Index	137

Chapter 1

Module Index

1.1 Modules

Here is a list of all modules:

Models	11
Environment	11
Gravity	11

Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

jeod	Namespace jeod	13
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Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

jeod::GravityControls	15
jeod::SphericalHarmonicsGravityControls	59
jeod::GravityIntegFrame	27
jeod::GravityInteraction	30
jeod::GravityManager	35
jeod::GravityMessages	41
jeod::GravitySource	45
jeod::SphericalHarmonicsGravitySource	75
jeod::SphericalHarmonicsDeltaCoeffs	49
jeod::SphericalHarmonicsTidalEffects	100
jeod::SphericalHarmonicsSolidBodyTides	96
jeod::SphericalHarmonicsDeltaCoeffsInit	53
jeod::SphericalHarmonicsTidalEffectsInit	105
jeod::SphericalHarmonicsSolidBodyTidesInit	98
jeod::SphericalHarmonicsDeltaControls	56
jeod::SphericalHarmonicsGravitySource_default_data	84
jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data	85
jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data	86
jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data	87
jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data	88
jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data	89
jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data	90
jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data	91
jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data	92
jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data	93
jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data	94
jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data	95
jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data	100

Chapter 4

Data Structure Index

4.1 Data Structures

Here are the data structures with brief descriptions:

jeod::GravityControls	
Specifies whether and how a GravitySource affects a vehicle	15
jeod::GravityIntegFrame	
Class that aids in determining whether gravity should be applied as a direct effect or a third body effect	27
jeod::GravityInteraction	
Specifies interactions between a vehicle and a set of gravitational bodies	30
jeod::GravityManager	
The master gravitational model for a simulation	35
jeod::GravityMessages	
Specifies the message IDs used in the gravity model	41
jeod::GravitySource	
Models the gravity for a specific planet;	45
jeod::SphericalHarmonicsDeltaCoeffs	
Base class for tidal and temporal gravity models	49
jeod::SphericalHarmonicsDeltaCoeffsInit	
Initialization data for a SphericalHarmonicsDeltaCoeffs instance	53
jeod::SphericalHarmonicsDeltaControls	
Provides controls for how a variational model affects a vehicle	56
jeod::SphericalHarmonicsGravityControls	
Specifies whether and how a SphericalHarmonicsGravitySource affects a vehicle	59
jeod::SphericalHarmonicsGravitySource	
Models the gravity for a specific planet using spherical harmonics	75
jeod::SphericalHarmonicsGravitySource_default_data	84
jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data	85
jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data	86
jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data	87
jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data	88
jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data	89
jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data	90
jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data	91
jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data	92
jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data	93
jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data	94
jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data	95

jeod::SphericalHarmonicsSolidBodyTides	
Models solid body tidal effects	96
jeod::SphericalHarmonicsSolidBodyTidesInit	
Initializes a solid body tides model	98
jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data	100
jeod::SphericalHarmonicsTidalEffects	
Models tidal effects as a delta on top of a gravity model	100
jeod::SphericalHarmonicsTidalEffectsInit	
Initializes a tidal gravity model	105

Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

class_declarations.hh	
Forward declarations of classes defined for the gravity model	109
earth_GEMT1.cc	109
earth_GEMT1.hh	110
earth_GGM02C.cc	110
earth_GGM02C.hh	111
earth_GGM05C.cc	111
earth_GGM05C.hh	112
earth_solid_tides.cc	112
earth_solid_tides.hh	113
earth_spherical.cc	113
earth_spherical.hh	114
gravity_controls.cc	
Define member functions for the GravityControls class	114
gravity_controls.hh	
Define the gravity controls	115
gravity_integ_frame.hh	
Define the gravity integration frame class	115
gravity_interaction.cc	
Define methods for the GravityInteraction class	116
gravity_interaction.hh	
Define the GravityInteraction class, used to represent the gravitational interaction between a DynBody and a set of planetary bodies	117
gravity_manager.cc	
Define member functions for the GravityManager class	117
gravity_manager.hh	
Define the Gravity Manager	118
gravity_messages.cc	
Implement the class GravityMessages	118
gravity_messages.hh	
Define the class GravityMessages, the class that specifies the message IDs used in the gravity model	119
gravity_source.cc	
Define member functions for the GravitySource class	120
gravity_source.hh	
Define the gravity body base (pure virtual) class	120

jupiter_spherical.cc	121
jupiter_spherical.hh	121
mars_MRO110B2.cc	122
mars_MRO110B2.hh	122
mars_spherical.cc	123
mars_spherical.hh	123
moon_GRAIL150.cc	124
moon_GRAIL150.hh	124
moon_LP150Q.cc	125
moon_LP150Q.hh	125
moon_spherical.cc	126
moon_spherical.hh	126
spherical_harmonics_calc_nonspherical.cc	
Define SphericalHarmonicsGravityControl calc_nonspherical method, which computes non-spherical gravitational acceleration of a gravitational body on a given position	127
spherical_harmonics_delta_coeffs.cc	
Define member functions for the SphericalHarmonicsDeltaCoeffs class	127
spherical_harmonics_delta_coeffs.hh	
Define the class SphericalHarmonicsDeltaCoeffs, the base class for tidal effects and temporal gravity sub-models	128
spherical_harmonics_delta_coeffs_init.hh	
Define the class SphericalHarmonicsDeltaCoeffsInit, the base initialization class for tidal effects and temporal gravity sub-models	128
spherical_harmonics_delta_controls.hh	
Define the gravity controls for the variational gravity models such as solid-body tides	129
spherical_harmonics_gravity_controls.cc	
Define member functions for the SphericalHarmonicsGravityControls class	129
spherical_harmonics_gravity_controls.hh	
Define the gravity controls	130
spherical_harmonics_gravity_source.cc	
Define member functions for the SphericalHarmonicsGravitySource class	130
spherical_harmonics_gravity_source.hh	
Define the spherical harmonics implementation of a gravity body	131
spherical_harmonics_gravity_source_default_data.hh	131
spherical_harmonics_solid_body_tides.cc	
Define member functions for the SphericalHarmonicsSolidBodyTides class	132
spherical_harmonics_solid_body_tides.hh	
Define the SphericalHarmonicsSolidBodyTides class, which models solid-body tidal effects	132
spherical_harmonics_solid_body_tides_init.hh	
Define the SphericalHarmonicsSolidBodyTidesInit class, which is the initialization class for the solid body tides model	133
spherical_harmonics_tidal_effects.cc	
Define member functions for the SphericalHarmonicsTidalEffects class	133
spherical_harmonics_tidal_effects.hh	
Define the class SphericalHarmonicsTidalEffects, which is the base class for solid-body and ocean tidal effects	134
spherical_harmonics_tidal_effects_init.hh	
Define the SphericalHarmonicsTidalEffectsInit class, the initialization class for tidal effects models	134
sun_spherical.cc	135
sun_spherical.hh	135

Chapter 6

Module Documentation

6.1 Models

Modules

- [Environment](#)

6.1.1 Detailed Description

6.2 Environment

Modules

- [Gravity](#)

6.2.1 Detailed Description

6.3 Gravity

Files

- file [class_declarations.hh](#)
Forward declarations of classes defined for the gravity model.
- file [gravity_controls.hh](#)
Define the gravity controls.
- file [gravity_integ_frame.hh](#)
Define the gravity integration frame class.
- file [gravity_interaction.hh](#)
Define the GravityInteraction class, used to represent the gravitational interaction between a DynBody and a set of planetary bodies.
- file [gravity_manager.hh](#)
Define the Gravity Manager.

- file [gravity_messages.hh](#)
Define the class GravityMessages, the class that specifies the message IDs used in the gravity model.
- file [gravity_source.hh](#)
Define the gravity body base (pure virtual) class.
- file [spherical_harmonics_delta_coeffs.hh](#)
Define the class SphericalHarmonicsDeltaCoeffs, the base class for tidal effects and temporal gravity sub-models.
- file [spherical_harmonics_delta_coeffs_init.hh](#)
Define the class SphericalHarmonicsDeltaCoeffsInit, the base initialization class for tidal effects and temporal gravity sub-models.
- file [spherical_harmonics_delta_controls.hh](#)
Define the gravity controls for the variational gravity models such as solid-body tides.
- file [spherical_harmonics_gravity_controls.hh](#)
Define the gravity controls.
- file [spherical_harmonics_gravity_source.hh](#)
Define the spherical harmonics implementation of a gravity body.
- file [spherical_harmonics_solid_body_tides.hh](#)
Define the SphericalHarmonicsSolidBodyTides class, which models solid-body tidal effects.
- file [spherical_harmonics_solid_body_tides_init.hh](#)
Define the SphericalHarmonicsSolidBodyTidesInit class, which is the initialization class for the solid body tides model.
- file [spherical_harmonics_tidal_effects.hh](#)
Define the class SphericalHarmonicsTidalEffects, which is the base class for solid-body and ocean tidal effects.
- file [spherical_harmonics_tidal_effects_init.hh](#)
Define the SphericalHarmonicsTidalEffectsInit class, the initialization class for tidal effects models.
- file [gravity_controls.cc](#)
Define member functions for the GravityControls class.
- file [gravity_interaction.cc](#)
Define methods for the GravityInteraction class.
- file [gravity_manager.cc](#)
Define member functions for the GravityManager class.
- file [gravity_messages.cc](#)
Implement the class GravityMessages.
- file [gravity_source.cc](#)
Define member functions for the GravitySource class.
- file [spherical_harmonics_calc_nonspherical.cc](#)
Define SphericalHarmonicsGravityControl calc_nonspherical method, which computes non-spherical gravitational acceleration of a gravitational body on a given position.
- file [spherical_harmonics_delta_coeffs.cc](#)
Define member functions for the SphericalHarmonicsDeltaCoeffs class.
- file [spherical_harmonics_gravity_controls.cc](#)
Define member functions for the SphericalHarmonicsGravityControls class.
- file [spherical_harmonics_gravity_source.cc](#)
Define member functions for the SphericalHarmonicsGravitySource class.
- file [spherical_harmonics_solid_body_tides.cc](#)
Define member functions for the SphericalHarmonicsSolidBodyTides class.
- file [spherical_harmonics_tidal_effects.cc](#)
Define member functions for the SphericalHarmonicsTidalEffects class.

6.3.1 Detailed Description

Chapter 7

Namespace Documentation

7.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [SphericalHarmonicsGravitySource_earth_GEMT1_default_data](#)
- class [SphericalHarmonicsGravitySource_earth_GGM02C_default_data](#)
- class [SphericalHarmonicsGravitySource_earth_GGM05C_default_data](#)
- class [SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data](#)
- class [SphericalHarmonicsGravitySource_earth_spherical_default_data](#)
- class [SphericalHarmonicsGravitySource_jupiter_spherical_default_data](#)
- class [SphericalHarmonicsGravitySource_mars_MRO110B2_default_data](#)
- class [SphericalHarmonicsGravitySource_mars_spherical_default_data](#)
- class [SphericalHarmonicsGravitySource_moon_GRAIL150_default_data](#)
- class [SphericalHarmonicsGravitySource_moon_LP150Q_default_data](#)
- class [SphericalHarmonicsGravitySource_moon_spherical_default_data](#)
- class [SphericalHarmonicsGravitySource_default_data](#)
- class [SphericalHarmonicsGravitySource_sun_spherical_default_data](#)
- class [GravityControls](#)
Specifies whether and how a [GravitySource](#) affects a vehicle.
- class [GravityIntegFrame](#)
Class that aids in determining whether gravity should be applied as a direct effect or a third body effect.
- class [GravityInteraction](#)
Specifies interactions between a vehicle and a set of gravitational bodies.
- class [GravityManager](#)
The master gravitational model for a simulation.
- class [GravityMessages](#)
Specifies the message IDs used in the gravity model.
- class [GravitySource](#)
Models the gravity for a specific planet;.
- class [SphericalHarmonicsDeltaCoeffs](#)
Base class for tidal and temporal gravity models.
- class [SphericalHarmonicsDeltaCoeffsInit](#)
Initialization data for a [SphericalHarmonicsDeltaCoeffs](#) instance.

- class [SphericalHarmonicsDeltaControls](#)
Provides controls for how a variational model affects a vehicle.
- class [SphericalHarmonicsGravityControls](#)
Specifies whether and how a [SphericalHarmonicsGravitySource](#) affects a vehicle.
- class [SphericalHarmonicsGravitySource](#)
Models the gravity for a specific planet using spherical harmonics.
- class [SphericalHarmonicsSolidBodyTides](#)
Models solid body tidal effects.
- class [SphericalHarmonicsSolidBodyTidesInit](#)
Initializes a solid body tides model.
- class [SphericalHarmonicsTidalEffects](#)
Models tidal effects as a delta on top of a gravity model.
- class [SphericalHarmonicsTidalEffectsInit](#)
Initializes a tidal gravity model.

Variables

- static constexpr double [speed_of_light_sq](#) = 89875517873681764.0
The speed of light squared, in m^2/s^2 .

7.1.1 Detailed Description

Namespace jeod.

7.1.2 Variable Documentation

7.1.2.1 speed_of_light_sq

```
constexpr double jeod::speed_of_light_sq = 89875517873681764.0 [static], [constexpr]
```

The speed of light squared, in m^2/s^2 .

Definition at line 58 of file gravity_controls.cc.

Referenced by jeod::GravityControls::calc_relativistic().

Chapter 8

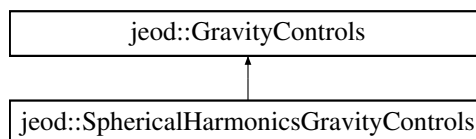
Data Structure Documentation

8.1 jeod::GravityControls Class Reference

Specifies whether and how a [GravitySource](#) affects a vehicle.

```
#include <gravity_controls.hh>
```

Inheritance diagram for jeod::GravityControls:



Public Member Functions

- [GravityControls](#) ()=default
- virtual [~GravityControls](#) ()=default
- [GravityControls](#) (const [GravityControls](#) &)=delete
- [GravityControls](#) & operator= (const [GravityControls](#) &)=delete
- virtual void [initialize_control](#) ([GravityManager](#) &grav_man)
Initialize this GravityControl.
- virtual void [reset_control](#) ([BaseDynManager](#) &dyn_manager)
Reset subscriptions for this GravityControl.
- virtual void [gravitation](#) (const double integ_pos[3], unsigned int integ_frame_idx, double body_grav_accel[3], double dgdx[3][3], double Pot[1])
Compute the gravitation at a given position toward a gravity body.
- virtual void [gravitation](#) (const [RefFrame](#) &point_of_interest, unsigned int integ_frame_idx, double body_grav_accel[3], double dgdx[3][3], double &pot)
Compute the gravitation at a given position toward a gravity body.

Static Public Member Functions

- static bool [accel_mag_less_ptr](#) (const [GravityControls](#) *a, const [GravityControls](#) *b)
Compares the magnitude of the two input gravity controls, returning true if a->grav_accel_magsq is less than b->grav_accel_magsq, false otherwise.

Data Fields

- `std::string source_name`
Planet name.
- `bool active {}`
Gravity for this body active?
- `bool spherical {}`
Ignore non-spherical effects?
- `bool gradient {}`
Compute gravity gradient matrix?
- `bool perturbing_only {}`
Compute only the perturbing gravity?
- `bool battin_method {}`
Compute third body gravity using Battin's method (Battin, Mathematics and Methods of Astrodynamics)?
- `bool relativistic {}`
Indicates that the relativistic correction to Newtonian gravitation is to be computed.
- `GravitySource * body {}`
Pointer to the GravitySource object named by planet_name.
- `double grav_accel [3] {}`
Gravitational acceleration toward the GravitySource at the location of the DynBody, including third body effects.
- `double grav_grad [3][3] {}`
Gradient of the gravitational acceleration.
- `double grav_pot {}`
Gravitational potential.
- `double grav_accel_magsq {}`
Square of the magnitude of grav_accel.

Protected Member Functions

- `virtual void calc_nonspherical (const double integ_pos[3], const double posn[3], const GravityIntegFrame &grav_source_frame, double body_grav_accel[3], double dgdx[3][3], double &pot)=0`
Nominally, compute the non-spherical contribution to gravity at a given position.
- `void calc_relativistic (const RefFrame &point_of_interest, const double rel_pos[3], const double rel_vel[3], double perturbing_accel[3])`
Calculates the relativistic correction to gravitational acceleration.
- `void calc_spherical (const double integ_pos[3], const double posn[3], const GravityIntegFrame &grav_source_frame, double body_grav_accel[3], double dgdx[3][3], double &pot)`
Calculate the spherical gravitational acceleration, either directly or as a third body acceleration.

Protected Attributes

- `GravityManager * grav_manager {}`
Pointer to the simulation-wide GravityManager object.
- `bool subscribed_to_inertial {}`
Indicates that a subscription to the planet-centered inertial frame of the planet associated with the gravity_source been issued.
- `bool subscribed_to_pfix {}`
Indicates that a subscription to the planet-centered, planet-fixed frame of the planet associated with the gravity_source been issued.
- `bool skip_spherical {}`
Some derived classes' calc_nonspherical method computes all contributions to gravitation, including spherical.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [GravityControls](#))

8.1.1 Detailed Description

Specifies whether and how a [GravitySource](#) affects a vehicle.

Definition at line 89 of file gravity_controls.hh.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 GravityControls() [1/2]

```
jeod::GravityControls::GravityControls ( ) [default]
```

8.1.2.2 ~GravityControls()

```
virtual jeod::GravityControls::~~GravityControls ( ) [virtual], [default]
```

8.1.2.3 GravityControls() [2/2]

```
jeod::GravityControls::GravityControls (
    const GravityControls & ) [delete]
```

8.1.3 Member Function Documentation

8.1.3.1 accel_mag_less_ptr()

```
static bool jeod::GravityControls::accel_mag_less_ptr (
    const GravityControls * a,
    const GravityControls * b ) [inline], [static]
```

Compares the magnitude of the two input gravity controls, returning true if `a->grav_accel_magsq` is less than `b->grav_accel_magsq`, false otherwise.

Returns

Result of comparison

Parameters

<i>a</i>	First control to be compared.
<i>b</i>	Second control to be compared.

Definition at line 228 of file gravity_controls.hh.

Referenced by `jeod::GravityInteraction::sort_controls()`.

8.1.3.2 calc_nonspherical()

```
virtual void jeod::GravityControls::calc_nonspherical (
    const double integ_pos[3],
    const double posn[3],
    const GravityIntegFrame & grav_source_frame,
    double body_grav_accel[3],
    double dgdx[3][3],
    double & pot ) [protected], [pure virtual]
```

Nominally, compute the non-spherical contribution to gravity at a given position.

Derived classes whose override of this function computes the full gravitation model, including the spherical contribution, should set the `skip_spherical` flag.

Parameters

in	<i>posn</i>	Inertial position of the point of interest relative to the gravitational body.
out	<i>body_grav_accel</i>	Acceleration at the point of interest due to the gravitational body.
out	<i>dgdx</i>	Gravity gradient at the point of interest.
out	<i>Pot</i>	Specific gravitational potential energy.

Implemented in `jeod::SphericalHarmonicsGravityControls`.

Referenced by `gravitation()`.

8.1.3.3 calc_relativistic()

```
void jeod::GravityControls::calc_relativistic (
    const RefFrame & point_of_interest,
    const double rel_pos[3],
    const double rel_vel[3],
    double perturbing_accel[3] ) [protected]
```

Calculates the relativistic correction to gravitational acceleration.

Parameters

<i>point_of_interest</i>	The point of interest, as a reference frame.
<i>rel_pos</i>	Displacement vector from the grav body to the POI.
<i>rel_vel</i>	Time derivative of <i>rel_pos</i> .
<i>perturbing_accel</i>	Output relativistic accel, sans the Newtonian term.

Implements equation 27 (Folkner) / equation 4 (Genova) to compute a parameteric post-Newtonian correction to gravitation. The referenced equation is of the form

$$\begin{aligned} \mathbf{a}_{A,\text{pm-pm}} = & \sum_{B \neq A} \frac{GM_B}{r_{AB}^3} (\mathbf{r}_B - \mathbf{r}_A) \left(1 + \frac{s_1}{c^2}\right) \\ & + \frac{1}{c^2} \sum_{B \neq A} \frac{GM_B}{r_{AB}^3} (\mathbf{v}_A - \mathbf{v}_B) \left((\mathbf{r}_A - \mathbf{r}_B) \cdot ((2 + 2\gamma)\mathbf{v}_A - (1 + 2\gamma)\mathbf{v}_B) \right) \\ & + \frac{3 + 4\gamma}{2c^2} \sum_{B \neq A} \frac{GM_B}{r_{AB}} \mathbf{a}_B \end{aligned}$$

where $\mathbf{a}_B$ is the net acceleration of gravitating body B toward the other gravitating bodies (typically taken to be the Newtonian gravitational acceleration) and

$$\begin{aligned} s_1 = & -2(\beta + \gamma) \sum_{C \neq A} \frac{GM_C}{r_{AC}} - (2\beta - 1) \sum_{C \neq A} \frac{GM_C}{r_{BC}} \\ & + \gamma v_a^2 + (1 + \gamma) v_b^2 - 2(1 + \gamma) \mathbf{v}_A \cdot \mathbf{v}_B \\ & - \frac{3}{2} \left(\frac{(\mathbf{r}_A - \mathbf{r}_B) \cdot \mathbf{v}_B}{r_B} \right)^2 \\ & + \frac{1}{2} (\mathbf{r}_B - \mathbf{r}_A) \cdot \mathbf{a}_B \end{aligned}$$

In terms of the referenced equation, the intent of this function is to compute the relativistic portion of the acceleration of body A toward body B . With this, the expression this function computes is

$$\Delta a_{A,B} = \frac{1}{c} \frac{GM_B}{r_{AB}} \left\{ \begin{aligned} & \frac{\mathbf{r}_B - \mathbf{r}_A}{r_{AB}^2} s_1 \\ & + \frac{\mathbf{v}_A - \mathbf{v}_B}{r_{AB}^2} \left((\mathbf{r}_A - \mathbf{r}_B) \cdot ((2 + 2\gamma)\mathbf{v}_A - (1 + 2\gamma)\mathbf{v}_B) \right) \\ & + \frac{3 + 4\gamma}{2} \mathbf{a}_B \end{aligned} \right\}$$

Note that the common factor $\frac{1}{c} \frac{GM_B}{r_{AB}}$ is unitless and thus each of the three terms in the braced expression has units of acceleration.

The referenced equations explicitly involve the parameterized post-Newtonian (PPN) factors β and γ . This function hardcodes both of those factors as one, consistent with general relativity. For example, the factor $(3 + 4\gamma)/2$ becomes 3.5 in the code. Magic numbers that involve β or γ , including a factor of one (e.g., $2\beta - 1$), are noted in comments.

References:

- William M. Folkner, et al., *Planetary and Lunar Ephemerides DE430 and DE431*, IPN Progress Report 42-196, 15 February 2014.
ftp://naif.jpl.nasa.gov/pub/naif/generic_kernels/spk/planets/de430_and_de431.pdf

- Antonio Genova, et al., *Solar system expansion and strong equivalence principle as seen by the NASA MESSENGER mission*, Nature Communications 9:289, 18 January 2018, DOI: 10.1038/s41467-017-02558-1.

<https://www.nature.com/articles/s41467-017-02558-1>

Definition at line 456 of file gravity_controls.cc.

References body, jeod::GravityManager::get_bodies(), grav_manager, jeod::GravitySource::inertial, jeod::GravitySource::mu, and jeod::speed_of_light_sq.

Referenced by gravitation().

8.1.3.4 calc_spherical()

```
void jeod::GravityControls::calc_spherical (
    const double integ_pos[3],
    const double posn[3],
    const GravityIntegFrame & grav_source_frame,
    double body_grav_accel[3],
    double dgdx[3][3],
    double & pot ) [protected]
```

Calculate the spherical gravitational acceleration, either directly or as a third body acceleration.

Parameters

<i>integ_pos</i>	Point of interest location, integ frame coordinates
<i>posn</i>	Vector from gravitational body to point of interest
<i>grav_source_frame</i>	Frame corresponding to the gravitational body
<i>body_grav_accel</i>	Acceleration at integ_pos due to the grav body
<i>dgdx</i>	Gravity gradient at integ_pos due to the grav body
<i>pot</i>	Gravitational potential at integ_pos due to the grav body.

Definition at line 291 of file gravity_controls.cc.

References battin_method, body, gradient, jeod::GravityIntegFrame::is_third_body, jeod::GravitySource::mu, and jeod::GravityIntegFrame::pos.

Referenced by gravitation().

8.1.3.5 gravitation() [1/2]

```
void jeod::GravityControls::gravitation (
    const double integ_pos[3],
    unsigned int integ_frame_idx,
    double body_grav_accel[3],
    double dgdx[3][3],
    double Pot[1] ) [virtual]
```

Compute the gravitation at a given position toward a gravity body.

Parameters

in	<i>integ_pos</i>	Point of interest, integ coords Units: M
in	<i>integ_frame_idx</i>	Integ frame index
out	<i>body_grav_accel</i>	Accel for given grav body Units: M/s ²
out	<i>dgdx</i>	Gradient for given grav body Units: 1/s ²
out	<i>Pot</i>	Potential

Definition at line 187 of file gravity_controls.cc.

References `body`, `calc_nonspherical()`, `calc_spherical()`, `jeod::GravitySource::frames`, `jeod::GravitySource::inertial`, `perturbing_only`, `jeod::GravityIntegFrame::pos`, `jeod::GravityIntegFrame::ref_frame`, `skip_spherical`, and `spherical`.

Referenced by `jeod::GravityManager::gravitation()`.

8.1.3.6 gravitation() [2/2]

```
void jeod::GravityControls::gravitation (
    const RefFrame & point_of_interest,
    unsigned int integ_frame_idx,
    double body_grav_accel[3],
    double dgdx[3][3],
    double & pot ) [virtual]
```

Compute the gravitation at a given position toward a gravity body.

Parameters

in	<i>point_of_interest</i>	Point of interest, as a reference frame.
in	<i>integ_frame_idx</i>	Integ frame index.
out	<i>body_grav_accel</i>	Accel for given grav body.
out	<i>dgdx</i>	Gradient for given grav body.
out	<i>pot</i>	Specific gravitational potential for given grav body.

Definition at line 232 of file gravity_controls.cc.

References `body`, `calc_nonspherical()`, `calc_relativistic()`, `calc_spherical()`, `jeod::GravitySource::frames`, `jeod::GravitySource::inertial`, `perturbing_only`, `jeod::GravityIntegFrame::pos`, `jeod::GravityIntegFrame::ref_frame`, `relativistic`, `skip_spherical`, and `spherical`.

8.1.3.7 initialize_control()

```
void jeod::GravityControls::initialize_control (
    GravityManager & grav_man ) [virtual]
```

Initialize this GravityControl.

Parameters

in	<i>grav_man</i>	Ref to Gravity Manager
----	-----------------	------------------------

Reimplemented in [jeod::SphericalHarmonicsGravityControls](#).

Definition at line 71 of file gravity_controls.cc.

References `body`, `jeod::GravityManager::find_grav_source()`, `grav_manager`, `jeod::GravitySource::inertial`, `jeod::GravityMessages::invalid_object`, `jeod::GravityMessages::missing_entry`, `jeod::GravitySource::name`, and `source_name`.

Referenced by `jeod::SphericalHarmonicsGravityControls::initialize_control()`.

8.1.3.8 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::GravityControls::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    GravityControls ) [private]
```

8.1.3.9 operator=()

```
GravityControls& jeod::GravityControls::operator= (
    const GravityControls & ) [delete]
```

8.1.3.10 reset_control()

```
void jeod::GravityControls::reset_control (
    BaseDynManager & dyn_manager ) [virtual]
```

Reset subscriptions for this GravityControl.

Parameters

in	<i>dyn_manager</i>	Ptr to dynamics manager
----	--------------------	-------------------------

Definition at line 119 of file gravity_controls.cc.

References `active`, `body`, `jeod::GravitySource::inertial`, `jeod::GravityMessages::null_pointer`, `jeod::GravitySource::pfix`, `source_name`, `spherical`, `subscribed_to_inertial`, and `subscribed_to_pfix`.

8.1.4 Field Documentation

8.1.4.1 active

```
bool jeod::GravityControls::active {}
```

Gravity for this body active?

trick_units(—)

Definition at line 101 of file gravity_controls.hh.

Referenced by jeod::GravityManager::gravitation(), reset_control(), jeod::GravityInteraction::sort_controls(), and jeod::SphericalHarmonicsGravityControls::update_deltacoeffs().

8.1.4.2 battin_method

```
bool jeod::GravityControls::battin_method {}
```

Compute third body gravity using Battin's method (Battin, Mathematics and Methods of Astrodynamics)?

trick_units(—)

Definition at line 122 of file gravity_controls.hh.

Referenced by calc_spherical().

8.1.4.3 body

```
GravitySource* jeod::GravityControls::body {}
```

Pointer to the [GravitySource](#) object named by planet_name.

Note

Users should not set this data member in the input file. trick_units(—)

Definition at line 135 of file gravity_controls.hh.

Referenced by calc_relativistic(), calc_spherical(), gravitation(), initialize_control(), jeod::SphericalHarmonicsGravityControls::initialize_control(), and reset_control().

8.1.4.4 gradient

```
bool jeod::GravityControls::gradient {}
```

Compute gravity gradient matrix?

trick_units(-)

Definition at line 111 of file gravity_controls.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), calc_spherical(), and jeod::SphericalHarmonicsGravityControls::check_validity().

8.1.4.5 grav_accel

```
double jeod::GravityControls::grav_accel[3] {}
```

Gravitational acceleration toward the [GravitySource](#) at the location of the DynBody, including third body effects.

trick_units(m/s2)

Definition at line 141 of file gravity_controls.hh.

Referenced by jeod::GravityManager::gravitation(), and jeod::GravityInteraction::sort_controls().

8.1.4.6 grav_accel_magsq

```
double jeod::GravityControls::grav_accel_magsq {}
```

Square of the magnitude of grav_accel.

trick_units(m2/s4)

Definition at line 156 of file gravity_controls.hh.

Referenced by jeod::GravityInteraction::sort_controls().

8.1.4.7 grav_grad

```
double jeod::GravityControls::grav_grad[3][3] {}
```

Gradient of the gravitational acceleration.

trick_units(1/s2)

Definition at line 146 of file gravity_controls.hh.

Referenced by jeod::GravityManager::gravitation().

8.1.4.8 grav_manager

```
GravityManager* jeod::GravityControls::grav_manager {} [protected]
```

Pointer to the simulation-wide [GravityManager](#) object.

Note

Users should not set this data member in the input file. `trick_units(-)`

Definition at line 163 of file `gravity_controls.hh`.

Referenced by `calc_relativistic()`, `initialize_control()`, and `jeod::SphericalHarmonicsGravityControls::initialize_↵ control()`.

8.1.4.9 grav_pot

```
double jeod::GravityControls::grav_pot {}
```

Gravitational potential.

`trick_units(m2/s2)`

Definition at line 151 of file `gravity_controls.hh`.

Referenced by `jeod::GravityManager::gravitation()`.

8.1.4.10 perturbing_only

```
bool jeod::GravityControls::perturbing_only {}
```

Compute only the perturbing gravity?

`trick_units(-)`

Definition at line 116 of file `gravity_controls.hh`.

Referenced by `gravitation()`.

8.1.4.11 relativistic

```
bool jeod::GravityControls::relativistic {}
```

Indicates that the relativistic correction to Newtonian gravitation is to be computed.

The public methods `enable_relativistic_correction` and `disable_relativistic_correction` set / clear this flag. `trick_↵ units(-)`

Definition at line 129 of file `gravity_controls.hh`.

Referenced by `gravitation()`.

8.1.4.12 skip_spherical

```
bool jeod::GravityControls::skip_spherical {} [protected]
```

Some derived classes' `calc_nonspherical` method computes all contributions to gravitation, including spherical.

Those classes need to set this flag. The flag is clear in the base class and in the [SphericalHarmonicsGravityControls](#) derived class. `trick_units(-)`

Definition at line 185 of file `gravity_controls.hh`.

Referenced by `gravitation()`.

8.1.4.13 source_name

```
std::string jeod::GravityControls::source_name
```

Planet name.

`trick_units(-)`

Definition at line 96 of file `gravity_controls.hh`.

Referenced by `jeod::SphericalHarmonicsGravityControls::check_validity()`, `initialize_control()`, and `reset_control()`.

8.1.4.14 spherical

```
bool jeod::GravityControls::spherical {}
```

Ignore non-spherical effects?

`trick_units(-)`

Definition at line 106 of file `gravity_controls.hh`.

Referenced by `jeod::SphericalHarmonicsGravityControls::check_validity()`, `gravitation()`, and `reset_control()`.

8.1.4.15 subscribed_to_inertial

```
bool jeod::GravityControls::subscribed_to_inertial {} [protected]
```

Indicates that a subscription to the planet-centered inertial frame of the planet associated with the `gravity_source` been issued.

Note

Users should not set this data member in the input file. `trick_units(-)`

Definition at line 170 of file `gravity_controls.hh`.

Referenced by `reset_control()`.

8.1.4.16 subscribed_to_pfix

```
bool jeod::GravityControls::subscribed_to_pfix {} [protected]
```

Indicates that a subscription to the planet-centered, planet-fixed frame of the planet associated with the gravity_↔ source been issued.

Note

Users should not set this data member in the input file. `trick_units(-)`

Definition at line 177 of file `gravity_controls.hh`.

Referenced by `reset_control()`.

The documentation for this class was generated from the following files:

- [gravity_controls.hh](#)
- [gravity_controls.cc](#)

8.2 jeod::GravityIntegFrame Class Reference

Class that aids in determining whether gravity should be applied as a direct effect or a third body effect.

```
#include <gravity_integ_frame.hh>
```

Public Member Functions

- [GravityIntegFrame](#) ()=default
- [~GravityIntegFrame](#) ()=default

Data Fields

- const EphemerisRefFrame * [ref_frame](#) {}
Reference frame.
- bool [is_third_body](#) {}
Is it a third body effect in this frame?
- double [pos](#) [3] {}
Position of the integration frame origin with respect to a body.
- double [accel](#) [3] {}
Acceleration of the frame origin with respect to the body.
- double [time](#) {9e99}
Timestamp of last update to this class.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [GravityIntegFrame](#))

8.2.1 Detailed Description

Class that aids in determining whether gravity should be applied as a direct effect or a third body effect.

Definition at line 83 of file gravity_integ_frame.hh.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 GravityIntegFrame()

```
jeod::GravityIntegFrame::GravityIntegFrame ( ) [default]
```

8.2.2.2 ~GravityIntegFrame()

```
jeod::GravityIntegFrame::~~GravityIntegFrame ( ) [default]
```

8.2.3 Member Function Documentation

8.2.3.1 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GravityIntegFrame::p (
    jeod ,
    GravityIntegFrame ) [private]
```

8.2.4 Field Documentation

8.2.4.1 accel

```
double jeod::GravityIntegFrame::accel[3] {}
```

Acceleration of the frame origin with respect to the body.

trick_units(m/s2)

Definition at line 104 of file gravity_integ_frame.hh.

8.2.4.2 is_third_body

```
bool jeod::GravityIntegFrame::is_third_body {}
```

Is it a third body effect in this frame?

trick_units(-)

Definition at line 94 of file gravity_integ_frame.hh.

Referenced by jeod::GravityControls::calc_spherical(), and jeod::GravitySource::initialize_state().

8.2.4.3 pos

```
double jeod::GravityIntegFrame::pos[3] {}
```

Position of the integration frame origin with respect to a body.

trick_units(m)

Definition at line 99 of file gravity_integ_frame.hh.

Referenced by jeod::GravityControls::calc_spherical(), and jeod::GravityControls::gravitation().

8.2.4.4 ref_frame

```
const EphemerisRefFrame* jeod::GravityIntegFrame::ref_frame {}
```

Reference frame.

trick_units(-)

Definition at line 89 of file gravity_integ_frame.hh.

Referenced by jeod::GravityControls::gravitation(), and jeod::GravitySource::initialize_state().

8.2.4.5 time

```
double jeod::GravityIntegFrame::time {9e99}
```

Timestamp of last update to this class.

trick_units(s)

Definition at line 109 of file gravity_integ_frame.hh.

Referenced by jeod::GravitySource::initialize_state().

The documentation for this class was generated from the following file:

- [gravity_integ_frame.hh](#)

8.3 jeod::GravityInteraction Class Reference

Specifies interactions between a vehicle and a set of gravitational bodies.

```
#include <gravity_interaction.hh>
```

Public Member Functions

- [GravityInteraction](#) ()
Construct a [GravityInteraction](#) instance.
- virtual [~GravityInteraction](#) ()
Destruct a [GravityInteraction](#) instance.
- [GravityInteraction](#) (const [GravityInteraction](#) &frame)=delete
- [GravityInteraction](#) & operator= (const [GravityInteraction](#) &frame)=delete
- virtual void [set_integ_frame](#) (const EphemerisRefFrame &ref_frame_in, const BaseDynManager &dyn_manager)
Set the integration frame and associated integration frame index.
- virtual void [add_control](#) ([GravityControls](#) *control)
Add a new [GravityControls](#) to the grav_controls list.
- virtual void [remove_control](#) ([GravityControls](#) *control)
Remove a [GravityControls](#) from the grav_controls list.
- virtual void [initialize_controls](#) (BaseDynManager &dyn_manager, [GravityManager](#) &grav_manager)
Initialize all [GravityControls](#) in the grav_controls list.
- virtual void [reset_controls](#) (BaseDynManager &dyn_manager)
Reset all [GravityControls](#) in the grav_controls list.
- virtual void [sort_controls](#) ()
Sort the [GravityControls](#) in the grav_controls list in increasing acceleration magnitude order.

Data Fields

- unsigned int [integ_frame_index](#) {9999}
The integration frame index number of the DynBody's integration frame.
- double [grav_accel](#) [3] {}
The total gravitational acceleration of the DynBody toward all planetary with which the vehicle interacts gravitationally.
- double [grav_grad](#) [3][3] {}
The gradient of the gravitational acceleration vector evaluated at the DynBody's position, expressed in the vehicle's integration frame.
- double [grav_pot](#) {}
The total gravitational potential at the location of the DynBody due to the gravity fields of all "active" gravitational bodies (i.e., planets).
- JeodPointerVector< [GravityControls](#) >::type [grav_controls](#)
The gravity controls list for a DynBody specifies the planetary bodies with which the DynBody interacts and specifies the nature of those interactions.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [GravityInteraction](#))

8.3.1 Detailed Description

Specifies interactions between a vehicle and a set of gravitational bodies.

Definition at line 96 of file gravity_interaction.hh.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 GravityInteraction() [1/2]

```
jeod::GravityInteraction::GravityInteraction ( )
```

Construct a [GravityInteraction](#) instance.

Definition at line 58 of file gravity_interaction.cc.

References [grav_controls](#).

8.3.2.2 ~GravityInteraction()

```
jeod::GravityInteraction::~~GravityInteraction ( ) [virtual]
```

Destruct a [GravityInteraction](#) instance.

Definition at line 68 of file gravity_interaction.cc.

References [grav_controls](#).

8.3.2.3 GravityInteraction() [2/2]

```
jeod::GravityInteraction::GravityInteraction (
    const GravityInteraction & frame ) [delete]
```

8.3.3 Member Function Documentation

8.3.3.1 add_control()

```
void jeod::GravityInteraction::add_control (
    GravityControls * control ) [virtual]
```

Add a new [GravityControls](#) to the [grav_controls](#) list.

Parameters

in	<i>control</i>	Control to be added
----	----------------	---------------------

Definition at line 91 of file gravity_interaction.cc.

References jeod::GravityMessages::duplicate_entry, and grav_controls.

8.3.3.2 initialize_controls()

```
void jeod::GravityInteraction::initialize_controls (
    BaseDynManager & dyn_manager,
    GravityManager & grav_manager ) [virtual]
```

Initialize all [GravityControls](#) in the grav_controls list.

Parameters

in	<i>dyn_manager</i>	Ref to Dyn Manager
in	<i>grav_manager</i>	Ref to Gravity Manager

Definition at line 135 of file gravity_interaction.cc.

References grav_controls, and reset_controls().

8.3.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::GravityInteraction::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    GravityInteraction ) [private]
```

8.3.3.4 operator=()

```
GravityInteraction& jeod::GravityInteraction::operator= (
    const GravityInteraction & frame ) [delete]
```

8.3.3.5 remove_control()

```
void jeod::GravityInteraction::remove_control (
    GravityControls * control ) [virtual]
```

Remove a [GravityControls](#) from the grav_controls list.

Parameters

in	<i>control</i>	GravityControls to be removed.
----	----------------	--

Definition at line 113 of file gravity_interaction.cc.

References `grav_controls`, and `jeod::GravityMessages::missing_entry`.

8.3.3.6 reset_controls()

```
void jeod::GravityInteraction::reset_controls (
    BaseDynManager & dyn_manager ) [virtual]
```

Reset all [GravityControls](#) in the `grav_controls` list.

Definition at line 150 of file gravity_interaction.cc.

References `grav_controls`.

Referenced by `initialize_controls()`.

8.3.3.7 set_integ_frame()

```
void jeod::GravityInteraction::set_integ_frame (
    const EphemerisRefFrame & ref_frame,
    const BaseDynManager & dyn_manager ) [virtual]
```

Set the integration frame and associated integration frame index.

Assumptions and Limitations

- Provided frame is a valid integration frame.

Parameters

in	<i>ref_frame</i>	Integration frame
in	<i>dyn_manager</i>	Dynamics manager

Definition at line 82 of file gravity_interaction.cc.

References `integ_frame_index`.

8.3.3.8 sort_controls()

```
void jeod::GravityInteraction::sort_controls ( ) [virtual]
```

Sort the [GravityControls](#) in the grav_controls list in increasing acceleration magnitude order.

Definition at line 163 of file gravity_interaction.cc.

References [jeod::GravityControls::accel_mag_less_ptr\(\)](#), [jeod::GravityControls::active](#), [jeod::GravityControls::grav_accel](#), [jeod::GravityControls::grav_accel_magsq](#), and [grav_controls](#).

8.3.4 Field Documentation

8.3.4.1 grav_accel

```
double jeod::GravityInteraction::grav_accel[3] {}
```

The total gravitational acceleration of the DynBody toward all planetary with which the vehicle interacts gravitationally.

The acceleration is expressed in the DynBody's integration frame. The gravitational acceleration of the integration frame itself toward the planetary bodies is excluded from this total acceleration. For example, for a vehicle integrated in Earth-centered inertial, the Sun component of the total gravitational acceleration is the Newtonian gravitation acceleration of the vehicle toward the Sun less the Newtonian gravitational acceleration of the Earth toward the Sun. `trick_units(m/s2)`

Definition at line 118 of file gravity_interaction.hh.

Referenced by [jeod::GravityManager::gravitation\(\)](#).

8.3.4.2 grav_controls

```
JeodPointerVector<GravityControls>::type jeod::GravityInteraction::grav_controls
```

The gravity controls list for a DynBody specifies the planetary bodies with which the DynBody interacts and specifies the nature of those interactions.

`trick_io(**)`

Definition at line 138 of file gravity_interaction.hh.

Referenced by [add_control\(\)](#), [jeod::GravityManager::gravitation\(\)](#), [GravityInteraction\(\)](#), [initialize_controls\(\)](#), [remove_control\(\)](#), [reset_controls\(\)](#), [sort_controls\(\)](#), and [~GravityInteraction\(\)](#).

8.3.4.3 grav_grad

```
double jeod::GravityInteraction::grav_grad[3][3] {}
```

The gradient of the gravitational acceleration vector evaluated at the DynBody's position, expressed in the vehicle's integration frame.

trick_units(1/s²)

Definition at line 124 of file gravity_interaction.hh.

Referenced by jeod::GravityManager::gravitation().

8.3.4.4 grav_pot

```
double jeod::GravityInteraction::grav_pot {}
```

The total gravitational potential at the location of the DynBody due to the gravity fields of all "active" gravitational bodies (i.e., planets).

trick_units(m²/s²)

Definition at line 131 of file gravity_interaction.hh.

Referenced by jeod::GravityManager::gravitation().

8.3.4.5 integ_frame_index

```
unsigned int jeod::GravityInteraction::integ_frame_index {9999}
```

The integration frame index number of the DynBody's integration frame.

This data member must be kept in strict synchronization with the DynBody's integration frame. trick_units(—)

Definition at line 105 of file gravity_interaction.hh.

Referenced by jeod::GravityManager::gravitation(), and set_integ_frame().

The documentation for this class was generated from the following files:

- [gravity_interaction.hh](#)
- [gravity_interaction.cc](#)

8.4 jeod::GravityManager Class Reference

The master gravitational model for a simulation.

```
#include <gravity_manager.hh>
```

Public Member Functions

- [GravityManager](#) ()
GravityManager constructor.
- [~GravityManager](#) ()
GravityManager destructor.
- [GravityManager](#) (const [GravityManager](#) &)=delete
- [GravityManager](#) & operator= (const [GravityManager](#) &)=delete
- [GravitySource](#) * [find_grav_source](#) (const std::string &source_name) const
Find the gravitational body with the given name.
- void [add_grav_source](#) ([GravitySource](#) &source)
Create a gravitational body, initialize it with the supplied gravity coefficients, and add it to the vector of bodies.
- void [initialize_model](#) (BaseDynManager &manager)
Perform base initialization.
- void [initialize_state](#) (const BaseDynManager &manager)
Pass the initialize_state method to each [GravitySource](#) object registered with the gravity manager.
- void [gravitation](#) (const double integ_pos[3], [GravityInteraction](#) &grav)
Compute the gravitational attraction of gravitational bodies on the provided dynamic body.
- void [gravitation](#) (const RefFrame &point, [GravityInteraction](#) &grav)
Compute the gravitational attraction of gravitational bodies on the provided dynamic body.
- const std::vector< [GravitySource](#) * > & [get_bodies](#) () const
Get the vector of gravitational bodies.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GravityManager](#))

Private Attributes

- JeodPointerVector< [GravitySource](#) >::type [sources](#)
The gravitational bodies.

8.4.1 Detailed Description

The master gravitational model for a simulation.

Definition at line 88 of file gravity_manager.hh.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 GravityManager() [1/2]

```
jeod::GravityManager::GravityManager ( )
```

[GravityManager](#) constructor.

Definition at line 57 of file gravity_manager.cc.

8.4.2.2 ~GravityManager()

```
jeod::GravityManager::~~GravityManager ( )
```

[GravityManager](#) destructor.

Definition at line 67 of file gravity_manager.cc.

References [sources](#).

8.4.2.3 GravityManager() [2/2]

```
jeod::GravityManager::GravityManager (
    const GravityManager & ) [delete]
```

8.4.3 Member Function Documentation

8.4.3.1 add_grav_source()

```
void jeod::GravityManager::add_grav_source (
    GravitySource & source )
```

Create a gravitational body, initialize it with the supplied gravity coefficients, and add it to the vector of bodies.

Parameters

<i>in</i>	<i>source</i>	Gravity source to be added
-----------	---------------	----------------------------

Definition at line 109 of file gravity_manager.cc.

References [jeod::GravityMessages::duplicate_entry](#), [find_grav_source\(\)](#), [jeod::GravityMessages::invalid_name](#), [jeod::GravitySource::name](#), and [sources](#).

8.4.3.2 find_grav_source()

```
GravitySource * jeod::GravityManager::find_grav_source (
    const std::string & source_name ) const
```

Find the gravitational body with the given name.

Returns

Pointer to found body

Parameters

in	<i>source_name</i>	Name of gravity source to be found
----	--------------------	------------------------------------

Definition at line 78 of file gravity_manager.cc.

References jeod::GravityMessages::invalid_name, jeod::GravitySource::name, and sources.

Referenced by add_grav_source(), and jeod::GravityControls::initialize_control().

8.4.3.3 get_bodies()

```
const std::vector<GravitySource *>& jeod::GravityManager::get_bodies ( ) const [inline]
```

Get the vector of gravitational bodies.

Warning

Do not modify the vector, or elements of it.

Definition at line 148 of file gravity_manager.hh.

Referenced by jeod::GravityControls::calc_relativistic().

8.4.3.4 gravitation() [1/2]

```
void jeod::GravityManager::gravitation (
    const double integ_pos[3],
    GravityInteraction & grav )
```

Compute the gravitational attraction of gravitational bodies on the provided dynamic body.

Assumptions and Limitations

- Only the gravitational bodies specified in the dynamic body's gravity controls have a bearing on the dynamic body's state.

Warning

This overload is deprecated.

Parameters

in	<i>integ_pos</i>	Dyn body location (integ frm) Units: M
in, out	<i>grav</i>	Gravity interaction

Definition at line 177 of file gravity_manager.cc.

References `jeod::GravityControls::active`, `jeod::GravityControls::grav_accel`, `jeod::GravityInteraction::grav_↵ accel`, `jeod::GravityInteraction::grav_controls`, `jeod::GravityControls::grav_grad`, `jeod::GravityInteraction::grav_↵ grad`, `jeod::GravityControls::grav_pot`, `jeod::GravityInteraction::grav_pot`, `jeod::GravityControls::gravitation()`, and `jeod::GravityInteraction::integ_frame_index`.

8.4.3.5 gravitation() [2/2]

```
void jeod::GravityManager::gravitation (
    const RefFrame & point,
    GravityInteraction & grav )
```

Compute the gravitational attraction of gravitational bodies on the provided dynamic body.

Assumptions and Limitations

- Only the gravitational bodies specified in the dynamic body's gravity controls have a bearing on the dynamic body's state.
- The supplied reference frame is assumed to be a direct child of the dynamic body's integration frame.

Parameters

in	<i>point</i>	Point of interest, as a reference frame.
in, out	<i>grav</i>	Gravity interaction

Definition at line 222 of file gravity_manager.cc.

References `jeod::GravityControls::active`, `jeod::GravityControls::grav_accel`, `jeod::GravityInteraction::grav_↵ accel`, `jeod::GravityInteraction::grav_controls`, `jeod::GravityControls::grav_grad`, `jeod::GravityInteraction::grav_↵ grad`, `jeod::GravityControls::grav_pot`, `jeod::GravityInteraction::grav_pot`, `jeod::GravityControls::gravitation()`, and `jeod::GravityInteraction::integ_frame_index`.

8.4.3.6 initialize_model()

```
void jeod::GravityManager::initialize_model (
    BaseDynManager & manager )
```

Perform base initialization.

Note

This method differs from the other `initialize_models` methods in that this method takes no coefficients as arguments. The `S_define` must call `add_grav_source` explicitly when this signature is used.

Parameters

<code>in, out</code>	<code>manager</code>	Dynamics manager
----------------------	----------------------	------------------

Definition at line 144 of file gravity_manager.cc.

8.4.3.7 initialize_state()

```
void jeod::GravityManager::initialize_state (
    const BaseDynManager & manager )
```

Pass the initialize_state method to each [GravitySource](#) object registered with the gravity manager.

Assumptions and Limitations

- Initialization phasing: The following must have been called prior to calling this method:
 - [GravityManager::initialize_model](#) to register the [GravityManager](#) object with the dynamics manager
 - [GravityManager::add_grav_source](#) to register all [GravitySource](#) objects in the simulation with the gravity manager.
 - [Planet::register_model](#) to associate the planet with a [GravitySource](#).
 - [DynamicsManager::activate_ephemerides](#) to identify which reference frames can serve as integration frames.

Parameters

<code>in, out</code>	<code>manager</code>	Dynamics manager
----------------------	----------------------	------------------

Definition at line 165 of file gravity_manager.cc.

References sources.

8.4.3.8 operator=()

```
GravityManager& jeod::GravityManager::operator= (
    const GravityManager & ) [delete]
```

8.4.3.9 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GravityManager::p (
    jeod ,
    GravityManager ) [private]
```

8.4.4 Field Documentation

8.4.4.1 sources

```
JeodPointerVector<GravitySource>::type jeod::GravityManager::sources [private]
```

The gravitational bodies.

trick_io(**)

Definition at line 97 of file gravity_manager.hh.

Referenced by add_grav_source(), find_grav_source(), initialize_state(), and ~GravityManager().

The documentation for this class was generated from the following files:

- [gravity_manager.hh](#)
- [gravity_manager.cc](#)

8.5 jeod::GravityMessages Class Reference

Specifies the message IDs used in the gravity model.

```
#include <gravity_messages.hh>
```

Public Member Functions

- [GravityMessages](#) ()=delete
- [GravityMessages](#) (const [GravityMessages](#) &)=delete
- [GravityMessages](#) & operator= (const [GravityMessages](#) &)=delete

Static Public Attributes

- static const char * [duplicate_entry](#) = "environment/gravity/" "duplicate_entry"
Issued when a duplicate entry is detected.
- static const char * [missing_entry](#) = "environment/gravity/" "missing_entry"
Issued when a missing entry is detected.
- static const char * [invalid_name](#) = "environment/gravity/" "invalid_name"
Error issued when a name is invalid (null or empty).
- static const char * [invalid_object](#) = "environment/gravity/" "invalid_object"
Error issued when an object is invalid (wrong type).
- static const char * [invalid_limit](#) = "environment/gravity/" "invalid_limit"
Issued when a limit is out of range.
- static const char * [domain_error](#) = "environment/gravity/" "domain_error"
Issued when a value is outside the known-to-be-valid range, e.g., a radial distance less than the planet's equatorial radius.
- static const char * [null_pointer](#) = "environment/gravity/" "null_pointer"
Error issued when a pointer is invalid (null or empty).

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [GravityMessages](#))

8.5.1 Detailed Description

Specifies the message IDs used in the gravity model.

Definition at line 82 of file gravity_messages.hh.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 GravityMessages() [1/2]

```
jeod::GravityMessages::GravityMessages ( ) [delete]
```

8.5.2.2 GravityMessages() [2/2]

```
jeod::GravityMessages::GravityMessages (
    const GravityMessages & ) [delete]
```

8.5.3 Member Function Documentation

8.5.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::GravityMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    GravityMessages ) [private]
```

8.5.3.2 operator=()

```
GravityMessages& jeod::GravityMessages::operator= (
    const GravityMessages & ) [delete]
```

8.5.4 Field Documentation

8.5.4.1 domain_error

```
char const * jeod::GravityMessages::domain_error = "environment/gravity/" "domain_error" [static]
```

Issued when a value is outside the known-to-be-valid range, e.g., a radial distance less than the planet's equatorial radius.

trick_units(—)

Definition at line 115 of file gravity_messages.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical().

8.5.4.2 duplicate_entry

```
char const * jeod::GravityMessages::duplicate_entry = "environment/gravity/" "duplicate_entry" [static]
```

Issued when a duplicate entry is detected.

trick_units(—)

Definition at line 89 of file gravity_messages.hh.

Referenced by jeod::GravityInteraction::add_control(), jeod::SphericalHarmonicsGravitySource::add_deltacoeff(), and jeod::GravityManager::add_grav_source().

8.5.4.3 invalid_limit

```
char const * jeod::GravityMessages::invalid_limit = "environment/gravity/" "invalid_limit" [static]
```

Issued when a limit is out of range.

trick_units(—)

Definition at line 109 of file gravity_messages.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::check_validity().

8.5.4.4 invalid_name

```
char const * jeod::GravityMessages::invalid_name = "environment/gravity/" "invalid_name" [static]
```

Error issued when a name is invalid (null or empty).

trick_units(—)

Definition at line 99 of file gravity_messages.hh.

Referenced by jeod::GravityManager::add_grav_source(), jeod::GravityManager::find_grav_source(), and jeod::↔SphericalHarmonicsTidalEffects::initialize().

8.5.4.5 invalid_object

```
char const * jeod::GravityMessages::invalid_object = "environment/gravity/" "invalid_object"  
[static]
```

Error issued when an object is invalid (wrong type).

trick_units(-)

Definition at line 104 of file gravity_messages.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::check_validity(), jeod::SphericalHarmonicsTidalEffects::initialize(), and jeod::GravityControls::initialize_control().

8.5.4.6 missing_entry

```
char const * jeod::GravityMessages::missing_entry = "environment/gravity/" "missing_entry"  
[static]
```

Issued when a missing entry is detected.

trick_units(-)

Definition at line 94 of file gravity_messages.hh.

Referenced by jeod::GravityControls::initialize_control(), and jeod::GravityInteraction::remove_control().

8.5.4.7 null_pointer

```
char const * jeod::GravityMessages::null_pointer = "environment/gravity/" "null_pointer" [static]
```

Error issued when a pointer is invalid (null or empty).

trick_units(-)

Definition at line 120 of file gravity_messages.hh.

Referenced by jeod::GravityControls::reset_control().

The documentation for this class was generated from the following files:

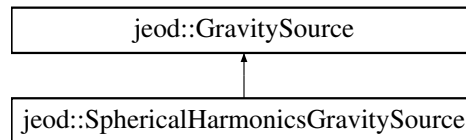
- [gravity_messages.hh](#)
- [gravity_messages.cc](#)

8.6 jeod::GravitySource Class Reference

Models the gravity for a specific planet;.

```
#include <gravity_source.hh>
```

Inheritance diagram for jeod::GravitySource:



Public Member Functions

- [GravitySource](#) ()
GravitySource constructor.
- virtual [~GravitySource](#) ()
GravitySource destructor.
- [GravitySource](#) (const [GravitySource](#) &)=delete
- [GravitySource](#) & operator= (const [GravitySource](#) &)=delete
- virtual void [initialize_state](#) (const std::vector< EphemerisRefFrame * > &integ_frames, const [GravityManager](#) &gravity_manager)
Initialize frame states for the gravity body.

Data Fields

- std::string [name](#) {}
The name of the source (i.e.
- EphemerisRefFrame * [inertial](#) {}
The pseudo-inertial frame associated with this gravity source.
- EphemerisRefFrame * [pfix](#) {}
Planetoid fixed frame.
- double [mu](#) {}
The planet's standard gravitational parameter, G times planet mass.
- [GravityIntegFrame](#) * [frames](#) {}
Relative states with respect to this body, for each integration frame.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [GravitySource](#))

8.6.1 Detailed Description

Models the gravity for a specific planet;.

Definition at line 89 of file gravity_source.hh.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 GravitySource() [1/2]

```
jeod::GravitySource::GravitySource ( )
```

[GravitySource](#) constructor.

Definition at line 51 of file gravity_source.cc.

8.6.2.2 ~GravitySource()

```
jeod::GravitySource::~~GravitySource ( ) [virtual]
```

[GravitySource](#) destructor.

Definition at line 59 of file gravity_source.cc.

References frames.

8.6.2.3 GravitySource() [2/2]

```
jeod::GravitySource::GravitySource (
    const GravitySource & ) [delete]
```

8.6.3 Member Function Documentation

8.6.3.1 initialize_state()

```
void jeod::GravitySource::initialize_state (
    const std::vector< EphemerisRefFrame * > & integ_frames,
    const GravityManager & gravity_manager ) [virtual]
```

Initialize frame states for the gravity body.

Parameters

in	<i>integ_frames</i>	All possible integration frames
in	<i>gravity_manager</i>	Gravity Manager

Definition at line 69 of file gravity_source.cc.

References frames, inertial, jeod::GravityIntegFrame::is_third_body, jeod::GravityIntegFrame::ref_frame, and jeod::GravityIntegFrame::time.

8.6.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::GravitySource::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    GravitySource ) [private]
```

8.6.3.3 operator=()

```
GravitySource& jeod::GravitySource::operator= (
    const GravitySource & ) [delete]
```

8.6.4 Field Documentation

8.6.4.1 frames

```
GravityIntegFrame* jeod::GravitySource::frames {}
```

Relative states with respect to this body, for each integration frame.

trick_units(-)

Definition at line 123 of file gravity_source.hh.

Referenced by jeod::GravityControls::gravitation(), initialize_state(), and ~GravitySource().

8.6.4.2 inertial

```
EphemerisRefFrame* jeod::GravitySource::inertial {}
```

The pseudo-inertial frame associated with this gravity source.

Used for most basic gravity calculations planet represented by this trick_units(-)

Definition at line 106 of file gravity_source.hh.

Referenced by jeod::GravityControls::calc_relativistic(), jeod::GravityControls::gravitation(), jeod::GravityControls::initialize_control(), initialize_state(), and jeod::GravityControls::reset_control().

8.6.4.3 mu

```
double jeod::GravitySource::mu {}
```

The planet's standard gravitational parameter, G times planet mass.

trick_units(m3/s2)

Definition at line 118 of file gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), jeod::GravityControls::calc_relativistic(), jeod::GravityControls::calc_spherical(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data::initialize(), and jeod::SphericalHarmonicsSolidBodyTides::update().

8.6.4.4 name

```
std::string jeod::GravitySource::name {""}
```

The name of the source (i.e.

associated planet or planetoid) The [GravitySource](#) object, the BasePlanet derived object that points to the [GravitySource](#) object, and the EphemerisPlanet that enables populating the planetoid's inertial RefFrame object must all have the exact same name. trick_units(-)

Definition at line 99 of file gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravitySource::add_deltacoeff(), jeod::GravityManager::add_grav_source(), jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), jeod::SphericalHarmonicsGravityControls::check_validity(), jeod::SphericalHarmonicsGravitySource::find_deltacoeff(), jeod::GravityManager::find_grav_source(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data::initialize(), and jeod::GravityControls::initialize_control().

8.6.4.5 pfix

```
EphemerisRefFrame* jeod::GravitySource::pfix {}
```

Planetoid fixed frame.

The Cartesian reference frame centered and fixed on the associated gravity source. Used for advanced (e.g. nonspherical gravity effects. `trick_units(-)`

Definition at line 113 of file `gravity_source.hh`.

Referenced by `jeod::SphericalHarmonicsGravityControls::calc_nonspherical()`, `jeod::SphericalHarmonicsTidalEffects::initialize()`, and `jeod::GravityControls::reset_control()`.

The documentation for this class was generated from the following files:

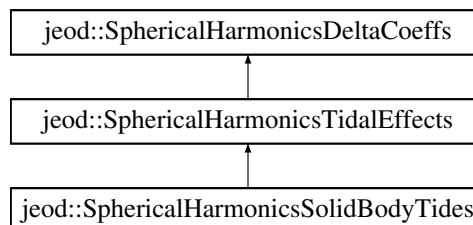
- [gravity_source.hh](#)
- [gravity_source.cc](#)

8.7 jeod::SphericalHarmonicsDeltaCoeffs Class Reference

Base class for tidal and temporal gravity models.

```
#include <spherical_harmonics_delta_coeffs.hh>
```

Inheritance diagram for `jeod::SphericalHarmonicsDeltaCoeffs`:



Public Member Functions

- [SphericalHarmonicsDeltaCoeffs\(\)](#)=default
- virtual [~SphericalHarmonicsDeltaCoeffs\(\)](#)
SphericalHarmonicsDeltaCoeffs destructor.
- virtual void [initialize](#) ([SphericalHarmonicsDeltaCoeffsInit](#) &var_init, BaseDynManager &dyn_manager)
Initialize the class.
- virtual void [update](#) ([SphericalHarmonicsGravityControls](#) &controls)
Pure virtual update method.

Data Fields

- [SphericalHarmonicsGravitySource](#) * [grav_source](#) {}
Pointer to the gravity body associated with this effect.
- double ** [delta_Cnm](#) {}
Normalized real (cosine) variational spherical harmonic coefficients.
- double ** [delta_Snm](#) {}
Normalized imaginary (sine) variational spherical harmonic coeffs.
- unsigned int [degree](#) {}
Coefficient degree to be used for this [SphericalHarmonicsDeltaCoeffs](#).
- unsigned int [order](#) {}
Coefficient order to be used for this [SphericalHarmonicsDeltaCoeffs](#).
- double [dC20](#) {}
delta C20 coefficient for first order effect

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [SphericalHarmonicsDeltaCoeffs](#))

8.7.1 Detailed Description

Base class for tidal and temporal gravity models.

Definition at line 88 of file `spherical_harmonics_delta_coeffs.hh`.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 [SphericalHarmonicsDeltaCoeffs\(\)](#)

```
jeod::SphericalHarmonicsDeltaCoeffs::SphericalHarmonicsDeltaCoeffs ( ) [default]
```

8.7.2.2 [~SphericalHarmonicsDeltaCoeffs\(\)](#)

```
jeod::SphericalHarmonicsDeltaCoeffs::~~SphericalHarmonicsDeltaCoeffs ( ) [virtual]
```

[SphericalHarmonicsDeltaCoeffs](#) destructor.

Definition at line 51 of file `spherical_harmonics_delta_coeffs.cc`.

References `degree`, `delta_Cnm`, and `delta_Snm`.

8.7.3 Member Function Documentation

8.7.3.1 [initialize\(\)](#)

```
void jeod::SphericalHarmonicsDeltaCoeffs::initialize (
    SphericalHarmonicsDeltaCoeffsInit & var_init,
    BaseDynManager & dyn_manager ) [virtual]
```

Initialize the class.

Parameters

<i>in</i>	<i>var_init</i>	Init structure
<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager

Reimplemented in [jeod::SphericalHarmonicsTidalEffects](#), and [jeod::SphericalHarmonicsSolidBodyTides](#).

Definition at line 62 of file `spherical_harmonics_delta_coeffs.cc`.

References `degree`, `jeod::SphericalHarmonicsDeltaCoeffsInit::degree`, `delta_Cnm`, `jeod::SphericalHarmonicsDeltaCoeffsInit::delta_Cnm`, `delta_Snm`, `jeod::SphericalHarmonicsDeltaCoeffsInit::delta_Snm`, `order`, and `jeod::SphericalHarmonicsDeltaCoeffsInit::order`.

Referenced by `jeod::SphericalHarmonicsGravitySource::add_deltacoeff()`, and `jeod::SphericalHarmonicsTidalEffects::initialize()`.

8.7.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SphericalHarmonicsDeltaCoeffs::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SphericalHarmonicsDeltaCoeffs ) [private]
```

8.7.3.3 update()

```
void jeod::SphericalHarmonicsDeltaCoeffs::update (
    SphericalHarmonicsGravityControls & controls ) [virtual]
```

Pure virtual update method.

Parameters

<i>in</i>	<i>controls</i>	Ignored
-----------	-----------------	---------

Reimplemented in [jeod::SphericalHarmonicsTidalEffects](#), and [jeod::SphericalHarmonicsSolidBodyTides](#).

Definition at line 96 of file `spherical_harmonics_delta_coeffs.cc`.

8.7.4 Field Documentation

8.7.4.1 dC20

```
double jeod::SphericalHarmonicsDeltaCoeffs::dC20 {}
```

delta C20 coefficient for first order effect

trick_units(-)

Definition at line 120 of file spherical_harmonics_delta_coeffs.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs(), and jeod::SphericalHarmonicsSolid↵
BodyTides::update().

8.7.4.2 degree

```
unsigned int jeod::SphericalHarmonicsDeltaCoeffs::degree {}
```

Coefficient degree to be used for this [SphericalHarmonicsDeltaCoeffs](#).

trick_units(-)

Definition at line 110 of file spherical_harmonics_delta_coeffs.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::add_deltacontrol(), initialize(), jeod::Spherical↵
HarmonicsTidalEffects::initialize(), and ~SphericalHarmonicsDeltaCoeffs().

8.7.4.3 delta_Cnm

```
double** jeod::SphericalHarmonicsDeltaCoeffs::delta_Cnm {}
```

Normalized real (cosine) variational spherical harmonic coefficients.

trick_units(-)

Definition at line 100 of file spherical_harmonics_delta_coeffs.hh.

Referenced by initialize(), jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs(), and ~Spherical↵
HarmonicsDeltaCoeffs().

8.7.4.4 delta_Snm

```
double** jeod::SphericalHarmonicsDeltaCoeffs::delta_Snm {}
```

Normalized imaginary (sine) variational spherical harmonic coeffs.

trick_units(-)

Definition at line 105 of file spherical_harmonics_delta_coeffs.hh.

Referenced by initialize(), jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs(), and ~Spherical↵
HarmonicsDeltaCoeffs().

8.7.4.5 grav_source

```
SphericalHarmonicsGravitySource* jeod::SphericalHarmonicsDeltaCoeffs::grav_source {}
```

Pointer to the gravity body associated with this effect.

trick_units(-)

Definition at line 95 of file spherical_harmonics_delta_coeffs.hh.

Referenced by `jeod::SphericalHarmonicsGravitySource::add_deltacoeff()`, `jeod::SphericalHarmonicsTidalEffects::initialize()`, and `jeod::SphericalHarmonicsSolidBodyTides::update()`.

8.7.4.6 order

```
unsigned int jeod::SphericalHarmonicsDeltaCoeffs::order {}
```

Coefficient order to be used for this [SphericalHarmonicsDeltaCoeffs](#).

trick_units(-)

Definition at line 115 of file spherical_harmonics_delta_coeffs.hh.

Referenced by `jeod::SphericalHarmonicsGravityControls::add_deltacontrol()`, `initialize()`, and `jeod::SphericalHarmonicsTidalEffects::initialize()`.

The documentation for this class was generated from the following files:

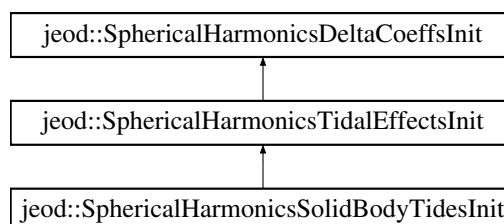
- [spherical_harmonics_delta_coeffs.hh](#)
- [spherical_harmonics_delta_coeffs.cc](#)

8.8 jeod::SphericalHarmonicsDeltaCoeffsInit Class Reference

Initialization data for a [SphericalHarmonicsDeltaCoeffs](#) instance.

```
#include <spherical_harmonics_delta_coeffs_init.hh>
```

Inheritance diagram for `jeod::SphericalHarmonicsDeltaCoeffsInit`:



Public Member Functions

- [SphericalHarmonicsDeltaCoeffsInit](#) ()=default
- virtual [~SphericalHarmonicsDeltaCoeffsInit](#) ()=default

Data Fields

- double ** [delta_Cnm](#) {}
Normalized real (cosine) variational spherical harmonic coefficients.
- double ** [delta_Snm](#) {}
Normalized imaginary (sine) variational spherical harmonic coeffs.
- unsigned int [degree](#) {}
Coefficient degree to be used for this [SphericalHarmonicsDeltaCoeffs](#).
- unsigned int [order](#) {}
Coefficient order to be used for this [SphericalHarmonicsDeltaCoeffs](#).

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [SphericalHarmonicsDeltaCoeffsInit](#))

8.8.1 Detailed Description

Initialization data for a [SphericalHarmonicsDeltaCoeffs](#) instance.

Definition at line 78 of file spherical_harmonics_delta_coeffs_init.hh.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 SphericalHarmonicsDeltaCoeffsInit()

```
jeod::SphericalHarmonicsDeltaCoeffsInit::SphericalHarmonicsDeltaCoeffsInit ( ) [default]
```

8.8.2.2 ~SphericalHarmonicsDeltaCoeffsInit()

```
virtual jeod::SphericalHarmonicsDeltaCoeffsInit::~SphericalHarmonicsDeltaCoeffsInit ( ) [virtual],  
[default]
```

8.8.3 Member Function Documentation

8.8.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SphericalHarmonicsDeltaCoeffsInit::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SphericalHarmonicsDeltaCoeffsInit ) [private]
```

8.8.4 Field Documentation

8.8.4.1 degree

```
unsigned int jeod::SphericalHarmonicsDeltaCoeffsInit::degree {}
```

Coefficient degree to be used for this [SphericalHarmonicsDeltaCoeffs](#).

trick_units(—)

Definition at line 95 of file spherical_harmonics_delta_coeffs_init.hh.

Referenced by jeod::SphericalHarmonicsDeltaCoeffs::initialize().

8.8.4.2 delta_Cnm

```
double** jeod::SphericalHarmonicsDeltaCoeffsInit::delta_Cnm {}
```

Normalized real (cosine) variational spherical harmonic coefficients.

trick_units(—)

Definition at line 85 of file spherical_harmonics_delta_coeffs_init.hh.

Referenced by jeod::SphericalHarmonicsDeltaCoeffs::initialize().

8.8.4.3 delta_Snm

```
double** jeod::SphericalHarmonicsDeltaCoeffsInit::delta_Snm {}
```

Normalized imaginary (sine) variational spherical harmonic coeffs.

trick_units(—)

Definition at line 90 of file spherical_harmonics_delta_coeffs_init.hh.

Referenced by jeod::SphericalHarmonicsDeltaCoeffs::initialize().

8.8.4.4 order

```
unsigned int jeod::SphericalHarmonicsDeltaCoeffsInit::order {}
```

Coefficient order to be used for this [SphericalHarmonicsDeltaCoeffs](#).

trick_units(-)

Definition at line 100 of file spherical_harmonics_delta_coeffs_init.hh.

Referenced by `jeod::SphericalHarmonicsDeltaCoeffs::initialize()`.

The documentation for this class was generated from the following file:

- [spherical_harmonics_delta_coeffs_init.hh](#)

8.9 jeod::SphericalHarmonicsDeltaControls Class Reference

Provides controls for how a variational model affects a vehicle.

```
#include <spherical_harmonics_delta_controls.hh>
```

Public Member Functions

- [SphericalHarmonicsDeltaControls](#) ()=default
- [~SphericalHarmonicsDeltaControls](#) ()=default

Data Fields

- [SphericalHarmonicsDeltaCoeffs](#) * [grav_effect](#) {}
Pointer to associated coefficient-altering gravitational effect.
- [SphericalHarmonicsGravitySource](#) * [grav_source](#) {}
Pointer to the gravity body associated with this effect.
- bool [active](#) {}
Is this variational gravity effect active for this body?
- bool [first_order_only](#) {true}
Calculate first-order term of this effect only; default to true for 2.0.
- unsigned int [degree](#) {}
Coefficient degree to be used for this gravity effect.
- unsigned int [order](#) {}
Coefficient order to be used for this gravity effect.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [SphericalHarmonicsDeltaControls](#))

8.9.1 Detailed Description

Provides controls for how a variational model affects a vehicle.

Definition at line 81 of file spherical_harmonics_delta_controls.hh.

8.9.2 Constructor & Destructor Documentation

8.9.2.1 SphericalHarmonicsDeltaControls()

```
jeod::SphericalHarmonicsDeltaControls::SphericalHarmonicsDeltaControls ( ) [default]
```

8.9.2.2 ~SphericalHarmonicsDeltaControls()

```
jeod::SphericalHarmonicsDeltaControls::~~SphericalHarmonicsDeltaControls ( ) [default]
```

8.9.3 Member Function Documentation

8.9.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SphericalHarmonicsDeltaControls::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SphericalHarmonicsDeltaControls ) [private]
```

8.9.4 Field Documentation

8.9.4.1 active

```
bool jeod::SphericalHarmonicsDeltaControls::active {}
```

Is this variational gravity effect active for this body?

trick_units(−)

Definition at line 98 of file spherical_harmonics_delta_controls.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs().

8.9.4.2 degree

```
unsigned int jeod::SphericalHarmonicsDeltaControls::degree {}
```

Coefficient degree to be used for this gravity effect.

trick_units(-)

Definition at line 108 of file spherical_harmonics_delta_controls.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs().

8.9.4.3 first_order_only

```
bool jeod::SphericalHarmonicsDeltaControls::first_order_only {true}
```

Calculate first-order term of this effect only; default to true for 2.0.

trick_units(-)

Definition at line 103 of file spherical_harmonics_delta_controls.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs().

8.9.4.4 grav_effect

```
SphericalHarmonicsDeltaCoeffs* jeod::SphericalHarmonicsDeltaControls::grav_effect {}
```

Pointer to associated coefficient-altering gravitational effect.

trick_units(-)

Definition at line 88 of file spherical_harmonics_delta_controls.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::add_deltacontrol(), and jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs().

8.9.4.5 grav_source

```
SphericalHarmonicsGravitySource* jeod::SphericalHarmonicsDeltaControls::grav_source {}
```

Pointer to the gravity body associated with this effect.

trick_units(-)

Definition at line 93 of file spherical_harmonics_delta_controls.hh.

8.9.4.6 order

```
unsigned int jeod::SphericalHarmonicsDeltaControls::order {}
```

Coefficient order to be used for this gravity effect.

trick_units(-)

Definition at line 113 of file spherical_harmonics_delta_controls.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs().

The documentation for this class was generated from the following file:

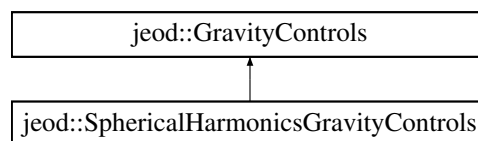
- [spherical_harmonics_delta_controls.hh](#)

8.10 jeod::SphericalHarmonicsGravityControls Class Reference

Specifies whether and how a [SphericalHarmonicsGravitySource](#) affects a vehicle.

```
#include <spherical_harmonics_gravity_controls.hh>
```

Inheritance diagram for jeod::SphericalHarmonicsGravityControls:



Public Member Functions

- [SphericalHarmonicsGravityControls](#) ()
SphericalHarmonicsGravityControls constructor.
- [~SphericalHarmonicsGravityControls](#) () override
SphericalHarmonicsGravityControls destructor.
- [SphericalHarmonicsGravityControls](#) (const [SphericalHarmonicsGravityControls](#) &)=delete
- [SphericalHarmonicsGravityControls](#) & operator= (const [SphericalHarmonicsGravityControls](#) &)=delete
- void [initialize_control](#) ([GravityManager](#) &grav_manager) override
Initialize this GravityControl.
- virtual void [add_deltacontrol](#) ([SphericalHarmonicsDeltaControls](#) *delta_control)
Add a new GravityDeltaControls to the var_effects list.
- unsigned int [get_degree](#) ()
Output the current functional degree.
- unsigned int [get_order](#) ()
Output the current functional order.
- void [get_degree_order](#) (unsigned int ¤t_degree, unsigned int ¤t_order)
Output the current functional degree and order.
- unsigned int [get_grad_degree](#) ()
Output the current functional gradient degree.

- unsigned int `get_grad_order` ()
Output the current functional gradient order.
- void `get_grad_degree_order` (unsigned int &curr_grad_degree, unsigned int &curr_grad_order)
Output the current functional gradient degree and order.
- void `set_degree` (unsigned int new_degree)
Update the functional degree.
- void `set_order` (unsigned int new_order)
Update the functional order.
- void `set_degree_order` (unsigned int new_degree, unsigned int new_order)
Update the functional degree and order.
- void `set_grad_degree` (unsigned int new_grad_degree)
Update the functional gradient degree.
- void `set_grad_order` (unsigned int new_grad_order)
Update the functional gradient order.
- void `set_grad_degree_order` (unsigned int new_grad_degree, unsigned int new_grad_order)
Update the functional gradient degree and order.
- void `disable_min_radius_warnings` ()
Disable minimum radius warnings for this spherical harmonics gravity control.

Data Fields

- `SphericalHarmonicsGravitySource * harmonics_source` {}
The GravitySource pointer from the base class, recast.
- double ** `Pnm` {}
Legendre polynomials used to calculate non-spherical attraction.
- unsigned int `delta_degree` {}
Coefficient degree to be used for totaling up all active delta_coeffs.
- unsigned int `delta_order` {}
Coefficient order to be used for totaling up all active delta_coeffs.
- double ** `delta_Cnm` {}
Array for collecting all active normalized real (cosine) variational spherical harmonic coefficients.
- double ** `delta_Snm` {}
Array for collecting all active normalized real (sine) variational spherical harmonic coefficients.
- double `total_dC20` {}
delta C20 coefficient for collecting first order effects of all active delta_coeffs.
- unsigned int `degree` {}
Non-spherical degree to be used.
- unsigned int `order` {}
Non-spherical order to be used.
- unsigned int `gradient_degree` {}
Non-spherical degree to be used for computing gradient.
- unsigned int `gradient_order` {}
Non-spherical order to be used for computing gradient.
- `JeodPointerVector< SphericalHarmonicsDeltaControls >::type var_effects`
List of controls for variational gravity effects like solid-body tides.

Protected Member Functions

- void [calc_nonspherical](#) (const double integ_pos[3], const double posn[3], const [GravityIntegFrame](#) &grav_↔ source_frame, double body_grav_accel[3], double dgdx[3][3], double &pot) override
Compute the gravitational acceleration at a given position toward a gravitational body assuming the body has a non-spherical mass distribution.
- virtual void [check_validity](#) ()
Check the validity of the gravity controls.
- virtual void [update_deltacoeffs](#) ()
Command all of the gravitational variation effects to update themselves.
- virtual void [sum_deltacoeffs](#) ()
Loop over all of the active gravitational variation effects models and aggregate their changes to the gravity coefficients into the top-level delta-coeffs "bin" for this gravity body.

Protected Attributes

- bool [min_radius_warn](#) {}
Indicates that the minimum radius threshold has been crossed and that a warning has been issued for such.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [SphericalHarmonicsGravityControls](#))

Additional Inherited Members

8.10.1 Detailed Description

Specifies whether and how a [SphericalHarmonicsGravitySource](#) affects a vehicle.

Definition at line 87 of file spherical_harmonics_gravity_controls.hh.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 SphericalHarmonicsGravityControls() [1/2]

```
jeod::SphericalHarmonicsGravityControls::SphericalHarmonicsGravityControls ( )
```

[SphericalHarmonicsGravityControls](#) constructor.

Definition at line 55 of file spherical_harmonics_gravity_controls.cc.

References [var_effects](#).

8.10.2.2 ~SphericalHarmonicsGravityControls()

```
jeod::SphericalHarmonicsGravityControls::~~SphericalHarmonicsGravityControls ( ) [override]
```

[SphericalHarmonicsGravityControls](#) destructor.

Definition at line 65 of file `spherical_harmonics_gravity_controls.cc`.

References `jeod::SphericalHarmonicsGravitySource::degree`, `delta_Cnm`, `delta_degree`, `delta_Snm`, `harmonics_↔source`, `Pnm`, and `var_effects`.

8.10.2.3 SphericalHarmonicsGravityControls() [2/2]

```
jeod::SphericalHarmonicsGravityControls::~SphericalHarmonicsGravityControls (
    const SphericalHarmonicsGravityControls & ) [delete]
```

8.10.3 Member Function Documentation

8.10.3.1 add_deltacontrol()

```
void jeod::SphericalHarmonicsGravityControls::add_deltacontrol (
    SphericalHarmonicsDeltaControls * delta_control ) [virtual]
```

Add a new GravityDeltaControls to the `var_effects` list.

Parameters

<code>in</code>	<code><i>delta_control</i></code>	Control to be added
-----------------	-----------------------------------	---------------------

Definition at line 145 of file `spherical_harmonics_gravity_controls.cc`.

References `jeod::SphericalHarmonicsDeltaCoeffs::degree`, `delta_Cnm`, `delta_degree`, `delta_order`, `delta_Snm`, `jeod::SphericalHarmonicsDeltaControls::grav_effect`, `jeod::SphericalHarmonicsDeltaCoeffs::order`, and `var_effects`.

8.10.3.2 calc_nonspherical()

```
void jeod::SphericalHarmonicsGravityControls::calc_nonspherical (
    const double integ_pos [3],
    const double posn[3],
    const GravityIntegFrame & grav_source_frame,
    double body_grav_accel[3],
```

```
double dgdx[3][3],  
double & pot ) [override], [protected], [virtual]
```

Compute the gravitational acceleration at a given position toward a gravitational body assuming the body has a non-spherical mass distribution.

Parameters

in	<i>posn</i>	Point of interest, inrtl coords Units: M
out	<i>body_grav_accel</i>	Accel for given grav body Units: M/s ²
out	<i>dgdxd</i>	Gradient for given grav body Units: 1/s ²
out	<i>Pot</i>	Potential

Implements [jeod::GravityControls](#).

Definition at line 54 of file spherical_harmonics_calc_nonspherical.cc.

References [jeod::SphericalHarmonicsGravitySource::alpha](#), [jeod::SphericalHarmonicsGravitySource::beta](#), [jeod::SphericalHarmonicsGravitySource::Cnm](#), [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravityMessages::domain_error](#), [jeod::SphericalHarmonicsGravitySource::eta](#), [jeod::GravityControls::gradient](#), [jeod::GravityControls::gradient_degree](#), [jeod::GravityControls::gradient_order](#), [jeod::GravityControls::harmonics_source](#), [jeod::SphericalHarmonicsGravitySource::int_to_double](#), [jeod::GravitySource::min_radius_warn](#), [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::nrdiag](#), [jeod::GravitySource::order](#), [jeod::GravitySource::pfix](#), [jeod::SphericalHarmonicsGravitySource::Pnm](#), [jeod::SphericalHarmonicsGravitySource::radius](#), [jeod::SphericalHarmonicsGravitySource::Snm](#), [jeod::SphericalHarmonicsGravitySource::sum_deltacoeffs\(\)](#), [jeod::SphericalHarmonicsGravitySource::tide_free](#), [jeod::SphericalHarmonicsGravitySource::tide_free_delta](#), [jeod::SphericalHarmonicsGravitySource::total_dC20](#), [jeod::SphericalHarmonicsGravitySource::update_deltacoeffs\(\)](#), [jeod::SphericalHarmonicsGravitySource::epsilon](#), [jeod::SphericalHarmonicsGravitySource::var_effects](#), [jeod::SphericalHarmonicsGravitySource::xi](#), and [jeod::SphericalHarmonicsGravitySource::zeta](#).

8.10.3.3 check_validity()

```
void jeod::SphericalHarmonicsGravityControls::check_validity ( ) [protected], [virtual]
```

Check the validity of the gravity controls.

Definition at line 322 of file spherical_harmonics_gravity_controls.cc.

References [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravityControls::gradient](#), [jeod::GravityControls::gradient_degree](#), [jeod::GravityControls::gradient_order](#), [jeod::GravityMessages::invalid_limit](#), [jeod::GravityMessages::invalid_object](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::order](#), [jeod::GravityControls::source_name](#), and [jeod::GravityControls::spherical](#).

Referenced by [initialize_control\(\)](#), [set_degree\(\)](#), [set_degree_order\(\)](#), [set_grad_degree\(\)](#), [set_grad_degree_order\(\)](#), [set_grad_order\(\)](#), and [set_order\(\)](#).

8.10.3.4 disable_min_radius_warnings()

```
void jeod::SphericalHarmonicsGravityControls::disable_min_radius_warnings ( ) [inline]
```

Disable minimum radius warnings for this spherical harmonics gravity control.

Definition at line 227 of file spherical_harmonics_gravity_controls.hh.

8.10.3.5 get_degree()

```
unsigned int jeod::SphericalHarmonicsGravityControls::get_degree ( )
```

Output the current functional degree.

Returns

Current degree

Definition at line 188 of file spherical_harmonics_gravity_controls.cc.

References degree.

8.10.3.6 get_degree_order()

```
void jeod::SphericalHarmonicsGravityControls::get_degree_order (
    unsigned int & current_degree,
    unsigned int & current_order )
```

Output the current functional degree and order.

Parameters

out	<i>current_degree</i>	Current degree
out	<i>current_order</i>	Current order

Definition at line 207 of file spherical_harmonics_gravity_controls.cc.

References degree, and order.

8.10.3.7 get_grad_degree()

```
unsigned int jeod::SphericalHarmonicsGravityControls::get_grad_degree ( )
```

Output the current functional gradient degree.

Returns

Current gradient degree

Definition at line 217 of file spherical_harmonics_gravity_controls.cc.

References gradient_degree.

8.10.3.8 `get_grad_degree_order()`

```
void jeod::SphericalHarmonicsGravityControls::get_grad_degree_order (
    unsigned int & curr_grad_degree,
    unsigned int & curr_grad_order )
```

Output the current functional gradient degree and order.

Parameters

out	<i>curr_grad_degree</i>	Current gradient degree
out	<i>curr_grad_order</i>	Current gradient order

Definition at line 236 of file spherical_harmonics_gravity_controls.cc.

References `gradient_degree`, and `gradient_order`.

8.10.3.9 get_grad_order()

```
unsigned int jeod::SphericalHarmonicsGravityControls::get_grad_order ( )
```

Output the current functional gradient order.

Returns

Current gradient order

Definition at line 226 of file spherical_harmonics_gravity_controls.cc.

References `gradient_order`.

8.10.3.10 get_order()

```
unsigned int jeod::SphericalHarmonicsGravityControls::get_order ( )
```

Output the current functional order.

Returns

Current order

Definition at line 197 of file spherical_harmonics_gravity_controls.cc.

References `order`.

8.10.3.11 initialize_control()

```
void jeod::SphericalHarmonicsGravityControls::initialize_control (
    GravityManager & grav_manager ) [override], [virtual]
```

Initialize this GravityControl.

Parameters

in	<i>grav_manager</i>	Ref to Gravity Manager
----	---------------------	------------------------

Reimplemented from [jeod::GravityControls](#).

Definition at line 86 of file `spherical_harmonics_gravity_controls.cc`.

References `jeod::GravityControls::body`, `check_validity()`, `jeod::SphericalHarmonicsGravitySource::degree`, `jeod::GravityControls::grav_manager`, `harmonics_source`, `jeod::GravityControls::initialize_control()`, and `Pnm`.

8.10.3.12 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SphericalHarmonicsGravityControls::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SphericalHarmonicsGravityControls ) [private]
```

8.10.3.13 operator=()

```
SphericalHarmonicsGravityControls& jeod::SphericalHarmonicsGravityControls::operator= (
    const SphericalHarmonicsGravityControls & ) [delete]
```

8.10.3.14 set_degree()

```
void jeod::SphericalHarmonicsGravityControls::set_degree (
    unsigned int new_degree )
```

Update the functional degree.

Parameters

in	<i>new_degree</i>	New desired degree
----	-------------------	--------------------

Definition at line 247 of file `spherical_harmonics_gravity_controls.cc`.

References `check_validity()`, and `degree`.

8.10.3.15 set_degree_order()

```
void jeod::SphericalHarmonicsGravityControls::set_degree_order (
    unsigned int new_degree,
    unsigned int new_order )
```

Update the functional degree and order.

Parameters

in	<i>new_degree</i>	New desired degree
in	<i>new_order</i>	New desired order

Definition at line 272 of file spherical_harmonics_gravity_controls.cc.

References [check_validity\(\)](#), [degree](#), and [order](#).

8.10.3.16 set_grad_degree()

```
void jeod::SphericalHarmonicsGravityControls::set_grad_degree (
    unsigned int new_grad_degree )
```

Update the functional gradient degree.

Parameters

in	<i>new_grad_degree</i>	New desired degree
----	------------------------	--------------------

Definition at line 285 of file spherical_harmonics_gravity_controls.cc.

References [check_validity\(\)](#), and [gradient_degree](#).

8.10.3.17 set_grad_degree_order()

```
void jeod::SphericalHarmonicsGravityControls::set_grad_degree_order (
    unsigned int new_grad_degree,
    unsigned int new_grad_order )
```

Update the functional gradient degree and order.

Parameters

in	<i>new_grad_degree</i>	New desired degree
in	<i>new_grad_order</i>	New desired order

Definition at line 310 of file spherical_harmonics_gravity_controls.cc.

References [check_validity\(\)](#), [gradient_degree](#), and [gradient_order](#).

8.10.3.18 set_grad_order()

```
void jeod::SphericalHarmonicsGravityControls::set_grad_order (
    unsigned int new_grad_order )
```

Update the functional gradient order.

Parameters

in	<i>new_grad_order</i>	New desired order
----	-----------------------	-------------------

Definition at line 297 of file spherical_harmonics_gravity_controls.cc.

References `check_validity()`, and `gradient_order`.

8.10.3.19 set_order()

```
void jeod::SphericalHarmonicsGravityControls::set_order (
    unsigned int new_order )
```

Update the functional order.

Parameters

in	<i>new_order</i>	New desired order
----	------------------	-------------------

Definition at line 259 of file spherical_harmonics_gravity_controls.cc.

References `check_validity()`, and `order`.

8.10.3.20 sum_deltacoeffs()

```
void jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs ( ) [protected], [virtual]
```

Loop over all of the active gravitational variation effects models and aggregate their changes to the gravity coefficients into the top-level delta-coeffs "bin" for this gravity body.

Definition at line 481 of file spherical_harmonics_gravity_controls.cc.

References `jeod::SphericalHarmonicsDeltaControls::active`, `jeod::SphericalHarmonicsDeltaCoeffs::dC20`, `jeod::SphericalHarmonicsDeltaControls::degree`, `jeod::SphericalHarmonicsDeltaCoeffs::delta_Cnm`, `delta_Cnm`, `jeod::SphericalHarmonicsGravitySource::delta_coeffs`, `delta_degree`, `delta_order`, `jeod::SphericalHarmonicsDeltaCoeffs::delta_Snm`, `delta_Snm`, `jeod::SphericalHarmonicsDeltaControls::first_order_only`, `jeod::SphericalHarmonicsDeltaControls::grav_effect`, `harmonics_source`, `jeod::SphericalHarmonicsDeltaControls::order`, `total_dC20`, and `var_effects`.

Referenced by `calc_nonspherical()`.

8.10.3.21 update_deltacoeffs()

```
void jeod::SphericalHarmonicsGravityControls::update_deltacoeffs ( ) [protected], [virtual]
```

Command all of the gravitational variation effects to update themselves.

Definition at line 462 of file spherical_harmonics_gravity_controls.cc.

References `jeod::GravityControls::active`, `jeod::SphericalHarmonicsGravitySource::delta_coeffs`, `harmonics_↔source`, and `var_effects`.

Referenced by `calc_nonspherical()`.

8.10.4 Field Documentation

8.10.4.1 degree

```
unsigned int jeod::SphericalHarmonicsGravityControls::degree {}
```

Non-spherical degree to be used.

NOTE: this data being public is deprecated and should not be relied upon in future releases. Use the provided accessor methods instead. `trick_units(-)`

Definition at line 146 of file spherical_harmonics_gravity_controls.hh.

Referenced by `calc_nonspherical()`, `check_validity()`, `get_degree()`, `get_degree_order()`, `set_degree()`, and `set_↔degree_order()`.

8.10.4.2 delta_Cnm

```
double** jeod::SphericalHarmonicsGravityControls::delta_Cnm {}
```

Array for collecting all active normalized real (cosine) variational spherical harmonic coefficients.

`trick_units(-)`

Definition at line 127 of file spherical_harmonics_gravity_controls.hh.

Referenced by `add_deltacontrol()`, `sum_deltacoeffs()`, and `~SphericalHarmonicsGravityControls()`.

8.10.4.3 `delta_degree`

```
unsigned int jeod::SphericalHarmonicsGravityControls::delta_degree {}
```

Coefficient degree to be used for totaling up all active `delta_coeffs`.

`trick_units(-)`

Definition at line 116 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `add_deltacontrol()`, `sum_deltacoeffs()`, and `~SphericalHarmonicsGravityControls()`.

8.10.4.4 `delta_order`

```
unsigned int jeod::SphericalHarmonicsGravityControls::delta_order {}
```

Coefficient order to be used for totaling up all active `delta_coeffs`.

`trick_units(-)`

Definition at line 121 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `add_deltacontrol()`, and `sum_deltacoeffs()`.

8.10.4.5 `delta_Snm`

```
double** jeod::SphericalHarmonicsGravityControls::delta_Snm {}
```

Array for collecting all active normalized real (sine) variational spherical harmonic coefficients.

`trick_units(-)`

Definition at line 133 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `add_deltacontrol()`, `sum_deltacoeffs()`, and `~SphericalHarmonicsGravityControls()`.

8.10.4.6 `gradient_degree`

```
unsigned int jeod::SphericalHarmonicsGravityControls::gradient_degree {}
```

Non-spherical degree to be used for computing gradient.

NOTE: this data being public is deprecated and should not be relied upon in future releases. Use the provided accessor methods instead. `trick_units(-)`

Definition at line 160 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `calc_nonspherical()`, `check_validity()`, `get_grad_degree()`, `get_grad_degree_order()`, `set_grad_degree()`, and `set_grad_degree_order()`.

8.10.4.7 gradient_order

```
unsigned int jeod::SphericalHarmonicsGravityControls::gradient_order {}
```

Non-spherical order to be used for computing gradient.

NOTE: this data being public is deprecated and should not be relied upon in future releases. Use the provided accessor methods instead. `trick_units(-)`

Definition at line 167 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `calc_nonspherical()`, `check_validity()`, `get_grad_degree_order()`, `get_grad_order()`, `set_grad_degree_order()`, and `set_grad_order()`.

8.10.4.8 harmonics_source

```
SphericalHarmonicsGravitySource* jeod::SphericalHarmonicsGravityControls::harmonics_source {}
```

The [GravitySource](#) pointer from the base class, recast.

Note

Users should not set this data member in the input file. `trick_units(-)`

Definition at line 106 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `calc_nonspherical()`, `check_validity()`, `initialize_control()`, `sum_deltacoeffs()`, `update_deltacoeffs()`, and `~SphericalHarmonicsGravityControls()`.

8.10.4.9 min_radius_warn

```
bool jeod::SphericalHarmonicsGravityControls::min_radius_warn {} [protected]
```

Indicates that the minimum radius threshold has been crossed and that a warning has been issued for such.

This prevents a spew of messages regarding such under-threshold conditions.

Note

Users should not set this data member in the input file unless you wish to disable all such messages. `trick_units(-)`

Definition at line 99 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `calc_nonspherical()`.

8.10.4.10 order

```
unsigned int jeod::SphericalHarmonicsGravityControls::order {}
```

Non-spherical order to be used.

NOTE: this data being public is deprecated and should not be relied upon in future releases. Use the provided accessor methods instead. `trick_units(-)`

Definition at line 153 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `calc_nonspherical()`, `check_validity()`, `get_degree_order()`, `get_order()`, `set_degree_order()`, and `set_order()`.

8.10.4.11 Pnm

```
double** jeod::SphericalHarmonicsGravityControls::Pnm {}
```

LeGendre polynomials used to calculate non-spherical attraction.

`trick_units(-)`

Definition at line 111 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `calc_nonspherical()`, `initialize_control()`, and `~SphericalHarmonicsGravityControls()`.

8.10.4.12 total_dC20

```
double jeod::SphericalHarmonicsGravityControls::total_dC20 {}
```

delta C20 coefficient for collecting first order effects of all active `delta_coefs`.

`trick_units(-)`

Definition at line 139 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `calc_nonspherical()`, and `sum_deltacoeffs()`.

8.10.4.13 var_effects

```
JeodPointerVector<SphericalHarmonicsDeltaControls>::type jeod::SphericalHarmonicsGravityControls::var_effects
```

List of controls for variational gravity effects like solid-body tides.

`trick_io(**)`

Definition at line 172 of file `spherical_harmonics_gravity_controls.hh`.

Referenced by `add_deltacontrol()`, `calc_nonspherical()`, `SphericalHarmonicsGravityControls()`, `sum_deltacoeffs()`, `update_deltacoeffs()`, and `~SphericalHarmonicsGravityControls()`.

The documentation for this class was generated from the following files:

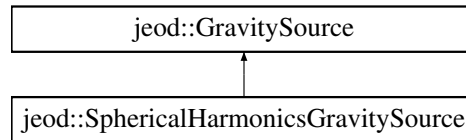
- [spherical_harmonics_gravity_controls.hh](#)
- [spherical_harmonics_calc_nonspherical.cc](#)
- [spherical_harmonics_gravity_controls.cc](#)

8.11 jeod::SphericalHarmonicsGravitySource Class Reference

Models the gravity for a specific planet using spherical harmonics.

```
#include <spherical_harmonics_gravity_source.hh>
```

Inheritance diagram for jeod::SphericalHarmonicsGravitySource:



Public Member Functions

- [SphericalHarmonicsGravitySource](#) ()
SphericalHarmonicsGravitySource constructor.
- [~SphericalHarmonicsGravitySource](#) () override
SphericalHarmonicsGravitySource destructor.
- [SphericalHarmonicsGravitySource](#) (const [SphericalHarmonicsGravitySource](#) &)=delete
- [SphericalHarmonicsGravitySource](#) & operator= (const [SphericalHarmonicsGravitySource](#) &)=delete
- virtual void [initialize_body](#) ()
Initialize Gottlieb gravity coefficients.
- int [find_deltacoeff](#) (const [SphericalHarmonicsDeltaCoeffs](#) &delta_coeff) const
Find the given variational gravity effect if already exists.
- void [add_deltacoeff](#) ([SphericalHarmonicsDeltaCoeffsInit](#) &var_init, [BaseDynManager](#) &dyn_manager, [SphericalHarmonicsDeltaCoeffs](#) &var_effect)
Add a gravitational variation effect (i.e., a delta coeffs) to the vector of effects.

Data Fields

- double [radius](#) {}
Spherical harmonics distance scale, typically the planet's mean equatorial radius.
- unsigned int [degree](#) {}
The degree of the spherical harmonics gravity coefficients.
- unsigned int [order](#) {}
The order of the spherical harmonics gravity coefficients.
- double ** [Cnm](#) {}
Normalized real (cosine) spherical harmonic coefficients.
- double ** [Snm](#) {}
Normalized imaginary (sine) spherical harmonic coefficients.
- bool [tide_free](#) {}
Is C20 coefficient free of the permanent tide effect?
- double [tide_free_delta](#) {}
Number to be added to C20 to remove the permanent tide.
- double * [a_by_rad](#) {}
(Planet radius/vehicle distance)ⁿ
- double * [alpha](#) {}
Gottlieb coefficient alpha.

- double * [beta](#) {}
Gottlieb coefficient beta.
- double ** [xi](#) {}
Gottlieb coefficient xi.
- double ** [eta](#) {}
Gottlieb coefficient eta.
- double ** [zeta](#) {}
Gottlieb coefficient zeta.
- double ** [epsilon](#) {}
Gottlieb coefficient epsilon.
- double * [nrdiag](#) {}
Gottlieb coefficient nrdiag.
- double * [int_to_double](#) {}
0 to degree+1 cast as doubles
- JeodPointerVector< [SphericalHarmonicsDeltaCoeffs](#) >::type [delta_coeffs](#)
List of all gravity coefficient altering effects such as solid-body tides.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [SphericalHarmonicsGravitySource](#))

8.11.1 Detailed Description

Models the gravity for a specific planet using spherical harmonics.

Definition at line 90 of file `spherical_harmonics_gravity_source.hh`.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 [SphericalHarmonicsGravitySource](#)() [1/2]

```
jeod::SphericalHarmonicsGravitySource::SphericalHarmonicsGravitySource ( )
```

[SphericalHarmonicsGravitySource](#) constructor.

Definition at line 57 of file `spherical_harmonics_gravity_source.cc`.

References [delta_coeffs](#).

8.11.2.2 ~SphericalHarmonicsGravitySource()

```
jeod::SphericalHarmonicsGravitySource::~~SphericalHarmonicsGravitySource ( ) [override]
```

[SphericalHarmonicsGravitySource](#) destructor.

Definition at line 67 of file `spherical_harmonics_gravity_source.cc`.

References `a_by_rad`, `alpha`, `beta`, `Cnm`, `degree`, `delta_coeffs`, `eta`, `int_to_double`, `nrdiag`, `Snm`, `upsilon`, `xi`, and `zeta`.

8.11.2.3 SphericalHarmonicsGravitySource() [2/2]

```
jeod::SphericalHarmonicsGravitySource::SphericalHarmonicsGravitySource (
    const SphericalHarmonicsGravitySource & ) [delete]
```

8.11.3 Member Function Documentation

8.11.3.1 add_deltacoeff()

```
void jeod::SphericalHarmonicsGravitySource::add_deltacoeff (
    SphericalHarmonicsDeltaCoeffsInit & var_init,
    BaseDynManager & dyn_manager,
    SphericalHarmonicsDeltaCoeffs & var_effect )
```

Add a gravitational variation effect (i.e., a delta coeffs) to the vector of effects.

Parameters

in	<i>var_init</i>	Effect init structure
in	<i>dyn_manager</i>	Dynamics manager
in	<i>var_effect</i>	Delta coeff to be added

Definition at line 240 of file `spherical_harmonics_gravity_source.cc`.

References `delta_coeffs`, `jeod::GravityMessages::duplicate_entry`, `find_deltacoeff()`, `jeod::SphericalHarmonicsDeltaCoeffs::grav_source`, `jeod::SphericalHarmonicsDeltaCoeffs::initialize()`, and `jeod::GravitySource::name`.

8.11.3.2 find_deltacoeff()

```
int jeod::SphericalHarmonicsGravitySource::find_deltacoeff (
    const SphericalHarmonicsDeltaCoeffs & delta_coeff ) const
```

Find the given variational gravity effect if already exists.

Returns

Index number of delta-coeff; -1 if not found

Parameters

in	<i>delta_coeff</i>	delta-coeff to be found
----	--------------------	-------------------------

Definition at line 211 of file spherical_harmonics_gravity_source.cc.

References `delta_coeffs`, and `jeod::GravitySource::name`.

Referenced by `add_deltacoeff()`.

8.11.3.3 initialize_body()

```
void jeod::SphericalHarmonicsGravitySource::initialize_body ( ) [virtual]
```

Initialize Gottlieb gravity coefficients.

Definition at line 86 of file spherical_harmonics_gravity_source.cc.

References `a_by_rad`, `alpha`, `beta`, `degree`, `eta`, `int_to_double`, `nrdiag`, `upsilon`, `xi`, and `zeta`.

8.11.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SphericalHarmonicsGravitySource::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SphericalHarmonicsGravitySource ) [private]
```

8.11.3.5 operator=()

```
SphericalHarmonicsGravitySource& jeod::SphericalHarmonicsGravitySource::operator= (
    const SphericalHarmonicsGravitySource & ) [delete]
```

8.11.4 Field Documentation

8.11.4.1 a_by_rad

```
double* jeod::SphericalHarmonicsGravitySource::a_by_rad {}
```

(Planet radius/vehicle distance)ⁿ

trick_units(-)

Definition at line 133 of file spherical_harmonics_gravity_source.hh.

Referenced by initialize_body(), and ~SphericalHarmonicsGravitySource().

8.11.4.2 alpha

```
double* jeod::SphericalHarmonicsGravitySource::alpha {}
```

Gottlieb coefficient alpha.

trick_units(-)

Definition at line 138 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), initialize_body(), and ~SphericalHarmonicsGravitySource().

8.11.4.3 beta

```
double* jeod::SphericalHarmonicsGravitySource::beta {}
```

Gottlieb coefficient beta.

trick_units(-)

Definition at line 143 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), initialize_body(), and ~SphericalHarmonicsGravitySource().

8.11.4.4 Cnm

```
double** jeod::SphericalHarmonicsGravitySource::Cnm {}
```

Normalized real (cosine) spherical harmonic coefficients.

trick_units(-)

Definition at line 113 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize(), and ~SphericalHarmonicsGravitySource().

8.11.4.5 degree

```
unsigned int jeod::SphericalHarmonicsGravitySource::degree {}
```

The degree of the spherical harmonics gravity coefficients.

trick_units(-)

Definition at line 103 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::check_validity(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize(), initialize_body(), jeod::SphericalHarmonicsGravityControls::initialize_control(), jeod::SphericalHarmonicsGravityControls::~SphericalHarmonicsGravityControls(), and ~SphericalHarmonicsGravitySource().

8.11.4.6 delta_coeffs

```
JeodPointerVector<SphericalHarmonicsDeltaCoeffs>::type jeod::SphericalHarmonicsGravitySource::delta_coeffs
```

List of all gravity coefficient altering effects such as solid-body tides.

trick_io(**)

Definition at line 179 of file spherical_harmonics_gravity_source.hh.

Referenced by add_deltacoeff(), find_deltacoeff(), SphericalHarmonicsGravitySource(), jeod::SphericalHarmonicsGravityControls::sum_deltacoeffs(), jeod::SphericalHarmonicsGravityControls::update_deltacoeffs(), and ~SphericalHarmonicsGravitySource().

8.11.4.7 eta

```
double** jeod::SphericalHarmonicsGravitySource::eta {}
```

Gottlieb coefficient eta.

trick_units(-)

Definition at line 153 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), initialize_body(), and ~SphericalHarmonicsGravitySource().

8.11.4.8 int_to_double

```
double* jeod::SphericalHarmonicsGravitySource::int_to_double {}
```

0 to degree+1 cast as doubles

trick_units(—)

Definition at line 173 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), initialize_body(), and ~SphericalHarmonicsGravitySource().

8.11.4.9 nrdiag

```
double* jeod::SphericalHarmonicsGravitySource::nrdiag {}
```

Gottlieb coefficient nrdiag.

trick_units(—)

Definition at line 168 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), initialize_body(), and ~SphericalHarmonicsGravitySource().

8.11.4.10 order

```
unsigned int jeod::SphericalHarmonicsGravitySource::order {}
```

The order of the spherical harmonics gravity coefficients.

trick_units(—)

Definition at line 108 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::check_validity(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), and jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize().

8.11.4.11 radius

```
double jeod::SphericalHarmonicsGravitySource::radius {}
```

Spherical harmonics distance scale, typically the planet's mean equatorial radius.

trick_units(m)

Definition at line 98 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data::initialize(), and jeod::SphericalHarmonicsSolidBodyTides::update().

8.11.4.12 Snm

```
double** jeod::SphericalHarmonicsGravitySource::Snm {}
```

Normalized imaginary (sine) spherical harmonic coefficients.

trick_units(-)

Definition at line 118 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize(), and ~SphericalHarmonicsGravitySource().

8.11.4.13 tide_free

```
bool jeod::SphericalHarmonicsGravitySource::tide_free {}
```

Is C20 coefficient free of the permanent tide effect?

trick_units(-)

Definition at line 123 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), and jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize().

8.11.4.14 tide_free_delta

```
double jeod::SphericalHarmonicsGravitySource::tide_free_delta {}
```

Number to be added to C20 to remove the permanent tide.

trick_units(—)

Definition at line 128 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize(), jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize(), and jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize().

8.11.4.15 upsilon

```
double** jeod::SphericalHarmonicsGravitySource::upsilon {}
```

Gottlieb coefficient upsilon.

trick_units(—)

Definition at line 163 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), initialize_body(), and ~SphericalHarmonicsGravitySource().

8.11.4.16 xi

```
double** jeod::SphericalHarmonicsGravitySource::xi {}
```

Gottlieb coefficient xi.

trick_units(—)

Definition at line 148 of file spherical_harmonics_gravity_source.hh.

Referenced by jeod::SphericalHarmonicsGravityControls::calc_nonspherical(), initialize_body(), and ~SphericalHarmonicsGravitySource().

8.11.4.17 zeta

```
double** jeod::SphericalHarmonicsGravitySource::zeta {}
```

Gottlieb coefficient zeta.

trick_units(−)

Definition at line 158 of file spherical_harmonics_gravity_source.hh.

Referenced by `jeod::SphericalHarmonicsGravityControls::calc_nonspherical()`, `initialize_body()`, and `~SphericalHarmonicsGravitySource()`.

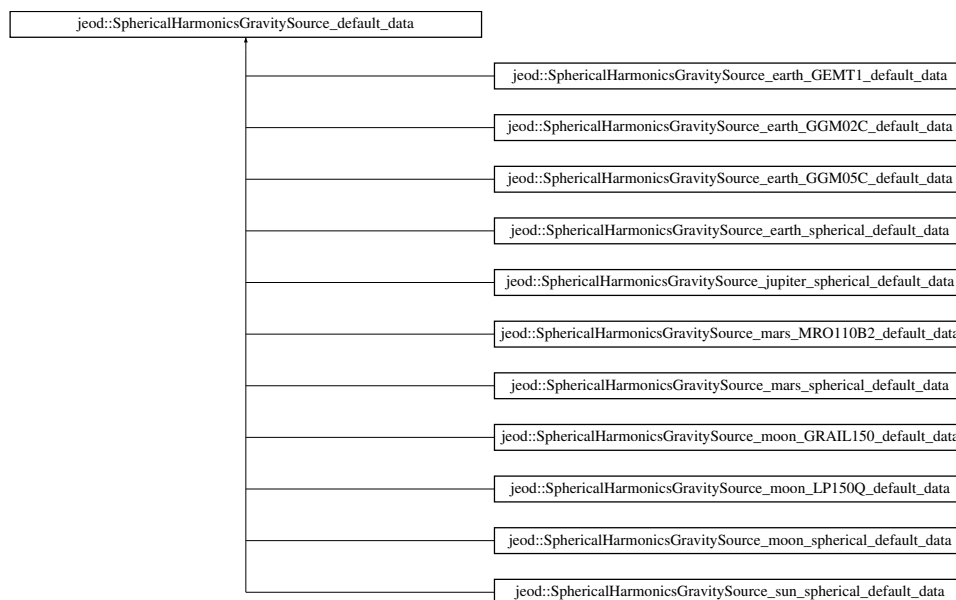
The documentation for this class was generated from the following files:

- [spherical_harmonics_gravity_source.hh](#)
- [spherical_harmonics_gravity_source.cc](#)

8.12 jeod::SphericalHarmonicsGravitySource_default_data Class Reference

```
#include <spherical_harmonics_gravity_source_default_data.hh>
```

Inheritance diagram for `jeod::SphericalHarmonicsGravitySource_default_data`:



Public Member Functions

- virtual void `initialize (SphericalHarmonicsGravitySource *)=0`
- virtual `~SphericalHarmonicsGravitySource_default_data ()=default`

8.12.1 Detailed Description

Definition at line 51 of file spherical_harmonics_gravity_source_default_data.hh.

8.12.2 Constructor & Destructor Documentation

8.12.2.1 ~SphericalHarmonicsGravitySource_default_data()

```
virtual jeod::SphericalHarmonicsGravitySource_default_data::~SphericalHarmonicsGravitySource↔
_default_data ( ) [virtual], [default]
```

8.12.3 Member Function Documentation

8.12.3.1 initialize()

```
virtual void jeod::SphericalHarmonicsGravitySource_default_data::initialize (
    SphericalHarmonicsGravitySource * ) [pure virtual]
```

Implemented in [jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data](#), [jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data](#), [jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data](#), [jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data](#), [jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data](#), [jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data](#), [jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data](#), [jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data](#), [jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data](#), [jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data](#) and [jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data](#).

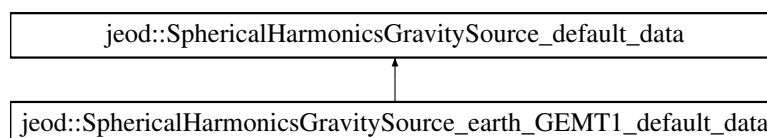
The documentation for this class was generated from the following file:

- [spherical_harmonics_gravity_source_default_data.hh](#)

8.13 jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data Class Reference ↩

```
#include <earth_GEMT1.hh>
```

Inheritance diagram for jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data:



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource](#) *) override

8.13.1 Detailed Description

Definition at line 55 of file `earth_GEMT1.hh`.

8.13.2 Member Function Documentation

8.13.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 60 of file `earth_GEMT1.cc`.

References [jeod::SphericalHarmonicsGravitySource::Cnm](#), [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::order](#), [jeod::SphericalHarmonicsGravitySource::radius](#), [jeod::SphericalHarmonicsGravitySource::Snm](#), [jeod::SphericalHarmonicsGravitySource::tide_free](#), and [jeod::SphericalHarmonicsGravitySource::tide_free_delta](#).

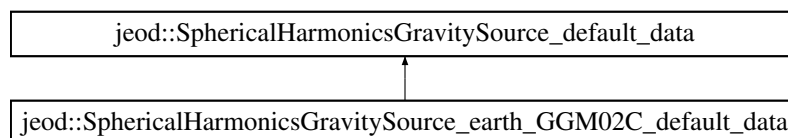
The documentation for this class was generated from the following files:

- [earth_GEMT1.hh](#)
- [earth_GEMT1.cc](#)

8.14 jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data Class Reference

```
#include <earth_GGM02C.hh>
```

Inheritance diagram for `jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data`:



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource](#) *) override

8.14.1 Detailed Description

Definition at line 55 of file earth_GGM02C.hh.

8.14.2 Member Function Documentation

8.14.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 56 of file earth_GGM02C.cc.

References [jeod::SphericalHarmonicsGravitySource::Cnm](#), [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::order](#), [jeod::SphericalHarmonicsGravitySource::radius](#), [jeod::SphericalHarmonicsGravitySource::Snm](#), [jeod::SphericalHarmonicsGravitySource::tide_free](#), and [jeod::SphericalHarmonicsGravitySource::tide_free_delta](#).

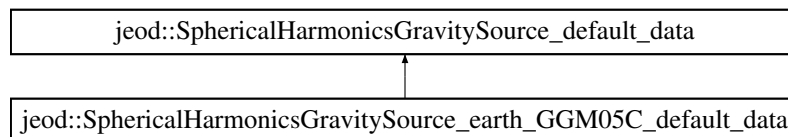
The documentation for this class was generated from the following files:

- [earth_GGM02C.hh](#)
- [earth_GGM02C.cc](#)

8.15 jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data Class Reference

```
#include <earth_GGM05C.hh>
```

Inheritance diagram for [jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data](#):



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource *](#)) override

8.15.1 Detailed Description

Definition at line 55 of file earth_GGM05C.hh.

8.15.2 Member Function Documentation

8.15.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 33 of file earth_GGM05C.cc.

References [jeod::SphericalHarmonicsGravitySource::Cnm](#), [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::order](#), [jeod::SphericalHarmonicsGravitySource::radius](#), [jeod::SphericalHarmonicsGravitySource::Snm](#), [jeod::SphericalHarmonicsGravitySource::tide_free](#), and [jeod::SphericalHarmonicsGravitySource::tide_free_delta](#).

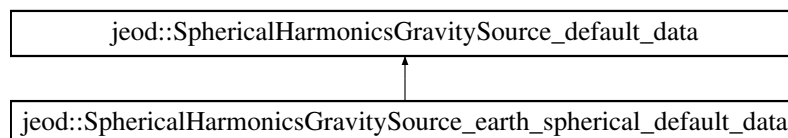
The documentation for this class was generated from the following files:

- [earth_GGM05C.hh](#)
- [earth_GGM05C.cc](#)

8.16 jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data Class Reference

```
#include <earth_spherical.hh>
```

Inheritance diagram for [jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data](#):



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource *](#)) override

8.16.1 Detailed Description

Definition at line 54 of file earth_spherical.hh.

8.16.2 Member Function Documentation

8.16.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 39 of file earth_spherical.cc.

References [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::order](#), and [jeod::SphericalHarmonicsGravitySource::radius](#).

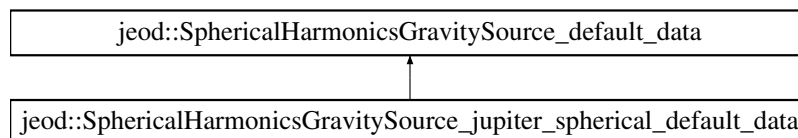
The documentation for this class was generated from the following files:

- [earth_spherical.hh](#)
- [earth_spherical.cc](#)

8.17 jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data Class Reference

```
#include <jupiter_spherical.hh>
```

Inheritance diagram for [jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data](#):



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource](#) *) override

8.17.1 Detailed Description

Definition at line 55 of file jupiter_spherical.hh.

8.17.2 Member Function Documentation

8.17.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 38 of file `jupiter_spherical.cc`.

References [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), and [jeod::SphericalHarmonicsGravitySource↵
::radius](#).

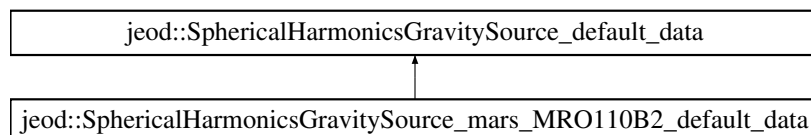
The documentation for this class was generated from the following files:

- [jupiter_spherical.hh](#)
- [jupiter_spherical.cc](#)

8.18 jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_↵ default_data Class Reference

```
#include <mars_MRO110B2.hh>
```

Inheritance diagram for `jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data`:



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource](#) *) override

8.18.1 Detailed Description

Definition at line 55 of file `mars_MRO110B2.hh`.

8.18.2 Member Function Documentation

8.18.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 50 of file mars_MRO110B2.cc.

References [jeod::SphericalHarmonicsGravitySource::Cnm](#), [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::order](#), [jeod::SphericalHarmonicsGravitySource::radius](#), [jeod::SphericalHarmonicsGravitySource::Snm](#), and [jeod::SphericalHarmonicsGravitySource::tide_free](#).

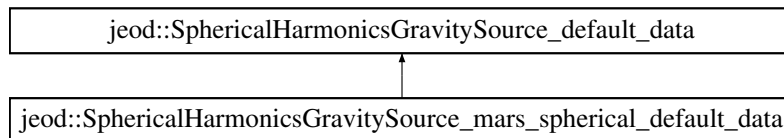
The documentation for this class was generated from the following files:

- [mars_MRO110B2.hh](#)
- [mars_MRO110B2.cc](#)

8.19 jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data Class Reference

```
#include <mars_spherical.hh>
```

Inheritance diagram for jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data:



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource](#) *) override

8.19.1 Detailed Description

Definition at line 55 of file mars_spherical.hh.

8.19.2 Member Function Documentation

8.19.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 39 of file mars_spherical.cc.

References [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), and [jeod::SphericalHarmonicsGravitySource↵
::radius](#).

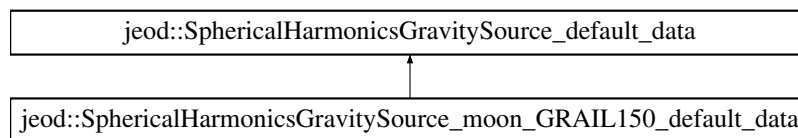
The documentation for this class was generated from the following files:

- [mars_spherical.hh](#)
- [mars_spherical.cc](#)

8.20 jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_↵ default_data Class Reference

```
#include <moon_GRAIL150.hh>
```

Inheritance diagram for [jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data](#):



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource *](#)) override

8.20.1 Detailed Description

Definition at line 54 of file moon_GRAIL150.hh.

8.20.2 Member Function Documentation

8.20.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 53 of file moon_GRAIL150.cc.

References [jeod::SphericalHarmonicsGravitySource::Cnm](#), [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::order](#), [jeod::SphericalHarmonicsGravitySource::radius](#), [jeod::SphericalHarmonicsGravitySource::Snm](#), [jeod::SphericalHarmonicsGravitySource::tide_free](#), and [jeod::SphericalHarmonicsGravitySource::tide_free_delta](#).

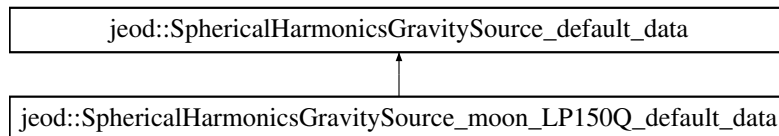
The documentation for this class was generated from the following files:

- [moon_GRAIL150.hh](#)
- [moon_GRAIL150.cc](#)

8.21 jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data Class Reference

```
#include <moon_LP150Q.hh>
```

Inheritance diagram for [jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data](#):



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource](#) *) override

8.21.1 Detailed Description

Definition at line 55 of file moon_LP150Q.hh.

8.21.2 Member Function Documentation

8.21.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 55 of file moon_LP150Q.cc.

References [jeod::SphericalHarmonicsGravitySource::Cnm](#), [jeod::SphericalHarmonicsGravitySource::degree](#), [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), [jeod::SphericalHarmonicsGravitySource::order](#), [jeod::SphericalHarmonicsGravitySource::radius](#), [jeod::SphericalHarmonicsGravitySource::Snm](#), [jeod::SphericalHarmonicsGravitySource::tide_free](#), and [jeod::SphericalHarmonicsGravitySource::tide_free_delta](#).

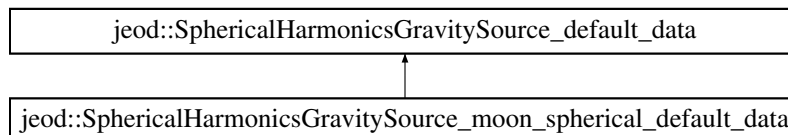
The documentation for this class was generated from the following files:

- [moon_LP150Q.hh](#)
- [moon_LP150Q.cc](#)

8.22 jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data Class Reference

```
#include <moon_spherical.hh>
```

Inheritance diagram for [jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data](#):



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource *](#)) override

8.22.1 Detailed Description

Definition at line 55 of file moon_spherical.hh.

8.22.2 Member Function Documentation

8.22.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 41 of file moon_spherical.cc.

References [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), and [jeod::SphericalHarmonicsGravitySource↵::radius](#).

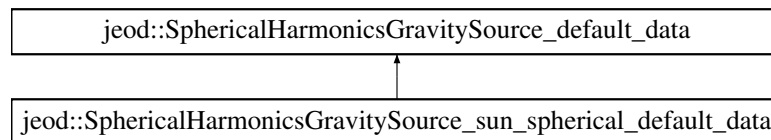
The documentation for this class was generated from the following files:

- [moon_spherical.hh](#)
- [moon_spherical.cc](#)

8.23 jeod::SphericalHarmonicsGravitySource_sun_spherical_default_↵data Class Reference

```
#include <sun_spherical.hh>
```

Inheritance diagram for jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data:



Public Member Functions

- void [initialize](#) ([SphericalHarmonicsGravitySource *](#)) override

8.23.1 Detailed Description

Definition at line 55 of file sun_spherical.hh.

8.23.2 Member Function Documentation

8.23.2.1 initialize()

```
void jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data::initialize (
    SphericalHarmonicsGravitySource * SphericalHarmonicsGravitySource_ptr ) [override],
[virtual]
```

Implements [jeod::SphericalHarmonicsGravitySource_default_data](#).

Definition at line 42 of file `sun_spherical.cc`.

References [jeod::GravitySource::mu](#), [jeod::GravitySource::name](#), and [jeod::SphericalHarmonicsGravitySource::radius](#).

The documentation for this class was generated from the following files:

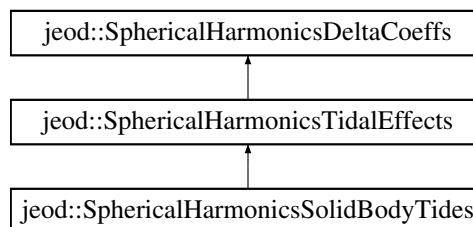
- [sun_spherical.hh](#)
- [sun_spherical.cc](#)

8.24 jeod::SphericalHarmonicsSolidBodyTides Class Reference

Models solid body tidal effects.

```
#include <spherical_harmonics_solid_body_tides.hh>
```

Inheritance diagram for `jeod::SphericalHarmonicsSolidBodyTides`:



Public Member Functions

- [SphericalHarmonicsSolidBodyTides](#) ()=default
- [~SphericalHarmonicsSolidBodyTides](#) () override=default
- void [initialize](#) ([SphericalHarmonicsDeltaCoeffsInit](#) &var_init, BaseDynManager &dyn_manager) override
Initialize the solid body tidal model.
- void [update](#) ([SphericalHarmonicsGravityControls](#) &controls) override
Update the solid-body tidal delta-coefficients.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [SphericalHarmonicsSolidBodyTides](#))

Additional Inherited Members

8.24.1 Detailed Description

Models solid body tidal effects.

Definition at line 89 of file spherical_harmonics_solid_body_tides.hh.

8.24.2 Constructor & Destructor Documentation

8.24.2.1 SphericalHarmonicsSolidBodyTides()

```
jeod::SphericalHarmonicsSolidBodyTides::SphericalHarmonicsSolidBodyTides ( ) [default]
```

8.24.2.2 ~SphericalHarmonicsSolidBodyTides()

```
jeod::SphericalHarmonicsSolidBodyTides::~~SphericalHarmonicsSolidBodyTides ( ) [override],
[default]
```

8.24.3 Member Function Documentation

8.24.3.1 initialize()

```
void jeod::SphericalHarmonicsSolidBodyTides::initialize (
    SphericalHarmonicsDeltaCoeffsInit & var_init,
    BaseDynManager & dyn_manager ) [override], [virtual]
```

Initialize the solid body tidal model.

Parameters

in	<i>var_init</i>	Effect init structure
in	<i>dyn_manager</i>	Dynamics manager

Reimplemented from [jeod::SphericalHarmonicsDeltaCoeffs](#).

Definition at line 58 of file spherical_harmonics_solid_body_tides.cc.

References [jeod::SphericalHarmonicsTidalEffects::initialize\(\)](#).

8.24.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SphericalHarmonicsSolidBodyTides::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SphericalHarmonicsSolidBodyTides ) [private]
```

8.24.3.3 update()

```
void jeod::SphericalHarmonicsSolidBodyTides::update (
    SphericalHarmonicsGravityControls & controls ) [override], [virtual]
```

Update the solid-body tidal delta-coefficients.

Parameters

in	<i>controls</i>	Gravity controls for planet
----	-----------------	-----------------------------

Reimplemented from [jeod::SphericalHarmonicsDeltaCoeffs](#).

Definition at line 69 of file `spherical_harmonics_solid_body_tides.cc`.

References `jeod::SphericalHarmonicsDeltaCoeffs::dC20`, `jeod::SphericalHarmonicsDeltaCoeffs::grav_source`, `jeod::SphericalHarmonicsTidalEffects::k2`, `jeod::GravitySource::mu`, `jeod::SphericalHarmonicsTidalEffects::num_tidal_bodies`, `jeod::SphericalHarmonicsTidalEffects::pfix`, `jeod::SphericalHarmonicsGravitySource::radius`, `jeod::SphericalHarmonicsTidalEffects::tidal_bodies`, and `jeod::SphericalHarmonicsTidalEffects::tidal_bodies_inertial`.

The documentation for this class was generated from the following files:

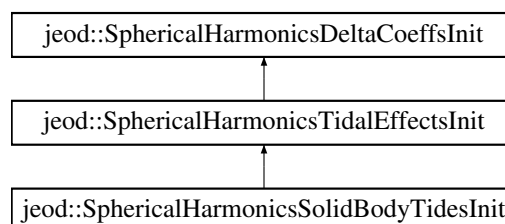
- [spherical_harmonics_solid_body_tides.hh](#)
- [spherical_harmonics_solid_body_tides.cc](#)

8.25 jeod::SphericalHarmonicsSolidBodyTidesInit Class Reference

Initializes a solid body tides model.

```
#include <spherical_harmonics_solid_body_tides_init.hh>
```

Inheritance diagram for `jeod::SphericalHarmonicsSolidBodyTidesInit`:



Public Member Functions

- [SphericalHarmonicsSolidBodyTidesInit \(\)](#)=default
- [~SphericalHarmonicsSolidBodyTidesInit \(\)](#) override=default

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2 p](#) (jeod, [SphericalHarmonicsSolidBodyTidesInit](#))

Additional Inherited Members

8.25.1 Detailed Description

Initializes a solid body tides model.

Definition at line 81 of file `spherical_harmonics_solid_body_tides_init.hh`.

8.25.2 Constructor & Destructor Documentation

8.25.2.1 SphericalHarmonicsSolidBodyTidesInit()

```
jeod::SphericalHarmonicsSolidBodyTidesInit::SphericalHarmonicsSolidBodyTidesInit ( ) [default]
```

8.25.2.2 ~SphericalHarmonicsSolidBodyTidesInit()

```
jeod::SphericalHarmonicsSolidBodyTidesInit::~~SphericalHarmonicsSolidBodyTidesInit ( ) [override],
[default]
```

8.25.3 Member Function Documentation

8.25.3.1 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::SphericalHarmonicsSolidBodyTidesInit::p (
    jeod ,
    SphericalHarmonicsSolidBodyTidesInit ) [private]
```

The documentation for this class was generated from the following file:

- [spherical_harmonics_solid_body_tides_init.hh](#)

8.26 jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data Class Reference

```
#include <earth_solid_tides.hh>
```

Public Member Functions

- void [initialize](#) ([SphericalHarmonicsSolidBodyTidesInit](#) *)

8.26.1 Detailed Description

Definition at line 55 of file `earth_solid_tides.hh`.

8.26.2 Member Function Documentation

8.26.2.1 initialize()

```
void jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data::initialize (
    SphericalHarmonicsSolidBodyTidesInit * SphericalHarmonicsSolidBodyTidesInit_ptr )
```

Definition at line 41 of file `earth_solid_tides.cc`.

References `jeod::SphericalHarmonicsTidalEffectsInit::k2`, and `jeod::SphericalHarmonicsTidalEffectsInit::tidal_body_names`.

The documentation for this class was generated from the following files:

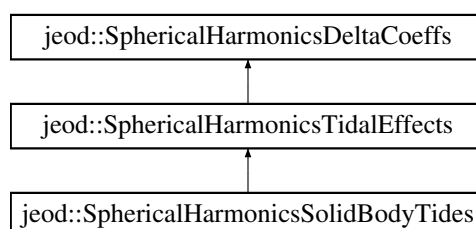
- [earth_solid_tides.hh](#)
- [earth_solid_tides.cc](#)

8.27 jeod::SphericalHarmonicsTidalEffects Class Reference

Models tidal effects as a delta on top of a gravity model.

```
#include <spherical_harmonics_tidal_effects.hh>
```

Inheritance diagram for `jeod::SphericalHarmonicsTidalEffects`:



Public Member Functions

- [SphericalHarmonicsTidalEffects](#) ()=default
- [~SphericalHarmonicsTidalEffects](#) () override
SphericalHarmonicsTidalEffects destructor.
- void [initialize](#) ([SphericalHarmonicsDeltaCoeffsInit](#) &var_init, BaseDynManager &dyn_manager) override
Initialize a SphericalHarmonicsTidalEffects object.
- void [update](#) ([SphericalHarmonicsGravityControls](#) &controls) override
Pure virtual update method.

Data Fields

- double [xp](#) {}
Copy of polar motion coefficient xp (from polar motion class).
- double [yp](#) {}
Copy of polar motion coefficient yp (from polar motion class).
- double [k2](#) {}
The love number.
- double ** [Knm](#) {}
A matrix of love numbers.
- unsigned int [num_tidal_bodies](#) {}
The number of tidal bodies named in tidal_bodies.

Protected Attributes

- Planet ** [tidal_bodies](#) {}
The tidal bodies.
- RefFrame ** [tidal_bodies_inertial](#) {}
Pointers to the tidal_bodies inertial reference frames.
- RefFrame * [pfix](#) {}
The planet fixed reference frame of the subject body.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [SphericalHarmonicsTidalEffects](#))

8.27.1 Detailed Description

Models tidal effects as a delta on top of a gravity model.

Definition at line 91 of file spherical_harmonics_tidal_effects.hh.

8.27.2 Constructor & Destructor Documentation

8.27.2.1 SphericalHarmonicsTidalEffects()

```
jeod::SphericalHarmonicsTidalEffects::SphericalHarmonicsTidalEffects ( ) [default]
```

8.27.2.2 ~SphericalHarmonicsTidalEffects()

```
jeod::SphericalHarmonicsTidalEffects::~~SphericalHarmonicsTidalEffects ( ) [override]
```

[SphericalHarmonicsTidalEffects](#) destructor.

Definition at line 65 of file `spherical_harmonics_tidal_effects.cc`.

8.27.3 Member Function Documentation

8.27.3.1 initialize()

```
void jeod::SphericalHarmonicsTidalEffects::initialize (
    SphericalHarmonicsDeltaCoeffsInit & gen_var_init,
    BaseDynManager & dyn_manager ) [override], [virtual]
```

Initialize a [SphericalHarmonicsTidalEffects](#) object.

This method overrides and calls the base class initialize method.

Parameters

in	<i>gen_var_init</i>	Effect init structure
in	<i>dyn_manager</i>	Dynamics manager

Reimplemented from [jeod::SphericalHarmonicsDeltaCoeffs](#).

Definition at line 78 of file `spherical_harmonics_tidal_effects.cc`.

References [jeod::SphericalHarmonicsDeltaCoeffs::degree](#), [jeod::SphericalHarmonicsDeltaCoeffs::grav_source](#), [jeod::SphericalHarmonicsDeltaCoeffs::initialize\(\)](#), [jeod::GravityMessages::invalid_name](#), [jeod::GravityMessages::invalid_object](#), [jeod::SphericalHarmonicsTidalEffectsInit::k2](#), [jeod::SphericalHarmonicsTidalEffectsInit::Kn](#), [jeod::SphericalHarmonicsTidalEffectsInit::Kn](#), [jeod::SphericalHarmonicsDeltaCoeffs::order](#), [jeod::GravitySource::pfix](#), [pfix](#), [tidal_bodies](#), [tidal_bodies_inertial](#), [jeod::SphericalHarmonicsTidalEffectsInit::tidal_body_names](#), [xp](#), [jeod::SphericalHarmonicsTidalEffectsInit::xp](#), [yp](#), and [jeod::SphericalHarmonicsTidalEffectsInit::yp](#).

Referenced by [jeod::SphericalHarmonicsSolidBodyTides::initialize\(\)](#).

8.27.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SphericalHarmonicsTidalEffects::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SphericalHarmonicsTidalEffects ) [private]
```

8.27.3.3 update()

```
void jeod::SphericalHarmonicsTidalEffects::update (
    SphericalHarmonicsGravityControls & controls ) [override], [virtual]
```

Pure virtual update method.

Parameters

in	<i>controls</i>	Gravity controls for planet
----	-----------------	-----------------------------

Reimplemented from [jeod::SphericalHarmonicsDeltaCoeffs](#).

Definition at line 174 of file `spherical_harmonics_tidal_effects.cc`.

8.27.4 Field Documentation

8.27.4.1 k2

```
double jeod::SphericalHarmonicsTidalEffects::k2 {}
```

The love number.

Only used for a first order tidal effect model. `trick_units(-)`

Definition at line 108 of file `spherical_harmonics_tidal_effects.hh`.

Referenced by `initialize()`, and `jeod::SphericalHarmonicsSolidBodyTides::update()`.

8.27.4.2 Knm

```
double** jeod::SphericalHarmonicsTidalEffects::Knm {}
```

A matrix of love numbers.

Used for higher order (not first-order) tidal effects. `trick_units(-)`

Definition at line 114 of file `spherical_harmonics_tidal_effects.hh`.

Referenced by `initialize()`.

8.27.4.3 num_tidal_bodies

```
unsigned int jeod::SphericalHarmonicsTidalEffects::num_tidal_bodies {}
```

The number of tidal bodies named in tidal_bodies.

trick_units(count)

Definition at line 119 of file spherical_harmonics_tidal_effects.hh.

Referenced by initialize(), and jeod::SphericalHarmonicsSolidBodyTides::update().

8.27.4.4 pfix

```
RefFrame* jeod::SphericalHarmonicsTidalEffects::pfix {} [protected]
```

The planet fixed reference frame of the subject body.

trick_units(-)

Definition at line 136 of file spherical_harmonics_tidal_effects.hh.

Referenced by initialize(), and jeod::SphericalHarmonicsSolidBodyTides::update().

8.27.4.5 tidal_bodies

```
Planet** jeod::SphericalHarmonicsTidalEffects::tidal_bodies {} [protected]
```

The tidal bodies.

Filled out at initialization. Length after init is num_tidal_bodies. trick_units(-)

Definition at line 126 of file spherical_harmonics_tidal_effects.hh.

Referenced by initialize(), and jeod::SphericalHarmonicsSolidBodyTides::update().

8.27.4.6 tidal_bodies_inertial

```
RefFrame** jeod::SphericalHarmonicsTidalEffects::tidal_bodies_inertial {} [protected]
```

Pointers to the tidal_bodies inertial reference frames.

trick_units(-)

Definition at line 131 of file spherical_harmonics_tidal_effects.hh.

Referenced by initialize(), and jeod::SphericalHarmonicsSolidBodyTides::update().

8.27.4.7 xp

```
double jeod::SphericalHarmonicsTidalEffects::xp {}
```

Copy of polar motion coefficient xp (from polar motion class).

trick_units(-)

Definition at line 98 of file spherical_harmonics_tidal_effects.hh.

Referenced by initialize().

8.27.4.8 yp

```
double jeod::SphericalHarmonicsTidalEffects::yp {}
```

Copy of polar motion coefficient yp (from polar motion class).

trick_units(-)

Definition at line 103 of file spherical_harmonics_tidal_effects.hh.

Referenced by initialize().

The documentation for this class was generated from the following files:

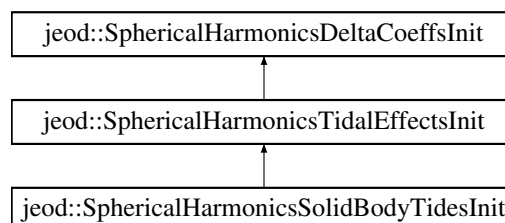
- [spherical_harmonics_tidal_effects.hh](#)
- [spherical_harmonics_tidal_effects.cc](#)

8.28 jeod::SphericalHarmonicsTidalEffectsInit Class Reference

Initializes a tidal gravity model.

```
#include <spherical_harmonics_tidal_effects_init.hh>
```

Inheritance diagram for jeod::SphericalHarmonicsTidalEffectsInit:

**Public Member Functions**

- [SphericalHarmonicsTidalEffectsInit](#) ()=default
- [~SphericalHarmonicsTidalEffectsInit](#) () override=default

Data Fields

- double `xp` {}
Copy of polar motion coefficient xp (from polar motion class).
- double `yp` {}
Copy of polar motion coefficient yp (from polar motion class).
- double `k2` {}
The love number.
- double ** `Knm` {}
A matrix of love numbers.
- `std::vector< std::string >` `tidal_body_names`
A named list of gravitational bodies contributing to this tidal effect.

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2` (jeod, `SphericalHarmonicsTidalEffectsInit`)

8.28.1 Detailed Description

Initializes a tidal gravity model.

Definition at line 83 of file `spherical_harmonics_tidal_effects_init.hh`.

8.28.2 Constructor & Destructor Documentation

8.28.2.1 `SphericalHarmonicsTidalEffectsInit()`

```
jeod::SphericalHarmonicsTidalEffectsInit::SphericalHarmonicsTidalEffectsInit ( ) [default]
```

8.28.2.2 `~SphericalHarmonicsTidalEffectsInit()`

```
jeod::SphericalHarmonicsTidalEffectsInit::~SphericalHarmonicsTidalEffectsInit ( ) [override],  
[default]
```

8.28.3 Member Function Documentation

8.28.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::SphericalHarmonicsTidalEffectsInit::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    SphericalHarmonicsTidalEffectsInit ) [private]
```

8.28.4 Field Documentation

8.28.4.1 k2

```
double jeod::SphericalHarmonicsTidalEffectsInit::k2 {}
```

The love number.

Only used for a first order tidal effect model `trick_units(-)`

Definition at line 100 of file `spherical_harmonics_tidal_effects_init.hh`.

Referenced by `jeod::SphericalHarmonicsTidalEffects::initialize()`, and `jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data::initialize()`.

8.28.4.2 Knm

```
double** jeod::SphericalHarmonicsTidalEffectsInit::Knm {}
```

A matrix of love numbers.

Used for higher order (not first) tidal effects `trick_units(-)`

Definition at line 105 of file `spherical_harmonics_tidal_effects_init.hh`.

Referenced by `jeod::SphericalHarmonicsTidalEffects::initialize()`.

8.28.4.3 tidal_body_names

```
std::vector<std::string> jeod::SphericalHarmonicsTidalEffectsInit::tidal_body_names
```

A named list of gravitational bodies contributing to this tidal effect.

`trick_units(-)`

Definition at line 110 of file `spherical_harmonics_tidal_effects_init.hh`.

Referenced by `jeod::SphericalHarmonicsTidalEffects::initialize()`, and `jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data::initialize()`.

8.28.4.4 xp

```
double jeod::SphericalHarmonicsTidalEffectsInit::xp {}
```

Copy of polar motion coefficient xp (from polar motion class).

trick_units(-)

Definition at line 90 of file spherical_harmonics_tidal_effects_init.hh.

Referenced by jeod::SphericalHarmonicsTidalEffects::initialize().

8.28.4.5 yp

```
double jeod::SphericalHarmonicsTidalEffectsInit::yp {}
```

Copy of polar motion coefficient yp (from polar motion class).

trick_units(-)

Definition at line 95 of file spherical_harmonics_tidal_effects_init.hh.

Referenced by jeod::SphericalHarmonicsTidalEffects::initialize().

The documentation for this class was generated from the following file:

- [spherical_harmonics_tidal_effects_init.hh](#)

Chapter 9

File Documentation

9.1 class_declarations.hh File Reference

Forward declarations of classes defined for the gravity model.

Namespaces

- [jeod](#)

Namespace jeod.

9.1.1 Detailed Description

Forward declarations of classes defined for the gravity model.

9.2 earth_GEMT1.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵
hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/earth_GEMT1.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- `#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_earth_GEMT1_default_data`

9.2.1 Macro Definition Documentation

9.2.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_earth_GEMT1_default_data
```

Definition at line 45 of file earth_GEMT1.cc.

9.3 earth_GEMT1.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.4 earth_GGM02C.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"  
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵  
hh"  
#include "utils/memory/include/jeod_alloc.hh"  
#include "utils/named_item/include/named_item.hh"  
#include "../include/earth_GGM02C.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_earth_GGM02C_default_data

9.4.1 Macro Definition Documentation

9.4.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_earth_GGM02C_default_data
```

Definition at line 41 of file earth_GGM02C.cc.

9.5 earth_GGM02C.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.6 earth_GGM05C.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"  
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵  
hh"  
#include "utils/memory/include/jeod_alloc.hh"  
#include "utils/named_item/include/named_item.hh"  
#include "../include/earth_GGM05C.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_earth_GGM05C_default_data

9.6.1 Macro Definition Documentation

9.6.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_earth_GGM05C_default_data
```

Definition at line 18 of file earth_GGM05C.cc.

9.7 earth_GGM05C.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data](#)

Namespaces

- [jeod](#)

Namespace jeod.

9.8 earth_solid_tides.cc File Reference

```
#include "environment/gravity/include/spherical_harmonics_delta_coeffs_↵
init.hh"
#include "environment/gravity/include/spherical_harmonics_solid_body_tides_↵
_init.hh"
#include "environment/gravity/include/spherical_harmonics_tidal_effects_↵
init.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/earth_solid_tides.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data

9.8.1 Macro Definition Documentation

9.8.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data
```

Definition at line 25 of file earth_solid_tides.cc.

9.9 earth_solid_tides.hh File Reference

Data Structures

- class [jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.10 earth_spherical.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"  
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵  
hh"  
#include "utils/named_item/include/named_item.hh"  
#include "../include/earth_spherical.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_earth_spherical_default_data

9.10.1 Macro Definition Documentation

9.10.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_earth_spherical_default_data
```

Definition at line 25 of file earth_spherical.cc.

9.11 earth_spherical.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.12 gravity_controls.cc File Reference

Define member functions for the GravityControls class.

```
#include <algorithm>
#include <cmath>
#include <cstdint>
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "environment/planet/include/planet.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame.hh"
#include "../include/gravity_controls.hh"
#include "../include/gravity_integ_frame.hh"
#include "../include/gravity_interaction.hh"
#include "../include/gravity_manager.hh"
#include "../include/gravity_messages.hh"
#include "../include/gravity_source.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Variables

- static constexpr double [jeod::speed_of_light_sq](#) = 89875517873681764.0
The speed of light squared, in m^2/s^2 .

9.12.1 Detailed Description

Define member functions for the GravityControls class.

9.13 gravity_controls.hh File Reference

Define the gravity controls.

```
#include <string>
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "gravity_source.hh"
```

Data Structures

- class [jeod::GravityControls](#)
Specifies whether and how a [GravitySource](#) affects a vehicle.

Namespaces

- [jeod](#)
Namespace jeod.

9.13.1 Detailed Description

Define the gravity controls.

9.14 gravity_integ_frame.hh File Reference

Define the gravity integration frame class.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::GravityIntegFrame](#)

Class that aids in determining whether gravity should be applied as a direct effect or a third body effect.

Namespaces

- [jeod](#)

Namespace jeod.

9.14.1 Detailed Description

Define the gravity integration frame class.

9.15 gravity_interaction.cc File Reference

Define methods for the GravityInteraction class.

```
#include <algorithm>
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "environment/planet/include/planet.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "../include/gravity_controls.hh"
#include "../include/gravity_interaction.hh"
#include "../include/gravity_manager.hh"
#include "../include/gravity_messages.hh"
#include "../include/gravity_source.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.15.1 Detailed Description

Define methods for the GravityInteraction class.

9.16 gravity_interaction.hh File Reference

Define the GravityInteraction class, used to represent the gravitational interaction between a DynBody and a set of planetary bodies.

```
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::GravityInteraction](#)

Specifies interactions between a vehicle and a set of gravitational bodies.

Namespaces

- [jeod](#)

Namespace jeod.

9.16.1 Detailed Description

Define the GravityInteraction class, used to represent the gravitational interaction between a DynBody and a set of planetary bodies.

Note that while each DynBody instance has a GravityInteraction data member, this class is defined as a part of the gravity model rather than the dyn_body model. This is because the coupling between this class and the other parts of the gravity model is much stronger than the coupling between this class and the dyn_body model.

9.17 gravity_manager.cc File Reference

Define member functions for the GravityManager class.

```
#include <cstddef>
#include <cstring>
#include <string>
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/gravity_controls.hh"
#include "../include/gravity_interaction.hh"
#include "../include/gravity_manager.hh"
#include "../include/gravity_messages.hh"
#include "../include/gravity_source.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.17.1 Detailed Description

Define member functions for the GravityManager class.

9.18 gravity_manager.hh File Reference

Define the Gravity Manager.

```
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::GravityManager](#)

The master gravitational model for a simulation.

Namespaces

- [jeod](#)

Namespace jeod.

9.18.1 Detailed Description

Define the Gravity Manager.

9.19 gravity_messages.cc File Reference

Implement the class GravityMessages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/gravity_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- `#define MAKE_GRAVITY_MESSAGE_CODE(id) JEOD_MAKE_MESSAGE_CODE(GravityMessages, "environment/gravity/", id)`

9.19.1 Detailed Description

Implement the class GravityMessages.

9.19.2 Macro Definition Documentation

9.19.2.1 MAKE_GRAVITY_MESSAGE_CODE

```
#define MAKE_GRAVITY_MESSAGE_CODE(  
    id ) JEOD_MAKE_MESSAGE_CODE(GravityMessages, "environment/gravity/", id)
```

Definition at line 43 of file gravity_messages.cc.

9.20 gravity_messages.hh File Reference

Define the class GravityMessages, the class that specifies the message IDs used in the gravity model.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::GravityMessages](#)
Specifies the message IDs used in the gravity model.

Namespaces

- [jeod](#)
Namespace jeod.

9.20.1 Detailed Description

Define the class GravityMessages, the class that specifies the message IDs used in the gravity model.

9.21 gravity_source.cc File Reference

Define member functions for the GravitySource class.

```
#include <cstdint>
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "environment/planet/include/planet.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "../include/gravity_integ_frame.hh"
#include "../include/gravity_source.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.21.1 Detailed Description

Define member functions for the GravitySource class.

9.22 gravity_source.hh File Reference

Define the gravity body base (pure virtual) class.

```
#include <string>
#include <vector>
#include "environment/ephemerides/ephem_interface/include/class_declarations.↵
hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "gravity_integ_frame.hh"
```

Data Structures

- class [jeod::GravitySource](#)

Models the gravity for a specific planet;.

Namespaces

- [jeod](#)

Namespace jeod.

9.22.1 Detailed Description

Define the gravity body base (pure virtual) class.

9.23 jupiter_spherical.cc File Reference

```
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵
hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/jupiter_spherical.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_jupiter_spherical_default_data

9.23.1 Macro Definition Documentation

9.23.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_jupiter_spherical_default_data
```

Definition at line 25 of file jupiter_spherical.cc.

9.24 jupiter_spherical.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.25 mars_MRO110B2.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵
hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/mars_MRO110B2.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_mars_MRO110B2_default_data

9.25.1 Macro Definition Documentation

9.25.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_mars_MRO110B2_default_data
```

Definition at line 35 of file mars_MRO110B2.cc.

9.26 mars_MRO110B2.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.27 mars_spherical.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵
hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/mars_spherical.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_mars_spherical_default_data

9.27.1 Macro Definition Documentation

9.27.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_mars_spherical_default_data
```

Definition at line 25 of file mars_spherical.cc.

9.28 mars_spherical.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_mars_spherical_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.29 moon_GRAIL150.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵
hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/moon_GRAIL150.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_moon_GRAIL150_default_data

9.29.1 Macro Definition Documentation

9.29.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_moon_GRAIL150_default_data
```

Definition at line 38 of file moon_GRAIL150.cc.

9.30 moon_GRAIL150.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.31 moon_LP150Q.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵
hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/moon_LP150Q.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_moon_LP150Q_default_data

9.31.1 Macro Definition Documentation

9.31.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_moon_LP150Q_default_data
```

Definition at line 40 of file moon_LP150Q.cc.

9.32 moon_LP150Q.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default_data](#)

Namespaces

- [jeod](#)

Namespace jeod.

9.33 moon_spherical.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵
hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/moon_spherical.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_moon_spherical_default_data

9.33.1 Macro Definition Documentation

9.33.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_moon_spherical_default_data
```

Definition at line 27 of file moon_spherical.cc.

9.34 moon_spherical.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_moon_spherical_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.35 spherical_harmonics_calc_nonspherical.cc File Reference

Define SphericalHarmonicsGravityControl calc_nonspherical method, which computes non-spherical gravitational acceleration of a gravitational body on a given position.

```
#include <cmath>
#include "environment/planet/include/planet.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "../include/gravity_messages.hh"
#include "../include/spherical_harmonics_delta_controls.hh"
#include "../include/spherical_harmonics_gravity_controls.hh"
#include "../include/spherical_harmonics_gravity_source.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.35.1 Detailed Description

Define SphericalHarmonicsGravityControl calc_nonspherical method, which computes non-spherical gravitational acceleration of a gravitational body on a given position.

9.36 spherical_harmonics_delta_coeffs.cc File Reference

Define member functions for the SphericalHarmonicsDeltaCoeffs class.

```
#include <cstddef>
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "../include/spherical_harmonics_delta_coeffs.hh"
#include "../include/spherical_harmonics_delta_coeffs_init.hh"
#include "../include/spherical_harmonics_delta_controls.hh"
#include "../include/spherical_harmonics_gravity_source.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.36.1 Detailed Description

Define member functions for the SphericalHarmonicsDeltaCoeffs class.

9.37 spherical_harmonics_delta_coeffs.hh File Reference

Define the class SphericalHarmonicsDeltaCoeffs, the base class for tidal effects and temporal gravity sub-models.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsDeltaCoeffs](#)
Base class for tidal and temporal gravity models.

Namespaces

- [jeod](#)
Namespace jeod.

9.37.1 Detailed Description

Define the class SphericalHarmonicsDeltaCoeffs, the base class for tidal effects and temporal gravity sub-models.

9.38 spherical_harmonics_delta_coeffs_init.hh File Reference

Define the class SphericalHarmonicsDeltaCoeffsInit, the base initialization class for tidal effects and temporal gravity sub-models.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsDeltaCoeffsInit](#)
Initialization data for a [SphericalHarmonicsDeltaCoeffs](#) instance.

Namespaces

- [jeod](#)
Namespace jeod.

9.38.1 Detailed Description

Define the class SphericalHarmonicsDeltaCoeffsInit, the base initialization class for tidal effects and temporal gravity sub-models.

9.39 spherical_harmonics_delta_controls.hh File Reference

Define the gravity controls for the variational gravity models such as solid-body tides.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsDeltaControls](#)
Provides controls for how a variational model affects a vehicle.

Namespaces

- [jeod](#)
Namespace jeod.

9.39.1 Detailed Description

Define the gravity controls for the variational gravity models such as solid-body tides.

9.40 spherical_harmonics_gravity_controls.cc File Reference

Define member functions for the SphericalHarmonicsGravityControls class.

```
#include <cmath>
#include <cstdint>
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/gravity_manager.hh"
#include "../include/gravity_messages.hh"
#include "../include/spherical_harmonics_delta_coeffs.hh"
#include "../include/spherical_harmonics_delta_controls.hh"
#include "../include/spherical_harmonics_gravity_controls.hh"
#include "../include/spherical_harmonics_gravity_source.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.40.1 Detailed Description

Define member functions for the SphericalHarmonicsGravityControls class.

9.41 spherical_harmonics_gravity_controls.hh File Reference

Define the gravity controls.

```
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "gravity_controls.hh"
#include "spherical_harmonics_gravity_source.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravityControls](#)
Specifies whether and how a [SphericalHarmonicsGravitySource](#) affects a vehicle.

Namespaces

- [jeod](#)
Namespace jeod.

9.41.1 Detailed Description

Define the gravity controls.

9.42 spherical_harmonics_gravity_source.cc File Reference

Define member functions for the SphericalHarmonicsGravitySource class.

```
#include <cmath>
#include <cstdlib>
#include <cstring>
#include <typeinfo>
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "utils/math/include/numerical.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/gravity_manager.hh"
#include "../include/gravity_messages.hh"
#include "../include/spherical_harmonics_delta_coeffs.hh"
#include "../include/spherical_harmonics_delta_coeffs_init.hh"
#include "../include/spherical_harmonics_gravity_source.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.42.1 Detailed Description

Define member functions for the SphericalHarmonicsGravitySource class.

9.43 spherical_harmonics_gravity_source.hh File Reference

Define the spherical harmonics implementation of a gravity body.

```
#include <vector>
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "gravity_source.hh"
#include "spherical_harmonics_delta_coeffs.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource](#)
Models the gravity for a specific planet using spherical harmonics.

Namespaces

- [jeod](#)
Namespace jeod.

9.43.1 Detailed Description

Define the spherical harmonics implementation of a gravity body.

9.44 spherical_harmonics_gravity_source_default_data.hh File Reference

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.45 spherical_harmonics_solid_body_tides.cc File Reference

Define member functions for the SphericalHarmonicsSolidBodyTides class.

```
#include <cmath>
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "environment/planet/include/planet.hh"
#include "utils/math/include/vector3.hh"
#include "utils/ref_frames/include/ref_frame.hh"
#include "../include/spherical_harmonics_delta_coeffs_init.hh"
#include "../include/spherical_harmonics_gravity_controls.hh"
#include "../include/spherical_harmonics_gravity_source.hh"
#include "../include/spherical_harmonics_solid_body_tides.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.45.1 Detailed Description

Define member functions for the SphericalHarmonicsSolidBodyTides class.

9.46 spherical_harmonics_solid_body_tides.hh File Reference

Define the SphericalHarmonicsSolidBodyTides class, which models solid-body tidal effects.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "spherical_harmonics_tidal_effects.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsSolidBodyTides](#)
Models solid body tidal effects.

Namespaces

- [jeod](#)
Namespace jeod.

9.46.1 Detailed Description

Define the SphericalHarmonicsSolidBodyTides class, which models solid-body tidal effects.

SphericalHarmonicsSolidBodyTides inherits directly from the SphericalHarmonicsTidalEffects class.

9.47 spherical_harmonics_solid_body_tides_init.hh File Reference

Define the SphericalHarmonicsSolidBodyTidesInit class, which is the initialization class for the solid body tides model.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "spherical_harmonics_tidal_effects_init.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsSolidBodyTidesInit](#)
Initializes a solid body tides model.

Namespaces

- [jeod](#)
Namespace jeod.

9.47.1 Detailed Description

Define the SphericalHarmonicsSolidBodyTidesInit class, which is the initialization class for the solid body tides model.

9.48 spherical_harmonics_tidal_effects.cc File Reference

Define member functions for the SphericalHarmonicsTidalEffects class.

```
#include <cstdint>
#include "dynamics/dyn_manager/include/base_dyn_manager.hh"
#include "environment/planet/include/planet.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame.hh"
#include "../include/gravity_messages.hh"
#include "../include/spherical_harmonics_delta_coeffs_init.hh"
#include "../include/spherical_harmonics_gravity_source.hh"
#include "../include/spherical_harmonics_tidal_effects.hh"
#include "../include/spherical_harmonics_tidal_effects_init.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.48.1 Detailed Description

Define member functions for the SphericalHarmonicsTidalEffects class.

9.49 spherical_harmonics_tidal_effects.hh File Reference

Define the class SphericalHarmonicsTidalEffects, which is the base class for solid-body and ocean tidal effects.

```
#include "environment/planet/include/class_declarations.hh"
#include "utils/ref_frames/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "spherical_harmonics_delta_coeffs.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsTidalEffects](#)
Models tidal effects as a delta on top of a gravity model.

Namespaces

- [jeod](#)
Namespace jeod.

9.49.1 Detailed Description

Define the class SphericalHarmonicsTidalEffects, which is the base class for solid-body and ocean tidal effects.

SphericalHarmonicsTidalEffects inherits directly from the SphericalHarmonicsDeltaCoeffs class.

9.50 spherical_harmonics_tidal_effects_init.hh File Reference

Define the SphericalHarmonicsTidalEffectsInit class, the initialization class for tidal effects models.

```
#include <string>
#include <vector>
#include "utils/sim_interface/include/jeod_class.hh"
#include "spherical_harmonics_delta_coeffs_init.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsTidalEffectsInit](#)
Initializes a tidal gravity model.

Namespaces

- [jeod](#)

Namespace jeod.

9.50.1 Detailed Description

Define the SphericalHarmonicsTidalEffectsInit class, the initialization class for tidal effects models.

9.51 sun_spherical.cc File Reference

```
#include "environment/gravity/include/gravity_source.hh"
#include "environment/gravity/include/spherical_harmonics_gravity_source.↵
hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/sun_spherical.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- `#define` [JEOD_FRIEND_CLASS](#) SphericalHarmonicsGravitySource_sun_spherical_default_data

9.51.1 Macro Definition Documentation

9.51.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS SphericalHarmonicsGravitySource_sun_spherical_default_data
```

Definition at line 28 of file sun_spherical.cc.

9.52 sun_spherical.hh File Reference

```
#include "spherical_harmonics_gravity_source_default_data.hh"
```

Data Structures

- class [jeod::SphericalHarmonicsGravitySource_sun_spherical_default_data](#)

Namespaces

- [jeod](#)

Namespace jeod.

Index

- ~GravityControls
 - jeod::GravityControls, [17](#)
- ~GravityIntegFrame
 - jeod::GravityIntegFrame, [28](#)
- ~GravityInteraction
 - jeod::GravityInteraction, [31](#)
- ~GravityManager
 - jeod::GravityManager, [36](#)
- ~GravitySource
 - jeod::GravitySource, [46](#)
- ~SphericalHarmonicsDeltaCoeffs
 - jeod::SphericalHarmonicsDeltaCoeffs, [50](#)
- ~SphericalHarmonicsDeltaCoeffsInit
 - jeod::SphericalHarmonicsDeltaCoeffsInit, [54](#)
- ~SphericalHarmonicsDeltaControls
 - jeod::SphericalHarmonicsDeltaControls, [57](#)
- ~SphericalHarmonicsGravityControls
 - jeod::SphericalHarmonicsGravityControls, [61](#)
- ~SphericalHarmonicsGravitySource
 - jeod::SphericalHarmonicsGravitySource, [76](#)
- ~SphericalHarmonicsGravitySource_default_data
 - jeod::SphericalHarmonicsGravitySource_default_data, [85](#)
- ~SphericalHarmonicsSolidBodyTides
 - jeod::SphericalHarmonicsSolidBodyTides, [97](#)
- ~SphericalHarmonicsSolidBodyTidesInit
 - jeod::SphericalHarmonicsSolidBodyTidesInit, [99](#)
- ~SphericalHarmonicsTidalEffects
 - jeod::SphericalHarmonicsTidalEffects, [102](#)
- ~SphericalHarmonicsTidalEffectsInit
 - jeod::SphericalHarmonicsTidalEffectsInit, [106](#)
- a_by_rad
 - jeod::SphericalHarmonicsGravitySource, [78](#)
- accel
 - jeod::GravityIntegFrame, [28](#)
- accel_mag_less_ptr
 - jeod::GravityControls, [17](#)
- active
 - jeod::GravityControls, [22](#)
 - jeod::SphericalHarmonicsDeltaControls, [57](#)
- add_control
 - jeod::GravityInteraction, [31](#)
- add_deltacoeff
 - jeod::SphericalHarmonicsGravitySource, [77](#)
- add_deltacontrol
 - jeod::SphericalHarmonicsGravityControls, [62](#)
- add_grav_source
 - jeod::GravityManager, [37](#)
- alpha
 - jeod::SphericalHarmonicsGravitySource, [79](#)
- battin_method
 - jeod::GravityControls, [23](#)
- beta
 - jeod::SphericalHarmonicsGravitySource, [79](#)
- body
 - jeod::GravityControls, [23](#)
- calc_nonspherical
 - jeod::GravityControls, [18](#)
 - jeod::SphericalHarmonicsGravityControls, [62](#)
- calc_relativistic
 - jeod::GravityControls, [18](#)
- calc_spherical
 - jeod::GravityControls, [20](#)
- check_validity
 - jeod::SphericalHarmonicsGravityControls, [64](#)
- class_declarations.hh, [109](#)
- Cnm
 - jeod::SphericalHarmonicsGravitySource, [79](#)
- dC20
 - jeod::SphericalHarmonicsDeltaCoeffs, [51](#)
- degree
 - jeod::SphericalHarmonicsDeltaCoeffs, [52](#)
 - jeod::SphericalHarmonicsDeltaCoeffsInit, [55](#)
 - jeod::SphericalHarmonicsDeltaControls, [57](#)
 - jeod::SphericalHarmonicsGravityControls, [71](#)
 - jeod::SphericalHarmonicsGravitySource, [79](#)
- delta_Cnm
 - jeod::SphericalHarmonicsDeltaCoeffs, [52](#)
 - jeod::SphericalHarmonicsDeltaCoeffsInit, [55](#)
 - jeod::SphericalHarmonicsGravityControls, [71](#)
- delta_coeffs
 - jeod::SphericalHarmonicsGravitySource, [80](#)
- delta_degree
 - jeod::SphericalHarmonicsGravityControls, [71](#)
- delta_order
 - jeod::SphericalHarmonicsGravityControls, [72](#)
- delta_Snm
 - jeod::SphericalHarmonicsDeltaCoeffs, [52](#)
 - jeod::SphericalHarmonicsDeltaCoeffsInit, [55](#)
 - jeod::SphericalHarmonicsGravityControls, [72](#)
- disable_min_radius_warnings
 - jeod::SphericalHarmonicsGravityControls, [64](#)
- domain_error
 - jeod::GravityMessages, [42](#)
- duplicate_entry
 - jeod::GravityMessages, [43](#)

- earth_GEMT1.cc, 109
 - JEOD_FRIEND_CLASS, 110
- earth_GEMT1.hh, 110
- earth_GGM02C.cc, 110
 - JEOD_FRIEND_CLASS, 111
- earth_GGM02C.hh, 111
- earth_GGM05C.cc, 111
 - JEOD_FRIEND_CLASS, 112
- earth_GGM05C.hh, 112
- earth_solid_tides.cc, 112
 - JEOD_FRIEND_CLASS, 113
- earth_solid_tides.hh, 113
- earth_spherical.cc, 113
 - JEOD_FRIEND_CLASS, 113
- earth_spherical.hh, 114
- Environment, 11
- eta
 - jeod::SphericalHarmonicsGravitySource, 80
- find_deltacoeff
 - jeod::SphericalHarmonicsGravitySource, 77
- find_grav_source
 - jeod::GravityManager, 37
- first_order_only
 - jeod::SphericalHarmonicsDeltaControls, 58
- frames
 - jeod::GravitySource, 47
- get_bodies
 - jeod::GravityManager, 38
- get_degree
 - jeod::SphericalHarmonicsGravityControls, 64
- get_degree_order
 - jeod::SphericalHarmonicsGravityControls, 65
- get_grad_degree
 - jeod::SphericalHarmonicsGravityControls, 65
- get_grad_degree_order
 - jeod::SphericalHarmonicsGravityControls, 65
- get_grad_order
 - jeod::SphericalHarmonicsGravityControls, 67
- get_order
 - jeod::SphericalHarmonicsGravityControls, 67
- gradient
 - jeod::GravityControls, 23
- gradient_degree
 - jeod::SphericalHarmonicsGravityControls, 72
- gradient_order
 - jeod::SphericalHarmonicsGravityControls, 72
- grav_accel
 - jeod::GravityControls, 24
 - jeod::GravityInteraction, 34
- grav_accel_magsq
 - jeod::GravityControls, 24
- grav_controls
 - jeod::GravityInteraction, 34
- grav_effect
 - jeod::SphericalHarmonicsDeltaControls, 58
- grav_grad
 - jeod::GravityControls, 24
- jeod::GravityInteraction, 34
- grav_manager
 - jeod::GravityControls, 24
- grav_pot
 - jeod::GravityControls, 25
 - jeod::GravityInteraction, 35
- grav_source
 - jeod::SphericalHarmonicsDeltaCoeffs, 52
 - jeod::SphericalHarmonicsDeltaControls, 58
- gravitation
 - jeod::GravityControls, 20, 21
 - jeod::GravityManager, 38, 39
- Gravity, 11
- gravity_controls.cc, 114
- gravity_controls.hh, 115
- gravity_integ_frame.hh, 115
- gravity_interaction.cc, 116
- gravity_interaction.hh, 117
- gravity_manager.cc, 117
- gravity_manager.hh, 118
- gravity_messages.cc, 118
 - MAKE_GRAVITY_MESSAGE_CODE, 119
- gravity_messages.hh, 119
- gravity_source.cc, 120
- gravity_source.hh, 120
- GravityControls
 - jeod::GravityControls, 17
- GravityIntegFrame
 - jeod::GravityIntegFrame, 28
- GravityInteraction
 - jeod::GravityInteraction, 31
- GravityManager
 - jeod::GravityManager, 36, 37
- GravityMessages
 - jeod::GravityMessages, 42
- GravitySource
 - jeod::GravitySource, 46
- harmonics_source
 - jeod::SphericalHarmonicsGravityControls, 73
- inertial
 - jeod::GravitySource, 47
- initialize
 - jeod::SphericalHarmonicsDeltaCoeffs, 50
 - jeod::SphericalHarmonicsGravitySource_default_data, 85
 - jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data, 86
 - jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data, 87
 - jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data, 88
 - jeod::SphericalHarmonicsGravitySource_earth_spherical_default_data, 89
 - jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default_data, 90
 - jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default_data, 90

- jeod::SphericalHarmonicsGravitySource_mars_spherical_data, 91
- jeod::SphericalHarmonicsGravitySource_moon_GRAIL150Q_data, 92
- jeod::SphericalHarmonicsGravitySource_moon_LP150Q_data, 93
- jeod::SphericalHarmonicsGravitySource_moon_spherical_data, 94
- jeod::SphericalHarmonicsGravitySource_sun_spherical_data, 95
- jeod::SphericalHarmonicsSolidBodyTides, 97
- jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_data, 100
- jeod::SphericalHarmonicsTidalEffects, 102
- initialize_body
 - jeod::SphericalHarmonicsGravitySource, 78
- initialize_control
 - jeod::GravityControls, 21
 - jeod::SphericalHarmonicsGravityControls, 67
- initialize_controls
 - jeod::GravityInteraction, 32
- initialize_model
 - jeod::GravityManager, 39
- initialize_state
 - jeod::GravityManager, 40
 - jeod::GravitySource, 46
- int_to_double
 - jeod::SphericalHarmonicsGravitySource, 80
- integ_frame_index
 - jeod::GravityInteraction, 35
- invalid_limit
 - jeod::GravityMessages, 43
- invalid_name
 - jeod::GravityMessages, 43
- invalid_object
 - jeod::GravityMessages, 43
- is_third_body
 - jeod::GravityIntegFrame, 28
- jeod, 13
 - speed_of_light_sq, 14
- jeod::GravityControls, 15
 - ~GravityControls, 17
 - accel_mag_less_ptr, 17
 - active, 22
 - battin_method, 23
 - body, 23
 - calc_nonspherical, 18
 - calc_relativistic, 18
 - calc_spherical, 20
 - gradient, 23
 - grav_accel, 24
 - grav_accel_magsq, 24
 - grav_grad, 24
 - grav_manager, 24
 - grav_pot, 25
 - gravitation, 20, 21
 - GravityControls, 17
 - initialize_control, 21
 - DEFAULT_CLASS_ESTABLISH_FRIENDS2, 22
 - operator=, 22
 - perturb_grad, 25
 - relativistic, 25
 - result_data, 22
 - skip_spherical, 25
 - source_data, 26
 - spherical, 26
 - subscribed_to_inertial, 26
 - subscribed_to_pfix, 26
- jeod::GravityIntegFrame, 27
 - ~GravityIntegFrame, 28
 - accel, 28
 - GravityIntegFrame, 28
 - is_third_body, 28
 - p, 28
 - pos, 29
 - ref_frame, 29
 - time, 29
- jeod::GravityInteraction, 30
 - ~GravityInteraction, 31
 - add_control, 31
 - grav_accel, 34
 - grav_controls, 34
 - grav_grad, 34
 - grav_pot, 35
 - GravityInteraction, 31
 - initialize_controls, 32
 - integ_frame_index, 35
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 32
 - operator=, 32
 - remove_control, 32
 - reset_controls, 33
 - set_integ_frame, 33
 - sort_controls, 33
- jeod::GravityManager, 35
 - ~GravityManager, 36
 - add_grav_source, 37
 - find_grav_source, 37
 - get_bodies, 38
 - gravitation, 38, 39
 - GravityManager, 36, 37
 - initialize_model, 39
 - initialize_state, 40
 - operator=, 40
 - p, 40
 - sources, 41
- jeod::GravityMessages, 41
 - domain_error, 42
 - duplicate_entry, 43
 - GravityMessages, 42
 - invalid_limit, 43
 - invalid_name, 43
 - invalid_object, 43
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 42
 - missing_entry, 44
 - null_pointer, 44
 - operator=, 42

- jeod::GravitySource, 45
 - ~GravitySource, 46
 - frames, 47
 - GravitySource, 46
 - inertial, 47
 - initialize_state, 46
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 47
 - mu, 47
 - name, 48
 - operator=, 47
 - pfix, 48
- jeod::SphericalHarmonicsDeltaCoeffs, 49
 - ~SphericalHarmonicsDeltaCoeffs, 50
 - dC20, 51
 - degree, 52
 - delta_Cnm, 52
 - delta_Snm, 52
 - grav_source, 52
 - initialize, 50
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 51
 - order, 53
 - SphericalHarmonicsDeltaCoeffs, 50
 - update, 51
- jeod::SphericalHarmonicsDeltaCoeffsInit, 53
 - ~SphericalHarmonicsDeltaCoeffsInit, 54
 - degree, 55
 - delta_Cnm, 55
 - delta_Snm, 55
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 54
 - order, 55
 - SphericalHarmonicsDeltaCoeffsInit, 54
- jeod::SphericalHarmonicsDeltaControls, 56
 - ~SphericalHarmonicsDeltaControls, 57
 - active, 57
 - degree, 57
 - first_order_only, 58
 - grav_effect, 58
 - grav_source, 58
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 57
 - order, 58
 - SphericalHarmonicsDeltaControls, 57
- jeod::SphericalHarmonicsGravityControls, 59
 - ~SphericalHarmonicsGravityControls, 61
 - add_deltacontrol, 62
 - calc_nonspherical, 62
 - check_validity, 64
 - degree, 71
 - delta_Cnm, 71
 - delta_degree, 71
 - delta_order, 72
 - delta_Snm, 72
 - disable_min_radius_warnings, 64
 - get_degree, 64
 - get_degree_order, 65
 - get_grad_degree, 65
 - get_grad_degree_order, 65
 - get_grad_order, 67
 - get_order, 67
 - gradient_degree, 72
 - gradient_order, 72
 - harmonics_source, 73
 - initialize_control, 67
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 68
 - min_radius_warn, 73
 - operator=, 68
 - order, 73
 - Pnm, 74
 - set_degree, 68
 - set_degree_order, 68
 - set_grad_degree, 69
 - set_grad_degree_order, 69
 - set_grad_order, 69
 - set_order, 70
 - SphericalHarmonicsGravityControls, 61, 62
 - sum_deltacoeffs, 70
 - total_dC20, 74
 - update_deltacoeffs, 70
 - var_effects, 74
- jeod::SphericalHarmonicsGravitySource, 75
 - ~SphericalHarmonicsGravitySource, 76
 - a_by_rad, 78
 - add_deltacoeff, 77
 - alpha, 79
 - beta, 79
 - Cnm, 79
 - degree, 79
 - delta_coeffs, 80
 - eta, 80
 - find_deltacoeff, 77
 - initialize_body, 78
 - int_to_double, 80
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 78
 - nrdiag, 81
 - operator=, 78
 - order, 81
 - radius, 81
 - Snm, 82
 - SphericalHarmonicsGravitySource, 76, 77
 - tide_free, 82
 - tide_free_delta, 82
 - upsilon, 83
 - xi, 83
 - zeta, 83
- jeod::SphericalHarmonicsGravitySource_default_data, 84
 - ~SphericalHarmonicsGravitySource_default_data, 85
 - initialize, 85
- jeod::SphericalHarmonicsGravitySource_earth_GEMT1_default_data, 85
 - initialize, 86
- jeod::SphericalHarmonicsGravitySource_earth_GGM02C_default_data, 86
 - initialize, 87
- jeod::SphericalHarmonicsGravitySource_earth_GGM05C_default_data, 87
 - initialize, 87

- initialize, 88
- jeod::SphericalHarmonicsGravitySource_earth_spherical_default, 88
- initialize, 89
- jeod::SphericalHarmonicsGravitySource_jupiter_spherical_default, 89
- initialize, 90
- jeod::SphericalHarmonicsGravitySource_mars_MRO110B2_default, 90
- initialize, 90
- jeod::SphericalHarmonicsGravitySource_mars_spherical_default, 91
- initialize, 91
- jeod::SphericalHarmonicsGravitySource_moon_GRAIL150_default, 92
- initialize, 92
- jeod::SphericalHarmonicsGravitySource_moon_LP150Q_default, 93
- initialize, 93
- jeod::SphericalHarmonicsGravitySource_moon_spherical_default, 94
- initialize, 94
- jeod::SphericalHarmonicsGravitySource_sun_spherical_default, 95
- initialize, 95
- jeod::SphericalHarmonicsSolidBodyTides, 96
- ~SphericalHarmonicsSolidBodyTides, 97
- initialize, 97
- JEOD_CLASS_ESTABLISH_FRIENDS2, 97
- SphericalHarmonicsSolidBodyTides, 97
- update, 98
- jeod::SphericalHarmonicsSolidBodyTidesInit, 98
- ~SphericalHarmonicsSolidBodyTidesInit, 99
- p, 99
- SphericalHarmonicsSolidBodyTidesInit, 99
- jeod::SphericalHarmonicsSolidBodyTidesInit_earth_solid_tides_earth_data, 100
- initialize, 100
- jeod::SphericalHarmonicsTidalEffects, 100
- ~SphericalHarmonicsTidalEffects, 102
- initialize, 102
- JEOD_CLASS_ESTABLISH_FRIENDS2, 102
- k2, 103
- KnM, 103
- num_tidal_bodies, 103
- prefix, 104
- SphericalHarmonicsTidalEffects, 101
- tidal_bodies, 104
- tidal_bodies_inertial, 104
- update, 103
- xp, 104
- yp, 105
- jeod::SphericalHarmonicsTidalEffectsInit, 105
- ~SphericalHarmonicsTidalEffectsInit, 106
- JEOD_CLASS_ESTABLISH_FRIENDS2, 106
- k2, 107
- KnM, 107
- SphericalHarmonicsTidalEffectsInit, 106
- tidal_body_names, 107
- update, 107
- yp, 108
- JEOD_CLASS_ESTABLISH_FRIENDS2
- jeod::GravityControls, 22
- jeod::GravityInteraction, 32
- jeod::GravityMessages, 42
- jeod::GravitySource, 47
- jeod::SphericalHarmonicsDeltaCoeffs, 51
- jeod::SphericalHarmonicsDeltaCoeffsInit, 54
- jeod::SphericalHarmonicsDeltaControls, 57
- jeod::SphericalHarmonicsGravityControls, 68
- jeod::SphericalHarmonicsGravitySource, 78
- jeod::SphericalHarmonicsSolidBodyTides, 97
- jeod::SphericalHarmonicsTidalEffects, 102
- jeod::SphericalHarmonicsTidalEffectsInit, 106
- JEOD_FRIEND_CLASS
- earth_GEMT1.cc, 110
- earth_GGM02C.cc, 111
- earth_GGM05C.cc, 112
- earth_solid_tides.cc, 113
- earth_spherical.cc, 113
- jupiter_spherical.cc, 121
- mars_MRO110B2.cc, 122
- mars_spherical.cc, 123
- moon_GRAIL150.cc, 124
- moon_LP150Q.cc, 125
- moon_spherical.cc, 126
- sun_spherical.cc, 135
- jupiter_spherical.cc, 121
- JEOD_FRIEND_CLASS, 121
- jupiter_spherical.hh, 121
- k2
- jeod::SphericalHarmonicsTidalEffects, 103
- jeod::SphericalHarmonicsTidalEffectsInit, 107
- KnM
- jeod::SphericalHarmonicsTidalEffects, 103
- jeod::SphericalHarmonicsTidalEffectsInit, 107
- MAKE_GRAVITY_MESSAGE_CODE
- gravity_messages.cc, 119
- mars_MRO110B2.cc, 122
- JEOD_FRIEND_CLASS, 122
- mars_MRO110B2.hh, 122
- mars_spherical.cc, 123
- JEOD_FRIEND_CLASS, 123
- mars_spherical.hh, 123
- min_radius_warn
- jeod::SphericalHarmonicsGravityControls, 73
- missing_entry
- jeod::GravityMessages, 44
- Models, 11
- moon_GRAIL150.cc, 124
- JEOD_FRIEND_CLASS, 124
- moon_GRAIL150.hh, 124
- moon_LP150Q.cc, 125
- JEOD_FRIEND_CLASS, 125
- moon_LP150Q.hh, 125

- moon_spherical.cc, 126
 - JEOD_FRIEND_CLASS, 126
- moon_spherical.hh, 126
- mu
 - jeod::GravitySource, 47
- name
 - jeod::GravitySource, 48
- nrdiag
 - jeod::SphericalHarmonicsGravitySource, 81
- null_pointer
 - jeod::GravityMessages, 44
- num_tidal_bodies
 - jeod::SphericalHarmonicsTidalEffects, 103
- operator=
 - jeod::GravityControls, 22
 - jeod::GravityInteraction, 32
 - jeod::GravityManager, 40
 - jeod::GravityMessages, 42
 - jeod::GravitySource, 47
 - jeod::SphericalHarmonicsGravityControls, 68
 - jeod::SphericalHarmonicsGravitySource, 78
- order
 - jeod::SphericalHarmonicsDeltaCoeffs, 53
 - jeod::SphericalHarmonicsDeltaCoeffsInit, 55
 - jeod::SphericalHarmonicsDeltaControls, 58
 - jeod::SphericalHarmonicsGravityControls, 73
 - jeod::SphericalHarmonicsGravitySource, 81
- p
 - jeod::GravityIntegFrame, 28
 - jeod::GravityManager, 40
 - jeod::SphericalHarmonicsSolidBodyTidesInit, 99
- perturbing_only
 - jeod::GravityControls, 25
- prefix
 - jeod::GravitySource, 48
 - jeod::SphericalHarmonicsTidalEffects, 104
- Pnm
 - jeod::SphericalHarmonicsGravityControls, 74
- pos
 - jeod::GravityIntegFrame, 29
- radius
 - jeod::SphericalHarmonicsGravitySource, 81
- ref_frame
 - jeod::GravityIntegFrame, 29
- relativistic
 - jeod::GravityControls, 25
- remove_control
 - jeod::GravityInteraction, 32
- reset_control
 - jeod::GravityControls, 22
- reset_controls
 - jeod::GravityInteraction, 33
- set_degree
 - jeod::SphericalHarmonicsGravityControls, 68
- set_degree_order
 - jeod::SphericalHarmonicsGravityControls, 68
- set_grad_degree
 - jeod::SphericalHarmonicsGravityControls, 69
- set_grad_degree_order
 - jeod::SphericalHarmonicsGravityControls, 69
- set_grad_order
 - jeod::SphericalHarmonicsGravityControls, 69
- set_integ_frame
 - jeod::GravityInteraction, 33
- set_order
 - jeod::SphericalHarmonicsGravityControls, 70
- skip_spherical
 - jeod::GravityControls, 25
- Snm
 - jeod::SphericalHarmonicsGravitySource, 82
- sort_controls
 - jeod::GravityInteraction, 33
- source_name
 - jeod::GravityControls, 26
- sources
 - jeod::GravityManager, 41
- speed_of_light_sq
 - jeod, 14
- spherical
 - jeod::GravityControls, 26
- spherical_harmonics_calc_nonspherical.cc, 127
- spherical_harmonics_delta_coeffs.cc, 127
- spherical_harmonics_delta_coeffs.hh, 128
- spherical_harmonics_delta_coeffs_init.hh, 128
- spherical_harmonics_delta_controls.hh, 129
- spherical_harmonics_gravity_controls.cc, 129
- spherical_harmonics_gravity_controls.hh, 130
- spherical_harmonics_gravity_source.cc, 130
- spherical_harmonics_gravity_source.hh, 131
- spherical_harmonics_gravity_source_default_data.hh, 131
- spherical_harmonics_solid_body_tides.cc, 132
- spherical_harmonics_solid_body_tides.hh, 132
- spherical_harmonics_solid_body_tides_init.hh, 133
- spherical_harmonics_tidal_effects.cc, 133
- spherical_harmonics_tidal_effects.hh, 134
- spherical_harmonics_tidal_effects_init.hh, 134
- SphericalHarmonicsDeltaCoeffs
 - jeod::SphericalHarmonicsDeltaCoeffs, 50
- SphericalHarmonicsDeltaCoeffsInit
 - jeod::SphericalHarmonicsDeltaCoeffsInit, 54
- SphericalHarmonicsDeltaControls
 - jeod::SphericalHarmonicsDeltaControls, 57
- SphericalHarmonicsGravityControls
 - jeod::SphericalHarmonicsGravityControls, 61, 62
- SphericalHarmonicsGravitySource
 - jeod::SphericalHarmonicsGravitySource, 76, 77
- SphericalHarmonicsSolidBodyTides
 - jeod::SphericalHarmonicsSolidBodyTides, 97
- SphericalHarmonicsSolidBodyTidesInit
 - jeod::SphericalHarmonicsSolidBodyTidesInit, 99
- SphericalHarmonicsTidalEffects

- [jeod::SphericalHarmonicsTidalEffects, 101](#)
- [SphericalHarmonicsTidalEffectsInit](#)
 - [jeod::SphericalHarmonicsTidalEffectsInit, 106](#)
- [subscribed_to_inertial](#)
 - [jeod::GravityControls, 26](#)
- [subscribed_to_pfix](#)
 - [jeod::GravityControls, 26](#)
- [sum_deltacoeffs](#)
 - [jeod::SphericalHarmonicsGravityControls, 70](#)
- [sun_spherical.cc, 135](#)
 - [JEOD_FRIEND_CLASS, 135](#)
- [sun_spherical.hh, 135](#)
- [tidal_bodies](#)
 - [jeod::SphericalHarmonicsTidalEffects, 104](#)
- [tidal_bodies_inertial](#)
 - [jeod::SphericalHarmonicsTidalEffects, 104](#)
- [tidal_body_names](#)
 - [jeod::SphericalHarmonicsTidalEffectsInit, 107](#)
- [tide_free](#)
 - [jeod::SphericalHarmonicsGravitySource, 82](#)
- [tide_free_delta](#)
 - [jeod::SphericalHarmonicsGravitySource, 82](#)
- [time](#)
 - [jeod::GravityIntegFrame, 29](#)
- [total_dC20](#)
 - [jeod::SphericalHarmonicsGravityControls, 74](#)
- [update](#)
 - [jeod::SphericalHarmonicsDeltaCoeffs, 51](#)
 - [jeod::SphericalHarmonicsSolidBodyTides, 98](#)
 - [jeod::SphericalHarmonicsTidalEffects, 103](#)
- [update_deltacoeffs](#)
 - [jeod::SphericalHarmonicsGravityControls, 70](#)
- [upsilon](#)
 - [jeod::SphericalHarmonicsGravitySource, 83](#)
- [var_effects](#)
 - [jeod::SphericalHarmonicsGravityControls, 74](#)
- [xi](#)
 - [jeod::SphericalHarmonicsGravitySource, 83](#)
- [xp](#)
 - [jeod::SphericalHarmonicsTidalEffects, 104](#)
 - [jeod::SphericalHarmonicsTidalEffectsInit, 107](#)
- [yp](#)
 - [jeod::SphericalHarmonicsTidalEffects, 105](#)
 - [jeod::SphericalHarmonicsTidalEffectsInit, 108](#)
- [zeta](#)
 - [jeod::SphericalHarmonicsGravitySource, 83](#)